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# 9 Probability & Statistics

## Introduction

Probability is a concept which measure the degree of certainty or uncertainty. Probability is basically originated to predict the chance of winning the game, mathematically. A big gambling industry depends upon the basic theory of probability. In general, it is a chance that something will happen in the future or in the given statement.

## Useful Terms and Definitions

### Sample Space

A sample space is the set of all possible outcomes of the experiment. It is denoted by $S$. Elements of sample space is referred as sample points. Some important sample space are given as follows :

**Coin or Unbiased Coin**

1. Tossing a single coin or unbiased coin,
$$S = \{H, T\}$$
Number of sample points = 2
2. Tossing of three coins
$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$
Number of sample points = 8
3. In general for tossing of $n$ coins $S = 2^n$,
where, $n$ = Number of coins

Coins

| 1st | 2nd | 3rd |
|---|---|---|
| H | H | H |
| H | H | T |
| H | T | H |
| H | T | T |
| T | H | H |
| T | H | T |
| T | T | H |
| T | T | T |

Fig. Sample space of 3 coins

**Dice or a Fair Die**

1. For throwing a dice $S = \{1, 2, 3, 4, 5, 6\}$
Number of sample point = 6
2. For throwing two dice
Sum of two dice (sample space and total favorable events)

$S =$

| Sum → 2 | 3 | 4 | 5 | 6 | 7 | Sum |
|---|---|---|---|---|---|---|
| (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) | |
| (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) | 8 |
| (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) | 9 |
| (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) | 10 |
| (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) | 11 |
| (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) | 12 |

Number of sample points = 36

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3. In general for throwing $n$ dice

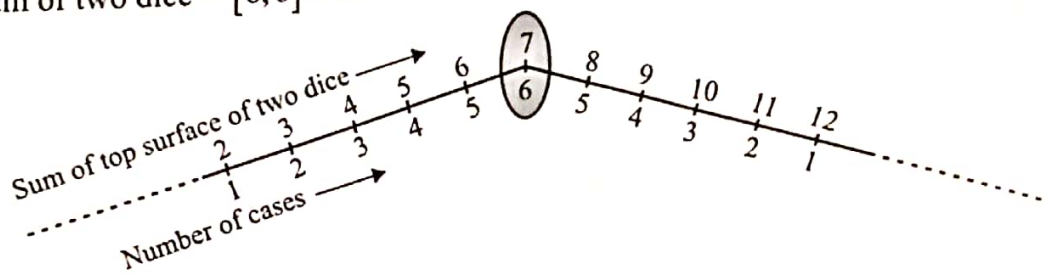
$$S = 6^n \qquad \because \quad n = \text{Number of dice}$$

**Triangle theory for dice :**

When two dices are thrown then the possibility of getting total sum on the top surface of the dice is shown below,

Minimum sum of two dice = $[1,1] = 2$

Maximum sum of two dice = $[6,6] = 12$




**Sample Space for a Pack of 52 playing Cards**

Total number of cards = 52 (except 2 jokers and one Blank card)

- 26 Black suits [13 Spades and 13 Clubs]
- 26 Red suits [13 Hearts and 13 Diamonds]

Total number of face cards = 12 (4 Kings, 4 Queens, 4 Jacks)

| Suits | Ace (A) | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Face Cards: Jack | Queen | King |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Club ♣ | Non-Face Cards 4 x 10 = 40 Cards | | | | | | | | | | ✓ | ✓ | ✓ |
| Diamond ♦ | | | | | | | | | | | ✓ | ✓ | ✓ |
| Heart ♥ | | | | | | | | | | | ✓ | ✓ | ✓ |
| Spades ♠ | | | | | | | | | | | ✓ | ✓ | ✓ |
| | | | | | | | | | | | 4 × 3 = 12 Cards | | |

**Random Experiment :**

A random experiment is an experiment or a process for which the outcomes cannot be predicted with certainty.

**Example :** In an experiment of throwing a dice and getting a number 1, 2, 3, 4, 5 or 6 are different events.

## Events :

A set of one or more outcome of an random experiment is called event.

**Types of Events :**

1. **Equally Likely Events :** Events are said to be equally likely when the chances are same for occurrence of all events.

   **Example :** When a dice is thrown any one number from 1 to 6 may occur. In this trial, the six events are equally likely.



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2. **Mutually Exhaustive Events :** Two or more events in any trial are known as exhaustive events. If one of them is necessarily (must) occurs.

**Example :** When a dice is thrown, there are six exhaustive events.

3. **Mutually Exclusive Events :** If the occurrence of anyone of the events in a trial prevents the occurrence of the other events, then the events are said to be mutually exclusive events.

**Example :** When a dice is thrown the event of getting faces numbered 1 to 6 are mutually exclusive.

**Key Point**

If $A$ and $B$ both are mutually exclusive events in same sample space then $P(A \cap B) = 0$.

4. **Independent Events :** If there are two or more event such that the occurrence of any one does not depend on occurrence of other, they are said to be independent event.

**Example :** Throwing two dice, event $A$ is face 4 in first dice and event $B$ face 3 in second dice, these both events are independent, and also not mutually exclusive because it can happen simultaneously.

**Key Point**

If $A$ and $B$ both are independent events in different sample space then $P(A \cap B) = P(A)P(B)$.

5. **Dependent Events :** When two events are dependent, the occurrence of one event influences the probability of another event.

**Example :** Throwing two dice, event $A$ is face 4 in first dice and event $B$ is sum total value 7. The value 4 in first dice gives the information that for total sum 7 it should be 3 in second dice.

## Classical Definition of Probability

In a random experiment, probability is the ratio of favourable events to corresponding sample space. The probability of event $E$ in the sample space $S$ is defined as,

$$P(E) = \frac{\text{Favourable events}}{\text{Total number of events}} = \frac{n(E)}{n(S)}$$

**Examples :**

(i) The typical example of classical probability would be fair dice roll because it is equally probable that you will land on any of the six numbers on the die : 1, 2, 3, 4, 5 or 6

(ii) Throwing a dice, the event $A$ that even number on the dice. Sample space for a dice $S = \{1, 2, 3, 4, 5, 6\}$ elements of event $A = \{2, 4, 6\}$.

The required probability, $P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2} = 0.5$

## Conditional Probability

It gives the probability of happening of any event if the another is already occurred.

$P\left(\frac{E_1}{E_2}\right)$ = Probability of getting the event $E_1$ when event $E_2$ is already occurred.

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)}$$

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## Baye's Theorem

An event A corresponds to a number of $n$ exhaustive event $B_1, B_2, B_3 \ldots\ldots, B_n$. If $P(B_i)$ and $P\left(\frac{A}{B_i}\right)$ are given, then

$$P\left(\frac{A}{B_i}\right)=\frac{P(B_i)P\left(\frac{A}{B_i}\right)}{\sum P(B_i)P\left(\frac{A}{B_i}\right)}$$

**Key Point**

**Results of Probability**

(i) $0 \le P(E) \le 1$

(ii) $P(\bar{E})=1-P(E)$

(iii) $P(E_1 \cup E_2)=P(E_1)+P(E_2)-P(E_1 \cap E_2)$ If mutually exclusive then $P(E_1 \cap E_2)=0$

Therefore, $P(E_1 \cup E_2)=P(E_1)+P(E_2)$

(iv) $P(E_1 \cup E_2 \cup E_3)=P(E_1)+P(E_2)+P(E_3)-P(E_2 \cap E_3)-P(E_1 \cap E_2)-P(E_1 \cap E_3)+P(E_1 \cap E_2 \cap E_3)$

(v) $P(\bar{E}_1 \cap \bar{E}_2)=P(\overline{E_1 \cup E_2})=1-P(E_1 \cup E_2)$

## Probability Density function

A continuous function $f(x)$ is called a probability density function of a random variable, if it satisfies the following conditions,

(i) $f(x) \ge 0$

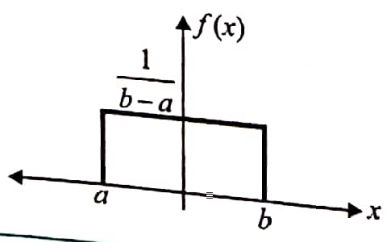
(ii) $\int_{-\infty}^{\infty} f(x)\,dx=1$

## Probability of Distribution

### Uniform Distribution

The probability density function of a uniform random variable on the interval $(a, b)$ is given by,

$$f(x)=\begin{cases}\frac{1}{b-a} & \text{if } a<x<b\\ 0 & \text{otherwise}\end{cases}$$

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**Key Point**

(i) Mean $\bar{x} = \frac{a+b}{2}$ (ii) Variable $\sigma^2 = \frac{(b-a)^2}{12}$

(iii) Standard deviation $\sigma = \sqrt{\text{Variance}}$

## Binomial Distribution

Binomial Distribution gives the probability of happening of event '$r$' times exactly in '$n$' trials

$$P(r) = {}^nC_r p^r q^{n-r}$$

where, $n$ = Number of trials

$r$ = Number of favourable events

$p$ = Probability of happening of event

$q$ = Probability of not happening of event $= 1 - p$

**Key Point**

(i) Mean = $np$ (ii) Variance = $npq$ (iii) Standard Deviation = $\sqrt{npq}$

## Poisson Distribution

Poisson Distribution is a particular limiting form of Binomial distribution when $p$ or $q$ is very small and '$n$' is large enough.

$$P(r) = \frac{m^r e^{-m}}{r!},\ r = 0,1,2,.....$$

where, $m = np$, $n$ = Number of trials

$p$ = Success case probability

$r$ = Number of the success trial

where, $m$ is mean of distribution

**Key Point**

The expected value (mean) and variance of a Poisson distributed random variable are approximately equal. i.e.

$$\text{Mean} \approx \text{Variance} = m$$

## Normal Distribution

Normal Distribution is a continuous distribution and it is derived as the limiting form of Binomial Distribution for large values of '$n$' and when neither '$p$' nor '$q$' is very small.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

where, $\mu$ = mean, $\sigma$ = Standard deviation

Probability of $x$ lying between $x_1$ and $x_2$ is given by the area under normal curve from $x_1$ and $x_2$ i.e.

$$P(x_1 \le x \le x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{\frac{-(x-\mu)^2}{2\sigma^2}}\, dx$$



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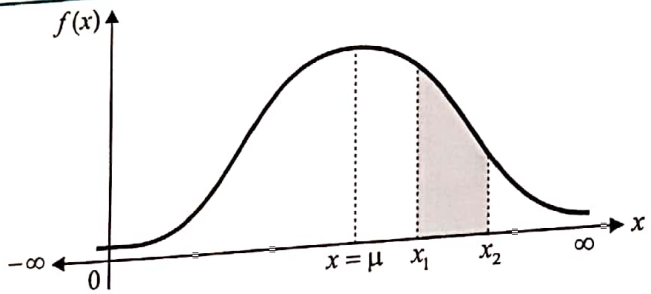
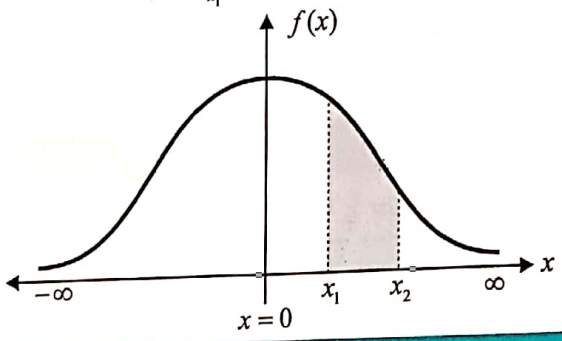



Probability when mean = 0,

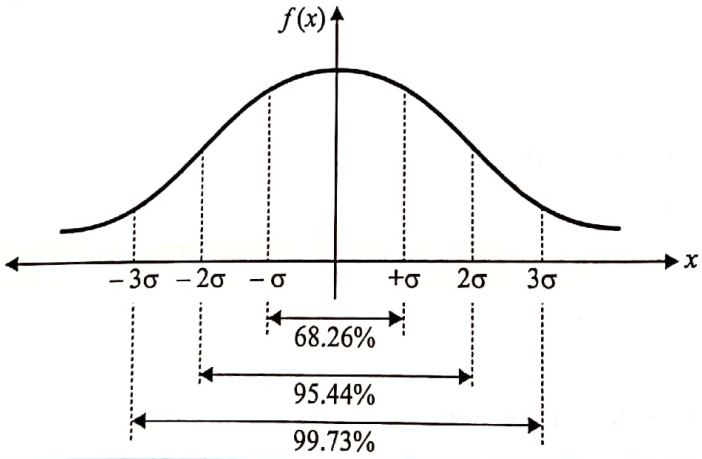
$$P(x_1 \le x \le x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{\frac{-x^2}{2\sigma^2}} dx$$




**Key Point**

(i) Normal distribution is symmetric about its mean

(ii) It is also referred as Gaussian distribution and bell shaped distribution curve.




## Exponential Distribution

A continuous random variable $x$ assuming non negative values is said to have exponential distributio with parameter $\lambda > 0$, if its probability density function is given by,

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \ge 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where, $\lambda$ is a parameter.

**Key Point**

(i) Mean $= \frac{1}{\lambda}$ (ii) Variance $= \frac{1}{\lambda^2}$ (iii) Standard derivation $= \frac{1}{\lambda}$

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# Random Variables and Statistics

## Random Variable

A random variable is defined as a real number $x$ connected with the outcomes of a random experiment.

1. **Discrete random variable :** A real valued function defined on a discrete sample space is called a discrete random variable.
2. **Continuous random variable :** A random variable is said to be continuous, if it can take all possible values between certain limits.

| Values and Parameters | Types of Random Variables | Formula |
|---|---|---|
| Expected Value | discrete random variable | $E(X)=\sum_{i=1}^{n} x_i P(x_i)$ |
| | continuous random variable | $E(X)=\int_{-\infty}^{\infty} x \cdot f(x)\,dx$ |
| Mean Square Value | discrete random variable | $E(X^2)=\sum_{i=1}^{n} x_i^2 P(x_i)$ |
| | continuous random variable | $E(X^2)=\int_{-\infty}^{\infty} x^2 \cdot f(x)\,dx$ |
| Variance | | $V(X)=\sigma^2=E\left[(X-\mu)^2\right]=E(X^2)-\left[E(X)\right]^2$ |
| **Statistics, Correlation and Regression** | | |
| Standard Deviation | | $\sigma_x=\sqrt{\text{Var}(X)}=\sqrt{\dfrac{\sum(x-\bar{x})^2 f_i}{n}}$ |
| Mean | For ungrouped data | $\bar{x}=\dfrac{\Sigma x_i}{n}$ |
| | For grouped data | $\bar{x}=\dfrac{\Sigma f_i x_i}{\Sigma f_i}$ |
| Median | For ungrouped data | $\text{Median}=\begin{cases} x_{k+1} & \text{When } n \text{ is odd and } n=2k+1 \\ \dfrac{x_k+x_{k+1}}{2} & \text{When } n \text{ is even and } n=2k \end{cases}$ |
| | For grouped data | $\text{Median}=L+\left(\dfrac{\frac{F}{2}-C}{f}\right)K$ where $L$ = Lower limit of the median class, $F$ = Total frequency, $f$ = Frequency of median class, $K$ = Width of median class, $C$ = Cumulative frequency up to the class preceding the median class |



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| | | |
|---|---|---|
| **Mode** | For ungrouped data | The mode is the value of the variables which occurs most often. |
| | For grouped data | $\text{Mode} = l + \left(\frac{F - F_{-1}}{2F - F_1 - F_{-1}}\right)K$<br>where, $l$ = Lower limit of class containing mode<br>$K$ = Size of model class or common width of the class<br>$F_1$ = Frequency after modal class<br>$F$ = Frequency of modal class<br>$F_{-1}$ = Frequency of before modal class |
| **Karl Pearson's Co-efficient of Correlation** | | $r = \frac{\sum xy}{\sqrt{\left(\sum x^2 \sum y^2\right)}}$<br>where, $x = X - M_x$ = deviation of variable $X$ measured from its mean $M_x$.<br>$y = Y - M_y$ = deviation of variable Y measured from its mean $M_y$. |
| **Spearman's formula for Rank Correlation Coefficient** | | $r = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$<br>where, $d_i$ = difference in rank of $i^{th}$ individual value<br>$n$ = number of individuals. |
| **Regression** | The regression line of $x$ on $y$ | $x - \bar{x} = r\frac{\sigma_x}{\sigma_y}(y - \bar{y})$ |
| | The regression line of $y$ on $x$ | $y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x}[x - \bar{x}]$ |
| | Angle between two line of regression | $\tan\theta = \left(\frac{1 - r^2}{r}\right)\left(\frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}\right)$ |

## Properties of Expectation

1. $E(c) = c$, $c$ is a constant.
2. $E(cX) = cE(X)$, $c$ is a constant.
3. $E(aX + b) = aE(X) + b$, $a$ and $b$ are constants.
4. $E(X + Y) = E(X) + E(Y)$
5. $E(X - Y) = E(X) - E(Y)$.

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## Properties of Variance

1. $V(c)=0$, $c$ is a constant.
2. $V(aX)=a^2V(X)$
3. $V(aX\pm b)=V(aX)+V(b)=a^2V(X)$, $a$ and $b$ are constants.
4. $V(aX+bY)=a^2V(X)+b^2V(Y)+2ab\,\text{Cov}(X, Y)$

## Covariance

The measures of the simultaneous variation between the random variable $X$ and $Y$ is called the covariance and written as $\text{Cov}(X, Y)$.

If $X$ and $Y$ are two random variables with respective expected values $E(X)$ and $E(Y)$, then

$$\text{Cov}(X, Y)=E(XY)-E(X)E(Y)$$

**Key Point**

(i) Rank of correlation is always less than equal to 1 $(r\le 1)$.

(ii) Mode = 3 median – 2 mean.

(iii) The covariance of two independent variable is equal to zero.

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## Electronics & Communication : EC

### 1988 IIT Kharagpur

**9.1** Events $A$ and $B$ are mutually exclusive and have nonzero probability. Which of the following statement ($s$) are true?

(A) $P(A \cup B) = P(A) + P(B)$

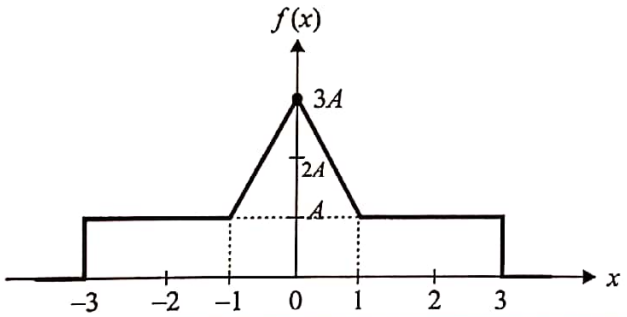
(B) $P(B^C) > P(A)$

(C) $P(A \cap B) = P(A)P(B)$

(D) $P(B^C) < P(A)$

### 1993 IIT Bombay

**9.2** The function shown in figure can represent a probability density function for $A$ _____.




### 2005 IIT Bombay

**9.3** A fair dice is rolled twice. The probability that an odd number will follow an even number is

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{1}{4}$

### 2006 IIT Kharagpur

**9.4** A probability density function is of the form $p(x) = Ke^{-a|x|}$, $x \in (-\infty, \infty)$. The value of $K$ is

(A) 0.5 (B) 1

(C) $0.5a$ (D) $a$

**9.5** Three companies $X$, $Y$ and $Z$ supply computers to a university. The percentage of computers supplied by them and the probability of those being defective are tabulated below

| Company | % of computers supplied | Probability of being defective |
|---|---|---|
| $X$ | 60 % | 0.01 |
| $Y$ | 30 % | 0.02 |
| $Z$ | 10 % | 0.03 |

Given that a computer is defective, the probability that it was supplied by $Y$ is

(A) 0.1 (B) 0.2

(C) 0.3 (D) 0.4

### 2007 IIT Kanpur

**9.6** If $E$ denotes expectation, the variance of a random variable $X$ is given by

(A) $E[X^2] - E^2[X]$

(B) $E[X^2] + E^2[X]$

(C) $E[X^2]$

(D) $E^2[X]$

**9.7** An examination consists of two papers, paper 1 and paper 2. The probability of failing in paper 1 is 0.3 and that in paper 2 is 0.2. Given that a student has failed in paper 2, the probability of failing in paper 1 is 0.6. The probability of a student failing in both papers is

(A) 0.5 (B) 0.18

(C) 0.12 (D) 0.06

### 2008 IISc Bangalore

**9.8** $P_x(x) = Me^{-2|x|} + Ne^{-3|x|}$ is the probability density function for the real random variable $X$ over the entire $x$ axis. $M$ and $N$ are both positive real numbers. The equation relating $M$ and $N$ is



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(A) $M+\frac{2}{3}N=1$ (B) $2M+\frac{1}{3}N=1$

(C) $M+N=1$ (D) $M+N=3$

## 2009 IIT Roorkee

9.9 A fair coin is tossed 10 times. What is the probability that only the first two tosses will yield heads?

(A) $\left(\frac{1}{2}\right)^2$ (B) ${}^{10}C_2\left(\frac{1}{2}\right)^2$

(C) $\left(\frac{1}{2}\right)^{10}$ (D) ${}^{10}C_2\left(\frac{1}{2}\right)^{10}$

## 2010 IIT Guwahati

9.10 A fair coin is tossed independently four times. The probability of the event "the number of time heads shown up is more than the number of times tails shown up" is

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$

(C) $\frac{1}{4}$ (D) $\frac{5}{16}$

## 2011 IIT Madras

9.11 A fair dice is tossed two times. The probability that the second toss results in a value that is higher than the first toss is

(A) $\frac{2}{36}$ (B) $\frac{2}{6}$

(C) $\frac{5}{12}$ (D) $\frac{1}{2}$

## 2012 IIT Delhi

9.12 A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

9.13 Two independent random variables $X$ and $Y$ are uniformly distributed in the interval $[-1, 1]$. The probability that max $[X, Y]$ is less than $\frac{1}{2}$ is

(A) $\frac{3}{4}$ (B) $\frac{9}{16}$

(C) $\frac{1}{4}$ (D) $\frac{2}{3}$

## 2014 IIT Kharagpur

9.14 In a housing society, half of the families have a single child per family while the remaining half have two children per family. The probability that a child picked at random, has a sibling is _____. **[Set - 01]**

9.15 An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is **[Set - 03]**

(A) 0.067 (B) 0.073

(C) 0.082 (D) 0.091

9.16 A fair coin is tossed repeatedly till both head and tail appear at least once. The average number of tosses required is_____. **[Set - 03]**

9.17 Let $X$ be a zero mean unit variance Gaussian random variable. $E[|x|]$ is equal to_____. **[Set - 04]**

9.18 If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier or future calls, the probability distribution function of the total number of calls in a fixed time interval will be **[Set - 04]**

(A) Poisson (B) Gaussian

(C) Exponential (D) Gamma



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9.19 Parcels from sender $S$ to receiver $R$ pass sequentially through two post-offices. Each post-office has a probability $\frac{1}{5}$ of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post-office is _____.
[Set - 04]

## 2015 IIT Kanpur

9.20 Suppose $A$ and $B$ are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$.

Let $\bar{A}$ and $\bar{B}$ be their complements. Which one of the following statements is FALSE? [Set - 01]

(A) $P(A \cap B) = P(A)P(B)$

(B) $P(A / B) = P(A)$

(C) $P(A \cup B) = P(A) + P(B)$

(D) $P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$

9.21 A fair die with faces [1, 2, 3, 4, 5, 6] is thrown repeatedly till '3' is observed for the first time. Let $X$ denote the number of times the die is thrown. The expected value of $X$ is _____.
[Set - 03]

## 2016 IISc Bangalore

9.22 The second moment of a Poisson-distributed random variable is 2. The mean of the random variable is _____.
[Set - 01]

9.23 The probability of getting a "head" in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a "head" is obtained. If the tosses are independent, then the probability of getting "head" for the first time in the fifth toss is _____.
[Set - 03]

## 2017 IIT Roorkee

9.24 Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (up to third decimal place) _____.
[Set - 01]

## 2018 IIT Guwahati

9.25 Let $X_1, X_2, X_3$ and $X_4$ be independent normal random variables with zero mean and unit variance. The probability that $X_4$ is the smallest among the four is _____.

## 2019 IIT Madras

9.26 If $X$ and $Y$ are random variable such that $E[2X + Y] = 0$ and $E[X + 2Y] = 33$, then $E[X] + E[Y] =$ _____

# Electrical Engineering : EE

## 2005 IIT Bombay

9.1 A fair coin is tossed three times in succession. If the first toss produces a head, then the probability of getting exactly two heads in three tosses is

(A) $\frac{1}{8}$ (B) $\frac{1}{2}$

(C) $\frac{3}{8}$ (D) $\frac{3}{4}$

9.2 If $P$ and $Q$ are two random events, then which of the following is true?

(A) Independence of $P$ and $Q$ implies that Probability $(P \cap Q) = 0$

(B) Probability $(P \cap Q) \geq$ Probability $(P)$ + Probability $(Q)$

(C) If $P$ and $Q$ are mutually exclusive then they must be independent

(D) Probability $(P \cap Q) \leq$ Probability $(P)$

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## 2006 IIT Kharagpur

9.3 Two fair dice are rolled and the sum $r$ of the numbers turned up is considered

(A) $\Pr(r>6)=\frac{1}{6}$

(B) $\Pr(r/3 \text{ is an integer})=\frac{5}{6}$

(C) $\Pr(r=8/(r/4) \text{ is an integer})=\frac{5}{9}$

(D) $\Pr(r=6/(r/5) \text{ is an integer})=\frac{1}{18}$

## 2007 IIT Kanpur

9.4 A loaded dice has following probability distribution of occurrences

| Dice value | Probability |
|---|---|
| 1 | 1/4 |
| 2 | 1/8 |
| 3 | 1/8 |
| 4 | 1/8 |
| 5 | 1/8 |
| 6 | 1/4 |

If three identical dice as the above are thrown, the probability of occurrence of values 1, 5 and 6 on the three dice is

(A) Same as that of occurrence of 3, 4, 5.

(B) Same as that of occurrence of 1, 2, 5.

(C) $\frac{1}{128}$

(D) $\frac{5}{8}$

## 2008 IISc Bangalore

9.5 $X$ is uniformly distributed random variable that take values between 0 and 1. The value of $E(X^3)$ will be,

(A) 0 (B) $\frac{1}{8}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

## 2009 IIT Roorkee

9.6 Assume for simplicity that $N$ people, all born in April (a month of 30 days) are collected in a room, consider the event of at least two people in the room being born on the same date of the month even if in different years e.g. 1980 and 1985. What is the smallest $N$ so that the probability of this exceeds 0.5 is?

(A) 20 (B) 7

(C) 15 (D) 16

## 2010 IIT Guwahati

9.7 A box contains 4 white balls and 3 red balls. In succession, two balls are randomly selected and removed from the box. Given that the first removed ball is white, the probability that the second removed ball is red is

(A) $\frac{1}{3}$ (B) $\frac{3}{7}$

(C) $\frac{1}{2}$ (D) $\frac{4}{7}$

## 2011 IIT Madras

9.8 A zero mean random signal is uniformly distributed between limits $-a$ and $+a$ and its mean square value is equal to its variance. Then the r.m.s value of the signal is

(A) $\frac{a}{\sqrt{3}}$ (B) $\frac{a}{\sqrt{2}}$

(C) $a\sqrt{2}$ (D) $a\sqrt{3}$

## 2012 IIT Delhi

9.9 A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$



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**9.10** Two independent random variables $X$ and $Y$ are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

(A) $\frac{3}{4}$ (B) $\frac{9}{16}$

(C) $\frac{1}{4}$ (D) $\frac{2}{3}$

## 2013 IIT Bombay

**9.11** A continuous random variable $X$ has a probability density function

$$f(x) = e^{-x},\ 0 < x < \infty.$$

Then $P\{X > 1\}$ is

(A) 0.368 (B) 0.5

(C) 0.632 (D) 1.0

## 2014 IIT Kharagpur

**9.12** A fair coin is tossed $n$ times. The probability that the difference between the number of heads and tails is $(n-3)$ is **[Set - 01]**

(A) $2^{-n}$ (B) 0

(C) ${}^{n}C_{n-3}2^{-n}$ (D) $2^{-n+3}$

**9.13** Consider a dice with the property that the probability of a face with $n$ dots showing up is proportional to $n$. The probability of the face with three dots showing up is _____. **[Set - 02]**

**9.14** Let $X$ be a random variable with probability density function

$$f(x) = \begin{cases} 0.2, & \text{for } |x| \le 1 \\ 0.1, & \text{for } 1 < |x| \le 4 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(0.5 < X < 5)$ is _____. **[Set - 02]**

**9.15** Lifetime of an electric bulb is a random variable with density $f(x) = kx^2$, where $x$ is measured in years. If the minimum and maximum lifetimes of bulb are 1 and 2 years respectively, then the value of $k$ is _____. **[Set - 03]**

## 2015 IIT Kanpur

**9.16** A random variable $X$ has probability density function $f(x)$ as given below :

$$f(x) = \begin{cases} a + bx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If the expected value $E[X] = \frac{2}{3}$, then $P[X < 0.5]$ is _____. **[Set - 01]**

**9.17** Two players, $A$ and $B$ alternately keep rolling a fair dice. The person to get a six first wins the game. Given that player $A$ starts the game, the probability that $A$ wins the game is **[Set - 01]**

(A) $\frac{5}{11}$ (B) $\frac{1}{2}$

(C) $\frac{7}{13}$ (D) $\frac{6}{11}$

**9.18** Two coins $R$ and $S$ are tossed. The 4 joint events $H_R H_S$, $T_R T_S$, $H_R T_S$, $T_R H_S$ have probabilities 0.28, 0.18, 0.30, 0.24 respectively, where $H$ represents head and $T$ represents tail. Which one of the following is TRUE? **[Set - 02]**

(A) The coin tosses are independent.

(B) $R$ is fair, $S$ is not.

(C) $S$ is fair, $R$ is not.

(D) The coin tosses are dependent.

## 2016 IISc Bangalore

**9.19** Candidates were asked to come to an interview with 3 pens each. Black, blue, green and red were the permitted pen colours that the candidate could bring. The probability that a candidate comes with all 3 pens having the same colour is _____. **[Set - 01]**



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9.20 Let the probability density function of a variable, $X$, be given as

$$f_X(x)=\frac{3}{2}e^{-3x}u(x)+ae^{4x}u(-x)$$

Where $u(x)$ is the unit step function.

Then the value of '$a$' and Prob $\{X \le 0\}$, respectively, are **[Set - 02]**

(A) $2, \frac{1}{2}$ (B) $4, \frac{1}{2}$

(C) $2, \frac{1}{4}$ (D) $4, \frac{1}{4}$

## 2017 IIT Roorkee

9.21 Assume that in a traffic junction, the cycle of the traffic signal lights is 2 minutes of green (vehicle does not stop) and 3 minutes of red (vehicle stops). Consider that the arrival time of vehicles at the junction is uniformly distributed over 5 minute cycle. The expected waiting time (in minutes) for the vehicle at the junction is _____. **[Set - 02]**

9.22 A urn contains 5 red ball and 5 black balls. In the first draw, one ball is picked at random and discarded without noticing its colour. The probability to get a red ball in the second draw is **[Set - 02]**

(A) $\frac{1}{2}$ (B) $\frac{4}{9}$

(C) $\frac{5}{9}$ (D) $\frac{6}{9}$

9.23 A person decides to toss a fair coin repeatedly until he gets a head. He will make at most 3 tosses. Let the random variable $Y$ denote the number of heads. The value of var$\{Y\}$, where var$\{\cdot\}$ denotes the variance, equals. **[Set - 02]**

(A) $\frac{7}{8}$ (B) $\frac{49}{64}$

(C) $\frac{7}{64}$ (D) $\frac{105}{64}$

## 2019 IIT Madras

9.24 The mean-square of a zero-mean random process is $kT/C$, where $k$ is Boltzmann's constant. $T$ is the absolute temperature, and $C$ is a capacitance. The standard deviation of the random process is

(A) $\frac{\sqrt{kT}}{C}$ (B) $\sqrt{\frac{kT}{C}}$

(C) $\frac{C}{kT}$ (D) $\frac{kT}{C}$

9.25 The probability of a resistor being defective is 0.02. There are 50 such resistors in a circuit. The probability of two or more defective resistors in the circuit (round off to two decimal places) is _____.

# Mechanical Engineering : ME

## 1993 IIT Bombay

9.1 If 20 percent managers are technocrats, the probability that a random committee of 5 managers consists of exactly 2 technocrats is

(A) 0.2048 (B) 0.4000

(C) 0.4096 (D) 0.9421

## 1996 IISc Bangalore

9.2 The probability of a defective piece being produced in a manufacturing process is 0.01. The probability that out of 5 successive pieces, only one is defective, is

(A) $(0.99)^4(0.01)$

(B) $(0.99)(0.01)^4$

(C) $5\times(0.99)(0.01)^4$

(D) $5\times(0.99)^4(0.01)$



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## 1997 IIT Madras

**9.3** A box contains 5 black balls and 3 red balls. A total of three balls are picked from the box one after another, without replacing them back. The probability of getting two black balls and one red ball is

(A) $\frac{3}{8}$ (B) $\frac{2}{15}$

(C) $\frac{15}{28}$ (D) $\frac{1}{2}$

## 1998 IIT Delhi

**9.4** The probability that two friends share the same birth-month is

(A) $\frac{1}{6}$ (B) $\frac{1}{12}$

(C) $\frac{1}{144}$ (D) $\frac{1}{24}$

## 1999 IIT Bombay

**9.5** Consider two events $E_1$ and $E_2$ such that $P(E_1)=\frac{1}{2}, P(E_2)=\frac{1}{3}$ and $P(E_1 \cap E_2)=\frac{1}{5}$. Which of the following statement is true?

(A) $P(E_1 \cup E_2)=\frac{2}{3}$

(B) $E_1$ and $E_2$ are independent

(C) $E_1$ and $E_2$ are not independent

(D) $P\left(\frac{E_1}{E_2}\right)=\frac{4}{5}$

## 2000 IIT Kharagpur

**9.6** $E_1$ and $E_2$ are events in a probability space satisfying the following constants $P(E_1)=P(E_2)$, $P(E_1 \cup E_2)=1$, $E_1$ and $E_2$ are independent then $P(E_1)=$ _____.

(A) 0 (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) 1

**9.7** In a manufacturing plant, the probability of making a defective bolt is 0.1. The mean and standard deviation of defective bolts in a total of 900 bolts are respectively

(A) 90 and 9 (B) 9 and 90

(C) 18 and 9 (D) 9 and 81

## 2001 IIT Kanpur

**9.8** An unbiased coin is tossed three times. The probability that the head turns up in exactly two cases is

(A) $\frac{1}{9}$ (B) $\frac{1}{8}$

(C) $\frac{2}{3}$ (D) $\frac{3}{8}$

**9.9** Seven car accidents occurred in a week, what is the probability that they all occurred on the same day?

(A) $\frac{1}{7^7}$ (B) $\frac{1}{7^6}$

(C) $\frac{1}{2^7}$ (D) $\frac{7}{2^7}$

## 2002 IISc Bangalore

**9.10** Two dice are thrown. What is the probability that the sum of the numbers on the two dice is eight?

(A) $\frac{5}{36}$ (B) $\frac{5}{18}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

**9.11** Four fair coins are tossed simultaneously. The probability that at least one head and at least one tail turn up is,

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$

(C) $\frac{7}{8}$ (D) $\frac{15}{16}$



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9.12 A regression model is used to express a variable $Y$ as a function of another variable $X$. This implies that

(A) There is a causal relationship between $Y$ and $X$.

(B) A value of $X$ may be used to estimate a value of $Y$.

(C) Value of $X$ exactly determine values of $Y$.

(D) There is no causal relationship between $Y$ and $X$.

## 2003 IIT Madras

9.13 A box contains 5 black and 5 red balls. Two balls are randomly picked one after another from the box, without replacement. The probability for both balls being red is

(A) $\frac{1}{90}$ (B) $\frac{1}{2}$

(C) $\frac{19}{90}$ (D) $\frac{2}{9}$

9.14 Let $P(E)$ denote the probability of an event $E$. Given $P(A) = 1, P(B) = \frac{1}{2}$, the values of $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$ respectively are

(A) $\frac{1}{4}, \frac{1}{2}$ (B) $\frac{1}{2}, \frac{1}{4}$

(C) $\frac{1}{2}, 1$ (D) $1, \frac{1}{2}$

## 2004 IIT Delhi

9.15 The following data about the flow of liquid was observed in a continuous chemical process plant

| Flow rate (litres/sec) | 7.5 to 7.7 | 7.7 to 7.9 | 7.9 to 8.1 | 8.1 to 8.3 | 8.3 to 8.5 | 8.5 to 8.7 |
|---|---|---|---|---|---|---|
| Frequency | 1 | 5 | 35 | 17 | 12 | 10 |

Mean flow rate of the liquid is

(A) 8.00 litres/sec (B) 8.06 litres/sec

(C) 8.16 litres/sec (D) 8.26 litres/sec

9.16 From a pack of regular playing cards, two cards are drawn at random. What is the probability that both cards will be Kings, if first card is NOT replaced?

(A) $\frac{1}{26}$ (B) $\frac{1}{52}$

(C) $\frac{1}{169}$ (D) $\frac{1}{221}$

## 2005 IIT Bombay

9.17 A lot has 10% defective items. Ten items are chosen randomly from this lot. The probability that exactly 2 of the chosen items are defective is

(A) 0.0036 (B) 0.1937

(C) 0.2234 (D) 0.3874

9.18 A single die is thrown twice. What is the probability that the sum is neither 8 nor 9?

(A) $\frac{1}{9}$ (B) $\frac{5}{36}$

(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

## 2006 IIT Kharagpur

9.19 A box contains 20 defective items and 80 non-defective items. If two items are selected at random without replacement, what will be the probability that both items are defective?

(A) $\frac{1}{5}$ (B) $\frac{1}{25}$

(C) $\frac{20}{99}$ (D) $\frac{19}{495}$

9.20 Consider the continuous random variable with probability density function

$f(t) = 1+t$ for $-1 \le t \le 0$

$= 1-t$ for $0 \le t \le 1$

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The standard deviation of the random variable is

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{6}}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

## 2007 IIT Kanpur

**9.21** Let $X$ and $Y$ be two independent random variables. Which one of the relations between expectation ($E$), variance (Var) and covariance (Cov) given below is False?

(A) $E(XY) = E(X)\,E(Y)$

(B) $\text{Cov}(X, Y) = 0$

(C) $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

(D) $E(X^2Y^2) = \left(E(X)\right)^2\left(E(Y)\right)^2$

## 2008 IISc Bangalore

**9.22** A coin is tossed 4 times. What is the probability of getting heads exactly 3 times?

(A) $\frac{1}{4}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

## 2009 IIT Roorkee

**9.23** If three coins are tossed simultaneously, the probability of getting at least one head is

(A) $\frac{1}{8}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{7}{8}$

**9.24** The standard deviation of a uniformly distributed random variable between 0 and 1 is

(A) $\frac{1}{\sqrt{12}}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{5}{\sqrt{12}}$ (D) $\frac{7}{\sqrt{12}}$

## 2010 IIT Guwahati

**9.25** A box contains 2 washers, 3 nuts and 4 bolts. Items are drawn from the box at random one at a time without replacement. The probability of drawing 2 washers first followed by 3 nuts and subsequently the 4 bolts is

(A) $\frac{2}{315}$ (B) $\frac{1}{630}$

(C) $\frac{1}{1260}$ (D) $\frac{1}{2520}$

## 2011 IIT Madras

**9.26** An unbiased coin is tossed five times. The outcome of each toss is either a head or a tail. The probability of getting at least one head is

(A) $\frac{1}{32}$ (B) $\frac{13}{32}$

(C) $\frac{16}{32}$ (D) $\frac{31}{32}$

## 2012 IIT Delhi

**9.27** A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is

(A) $\frac{1}{20}$ (B) $\frac{1}{12}$

(C) $\frac{3}{10}$ (D) $\frac{1}{2}$

**9.28** An automobile plant contracted to buy shock absorbers from two suppliers $X$ and $Y$. $X$ supplies 60% and $Y$ supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test.



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The ones that pass the quality test are considered reliable. Of $X$'s shock absorbers, 96% are reliable. Of $Y$'s shock absorbers, 72% are reliable.
The probability that a randomly chosen shock absorber, which is found to be reliable, is made by $Y$ is
(A) 0.288 (B) 0.334
(C) 0.667 (D) 0.720

## 2013 IIT Bombay

9.29 The probability that a student knows the correct answer to a multiple choice question is 2/3. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is 1/4. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is
(A) $\frac{2}{3}$ (B) $\frac{3}{4}$
(C) $\frac{5}{6}$ (D) $\frac{8}{9}$

9.30 Let $X$ be a normal random variable with mean 1 and variance 4. The probability $P\{X<0\}$ is
(A) 0.5
(B) greater than zero and less than 0.5
(C) greater than 0.5 and less than 1.0
(D) 1.0

9.31 Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?
(A) $\frac{13}{90}$ (B) $\frac{12}{90}$
(C) $\frac{78}{90}$ (D) $\frac{77}{90}$

## 2014 IIT Kharagpur

9.32 In the following table, $X$ is a discrete random variable and $P(x)$ is the probability density. The standard deviation of $X$ is **[Set - 01]**

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | 0.3 | 0.6 | 0.1 |

(A) 0.18 (B) 0.36
(C) 0.54 (D) 0.6

9.33 A box contains 25 parts of which 10 are defective. Two parts are being drawn simultaneously in a random manner from the box. The probability of both the parts being good is **[Set - 02]**
(A) $\frac{7}{20}$ (B) $\frac{42}{125}$
(C) $\frac{25}{29}$ (D) $\frac{5}{9}$

9.34 Consider an unbiased cubic dice with opposite faces coloured identically and each face coloured red, blue or green such that each colour appears only two times on the dice. If the dice is thrown thrice, the probability of obtaining red colour on top face of the dice at least twice is _____. **[Set - 02]**

9.35 A group consists of equal number of men and women. Of this group 20% of the men and 50% of the women are unemployed. If a person is selected at random from this group, the probability of the selected person being employed is ____. **[Set - 03]**

9.36 A machine produces 0, 1 or 2 defective pieces in a day with associated probability of $\frac{1}{6}, \frac{2}{3}$ and $\frac{1}{6}$, respectively.

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The mean value and the variance of the number of defective pieces produced by the machine in a day respectively, are

**[Set - 03]**

(A) 1 and $\frac{1}{3}$ (B) $\frac{1}{3}$ and 1

(C) 1 and $\frac{4}{3}$ (D) $\frac{1}{3}$ and $\frac{4}{3}$

**9.37** A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch accepted is _____.

**[Set - 03]**

**9.38** A nationalized bank has found that the daily balance available in its savings accounts follows a normal distribution with a mean of Rs. 500 and a standard deviation of Rs. 50. The percentage of savings account holders, who maintain an average daily balance more than Rs. 500 is _____. **[Set - 04]**

**9.39** The number of accidents occurring in a plant in a month follows Poisson distribution with mean as 5.2. The probability of occurrence of less than 2 accidents in the plant during a randomly selected month is **[Set - 04]**

(A) 0.029 (B) 0.034

(C) 0.039 (D) 0.044

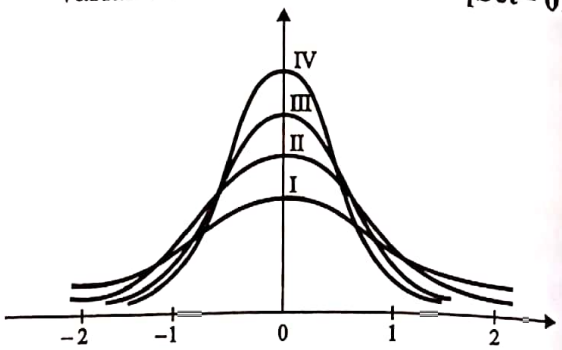
**9.40** Demand during lead time with associated probabilities is shown below

| Demand | 50 | 70 | 75 | 80 | 85 |
|---|---|---|---|---|---|
| Probability | 0.15 | 0.14 | 0.21 | 0.20 | 0.30 |

Expected demand during lead time is _____. **[Set - 04]**

## 2015 IIT Kanpur

**9.41** Among the four normal distribution with probability density functions as shown below, which one has the lowest variance? **[Set - 01]**




(A) I (B) II

(C) III (D) IV

**9.42** If $P(X)=\frac{1}{4}$ $P(Y)=\frac{1}{3}$ and $P(X\cap Y)=\frac{1}{12}$, the value of $P\left(\frac{Y}{X}\right)$ is **[Set - 03]**

(A) $\frac{1}{4}$ (B) $\frac{4}{25}$

(C) $\frac{1}{3}$ (D) $\frac{29}{50}$

## 2016 IISc Bangalore

**9.43** Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is $\mu$. The standard deviation for this distribution is given by **[Set - 01]**

(A) $\sqrt{\mu}$ (B) $\mu^2$

(C) $\mu$ (D) $\frac{1}{\mu}$

**9.44** The probability that a screw manufactured by a company is defective is 0.1. The company sells screws in packets containing 5 screws and gives a guarantee of replacement if one or more screws in the packet are found to be defective. The probability that a packet would have to be replaced is _____. **[Set - 02]**

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9.45 The area (in percentage) under standard normal distribution curve of random variable $Z$ within limits from $-3$ to $+3$ is _____. **[Set - 03]**

9.46 Three cards were drawn from a pack of 52 cards. The probability that they are a king, a queen, and a jack is **[Set - 03]**

(A) $\frac{16}{5525}$ (B) $\frac{64}{2197}$

(C) $\frac{3}{13}$ (D) $\frac{8}{16575}$

## 2017 IIT Roorkee

9.47 A sample of 15 data is as follows: 17, 18, 17, 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3. The mode of the data is **[Set - 02]**

(A) 4 (B) 13

(C) 17 (D) 20

## 2018 IIT Guwahati

9.48 Four red balls, four green balls and four blue balls are put in a box. Three balls are pulled out of the box at random one after another without replacement. The probability that all the three balls are red is **[Set - 01]**

(A) $\frac{1}{72}$ (B) $\frac{1}{55}$

(C) $\frac{1}{36}$ (D) $\frac{1}{27}$

9.49 A six-faced fair dice is rolled five times. Then probability (in %) of obtaining 'ONE' at least four times is **[Set - 01]**

(A) 33.3 (B) 3.33

(C) 0.33 (D) 0.0033

9.50 Let $X_1, X_2$ be two normal random variable with means $\mu_1, \mu_2$ and standard deviation $\sigma_1, \sigma_2$ respectively. Consider $Y = X_1 - X_2$; $\mu_1 = \mu_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = 2$ then **[Set - 02]**

(A) $Y$ is normal distributed random with mean 0 and variance 1.

(B) $Y$ is normal distributed with mean 0 and variance 5.

(C) $Y$ has mean 0 and variance 5, but is NOT normally distributed.

(D) $Y$ has mean 0 and variance 1, but is NOT normally distributed.

9.51 Let $X_1$ and $X_2$ be two independent exponentially distributed random variables with means 0.5 and 0.25 respectively. Then $Y = \min(X_1, X_2)$ is **[Set - 02]**

(A) Exponentially distributed with mean 1/6.

(B) Exponentially distributed with mean 2.

(C) Normally distributed with mean 3/4.

(D) Normally distributed with mean 1/6.

## 2019 IIT Madras

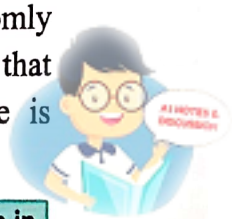
9.52 The variable $X$ takes a value between 0 and 10 with uniform probability distribution. The variable $Y$ takes a value between 0 and 20 with uniform probability distribution. The probability of the sum of variable $(X + Y)$ being greater than 20 is

(A) 0.25 (B) 0.5

(C) 0 (D) 0.33

9.53 If $x$ is the mean of data 3, $x$, 2, 4 then the mode is _____.

9.54 The probability that a part manufactured by a company will be deflective is 0.05. If 15 such parts are selected randomly and inspected, then the probability that atleast two parts will be defective is _____.





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9.8 Three values of $x$ and $y$ are to be fitted in a straight line in the form $y = a + bx$ by the method of least squares. Given $\Sigma x = 6$, $\Sigma y = 21$, $\Sigma x^2 = 14$ and $\Sigma xy = 46$, the values of $a$ and $b$ are respectively

(A) 2 and 3 (B) 1 and 2
(C) 2 and 1 (D) 3 and 2

9.9 A person on a trip has a choice between private car and public transport. The probability of using a private car is 0.45. While using the public transport, further choices available are bus and metro, out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (rounded up to two decimals) of using a car, bus and metro, respectively would be

(A) 0.45, 0.30 and 0.25
(B) 0.45, 0.25 and 0.30
(C) 0.45, 0.55 and 0.00
(D) 0.45, 0.35 and 0.20

## 2009 IIT Roorkee

9.10 The standard normal probability function can be approximated as

$$F(x_N) = \frac{1}{1+\exp\left(-1.7255x_N\left|x_N\right|^{0.12}\right)}$$

Where $x_N$ = standard normal deviate. If mean and standard deviation of annual precipitation are 102 cm and 27 cm respectively, the probability that the annual precipitation will be between 90 cm and 102 cm is

(A) 66.7% (B) 50.0%
(C) 33.3% (D) 16.7%

## 2010 IIT Guwahati

9.11 Two coins are simultaneously tossed. The probability of two heads simultaneously appearing is

(A) $\frac{1}{8}$ (B) $\frac{1}{6}$
(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

## 2011 IIT Madras

9.12 There are two containers, with one containing 4 Red and 3 Green balls and the other containing 3 Blue and 4 Green balls. One ball is drawn at random from each container. The probability that one of the balls is Red and the other is Blue will be

(A) $\frac{1}{7}$ (B) $\frac{9}{49}$
(C) $\frac{12}{49}$ (D) $\frac{3}{7}$

## 2012 IIT Delhi

9.13 In an experiment, positive and negative values are equally likely to occur. The probability of obtaining at most one negative value in five trials is

(A) $\frac{1}{32}$ (B) $\frac{2}{32}$
(C) $\frac{3}{32}$ (D) $\frac{6}{32}$

9.14 The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000 mm and 200 mm, respectively. The probability that the annual precipitation will be more than 1200 mm is

(A) < 50 % (B) 50 %
(C) 75 % (D) 100 %

## 2013 IIT Bombay

9.15 Find the value of $\lambda$ such that function $f(x)$ is valid probability density function

$f(x) = \lambda(x-1)(2-x)$; for $1 \le x \le 2$
$= 0$ ; otherwise

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**9.16** A 1 hour rainfall of 10 cm has return period of 50 year. The probability that 1 hour of rainfall of 10 cm or more will occur in each of two successive years is

(A) 0.04 (B) 0.2

(C) 0.2 (D) 0.0004

## 2014 IIT Kharagpur

**9.17** The probability density function of evaporation $E$ on any day during a year in a watershed is given by **[Set - 01]**

$$f(E)=\begin{cases}\frac{1}{5}; & 0\le E\le 5\,\text{mm/day}\\ 0; & \text{otherwise}\end{cases}$$

The probability that $E$ lies in between 2 and 4 mm/day in the watershed is (in decimal) _____.

**9.18** A traffic officer imposes on an average 5 number of penalties daily on traffic violators. Assume that the number of penalties on different days is independent and follows a Poisson distribution. The probability that there will be less than 4 penalties in a day is _____. **[Set - 01]**

**9.19** A fair (unbiased) coin was tossed four times in succession and resulted in the following outcomes :

(i) Head (ii) Head (iii) Head (iv) Head.

The probability of obtaining a 'Tail' when the coin is tossed again is **[Set - 02]**

(A) 0 (B) $\frac{1}{2}$

(C) $\frac{4}{5}$ (D) $\frac{1}{5}$

**9.20** If $X$ is a continuous, real valued random variable defined over the interval $(-\infty,+\infty)$ and its occurrence is defined by the density function given as,

$$f(x)=\frac{1}{\sqrt{2\pi}\times b}e^{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2}$$

where '$a$' and '$b$' are the statistical attributes of the random variable $X$. The value of the integral **[Set - 02]**

$$\int_{-\infty}^{a}\frac{1}{\sqrt{2\pi}\times b}e^{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2}dx \text{ is}$$

(A) 1 (B) 0.5

(C) $\pi$ (D) $\frac{\pi}{2}$

**9.21** An observer counts 240 veh/h at a specific highway location. Assume that the vehicle arrival at the locations is Poisson distributed, the probability of having one vehicle arriving over a 30-second time interval is_____.

**[Set - 02]**

## 2015 IIT Kanpur

**9.22** Consider the following probability mass function (p.m.f) of a random variable $X$

$$p(X,q)=\begin{cases}q, & \text{if } X=0\\ 1-q, & \text{if } X=1\\ 0, & \text{otherwise}\end{cases}$$

If $q=0.4$, the variance of $X$ is_____.

**[Set - 01]**

**9.23** For probability density function of a random variable, $x$ is

$$f(x)=\frac{x}{4}(4-x^2);\ \text{for } 0\le x\le 2$$

$$=0 \quad ;\ \text{otherwise}$$

The mean $\mu_x$ of the random variable is _____. **[Set - 02]**

## 2016 IISc Bangalore

**9.24** Probability density function of a random variable $X$ is given below **[Set - 01]**

$$f(x)=\begin{cases}0.25; & \text{if } 1\le x\le 5\\ 0 \ ; & \text{otherwise}\end{cases}$$



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$P(X \le 4)$ is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{8}$

9.25 The spot speeds (expressed in km/hr) observed at a road section are 66, 62, 45, 79, 32, 51, 56, 60, 53, and 49. The median speed (expressed in km/hr) is _____. **[Set - 02]**

[Note : Answer with one decimal accuracy].

9.26 $X$ and $Y$ are two random independent events. It is known that $P(X) = 0.40$ and $P(X \cup Y^C) = 0.7$. Which one of the following is the value of $P(X \cup Y)$?

**[Set - 02]**

(A) 0.7 (B) 0.5

(C) 0.4 (D) 0.3

9.27 If $f(x)$ and $g(x)$ are two probability density functions,

$$f(x) = \begin{cases} \frac{x}{a} + 1 & ; \quad -a \le x < 0 \\ -\frac{x}{a} + 1 & ; \quad 0 \le x \le a \\ 0 & ; \quad \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} -\frac{x}{a} & ; \quad -a \le x < 0 \\ \frac{x}{a} & ; \quad 0 \le x \le a \\ 0 & ; \quad \text{otherwise} \end{cases}$$

Which one of the following statement is true? **[Set - 02]**

(A) Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are same.

(B) Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are different.

(C) Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are same.

(D) Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are different.

## 2017 IIT Roorkee

**9.28** A two-faced fair coin has its faces designated as head ($H$) and tail ($T$). This coin is tossed three times in succession to record the following outcomes : $H, H, H$. If the coin is tossed one more time, the probability (up to one decimal place) of obtaining $H$ again, given the previous realizations of $H$, $H$ and $H$, would be _____. **[Set - 02]**

**9.29** The number of parameters in the univariate exponential and Gaussian distributions, respectively are

(A) 2 and 2 (B) 1 and 2

(C) 2 and 1 (D) 1 and 1

**9.30** For the function $f(x) = a + bx$, $0 \le x \le 1$ to be a valid probability density function, which one of the following statements is correct?

(A) $a = 1$, $b = 4$ (B) $a = 0.5$, $b = 1$

(C) $a = 0, b = 1$ (D) $a = 1$, $b = -1$



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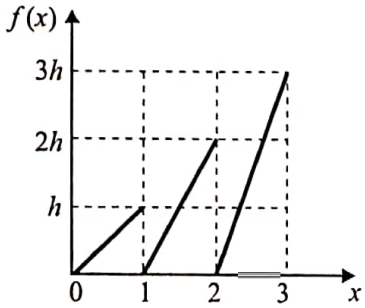
## 2018 IIT Guwahati

**9.31** The frequency distribution of the compressive strength of 20 concrete cube specimens is given in the table.

| $f$ (MPa) | Number of specimens with compressive strength equal to $f$ |
|---|---|
| 23 | 4 |
| 28 | 2 |
| 22.5 | 5 |
| 31 | 5 |
| 29 | 4 |

If $\mu$ is the mean strength of the specimens and $\sigma$ is the standard deviation, the number of specimens (out of 20) with compressive strength less than $\mu - 3\sigma$ is _____. **[Set - 01]**

**9.32** The graph of a function $f(x)$ is shown in the figure




For $f(x)$ to be a valid probability density function, the value of $h$ is

**[Set - 02]**

(A) 1/3 (B) 2/3

(C) 1 (D) 3

**9.33** Probability (up to one decimal place) of consecutively picking 3 red balls without replacement from a box containing 5 red balls and 1 white ball is _____.

**[Set - 02]**

**9.34** A probability distribution with right skew is shown in the figure.

The correct statement for the probability distriBbution is **[Set - 01]**

(A) Mean is equal to mode.

(B) Mean is greater than median but less than mode.

(C) Mean is greater than median and mode.

(D) Mode is greater than median.

## 2019 IIT Madras

**9.35** The probability that the annual maximum flood discharge will exceed 25000 $m^3/s$, at least once in next 5 years is found to be 0.25. The return period of this flood event (in years, round off to 1 decimal place) is _____.

**9.36** The probability density function of a continuous random variable distributed uniformly between $x$ and $y$ (for $y > x$) is

(A) $y - x$ (B) $\frac{1}{y-x}$

(C) $x - y$ (D) $\frac{1}{x-y}$

# Instrumentation Engineering : IN

## 2005 IIT Bombay

**9.1** The probability that there are 53 Sundays in a randomly chosen leap year is

(A) $\frac{1}{7}$ (B) $\frac{1}{14}$

(C) $\frac{1}{28}$ (D) $\frac{2}{7}$



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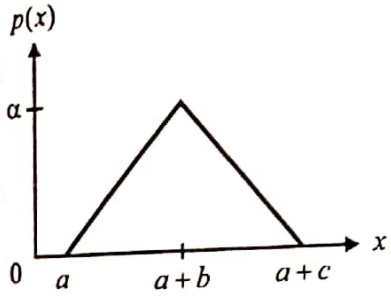
## 2006 IIT Kharagpur

9.2 You have gone to a cyber-cafe with a friend. You found that the cyber-cafe has only three terminals. All terminals are unoccupied. You and your friend have to make a random choice of selecting a terminal. What is the probability that both of you will NOT select the same terminal?

(A) $\frac{1}{9}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) 1

9.3 Probability density function differential function $p(x)$ of a random variable $x$ is as shown below. The value of $\alpha$ is




(A) $\frac{2}{c}$ (B) $\frac{1}{c}$

(C) $\frac{2}{(b+c)}$ (D) $\frac{1}{(b+c)}$

9.4 Two dices are rolled simultaneously. The probability that the sum of digits on the top surface of the two dices is even, is

(A) 0.5 (B) 0.25

(C) 0.167 (D) 0.125

## 2007 IIT Kanpur

9.5 Assume that the duration in minutes of a telephone conversation follows the exponential distribution $f(x)=\frac{1}{5}e^{\frac{-x}{5}}$, $x \geq 0$. The probability that the conversation will exceed five minutes is

(A) $\frac{1}{e}$ (B) $1-\frac{1}{e}$

(C) $\frac{1}{e^2}$ (D) $1-\frac{1}{e^2}$

## 2008 IISc Bangalore

9.6 A random variable is uniformly distributed over the interval 2 to 10. Its variance will be

(A) $\frac{16}{3}$ (B) 6

(C) $\frac{256}{9}$ (D) 36

9.7 Consider a Gaussian distributed random variable with zero mean and standard deviation $\sigma$. The value of its cumulative distribution function at the origin will be

(A) 0 (B) 0.5

(C) 1 (D) $10\sigma$

9.8 $P_X(x) = Me^{-2|x|} + Ne^{-3|x|}$ is the probability density function for the real random variable $X$ over the entire $x$-axis. $M$ and $N$ are both positive real numbers. The equation relating $M$ and $N$ is

(A) $M+\frac{2}{3}N=1$

(B) $2M+\frac{1}{3}N=1$

(C) $M+N=1$

(D) $M+N=3$

## 2009 IIT Roorkee

9.9 A screening test is carried out to detect a certain disease. It is found that 12 % of the positive reports and 15% of the negative reports are incorrect. Assuming that the probability of a person getting a positive report is 0.01, the probability that a person tested gets an incorrect report is

(A) 0.0027 (B) 0.0173

(C) 0.1497 (D) 0.2100



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## 2010 IIT Guwahati

**9.10** The diameters of 10000 ball bearings were measured. The mean diameter and standard deviation were found to be 10 mm and 0.05 mm respectively. Assuming Gaussian distribution of measurements, it can be expected that the number of measurements more than 10.15 mm will be

(A) 230 (B) 115
(C) 15 (D) 2

## 2011 IIT Madras

**9.11** The box 1 contains chips numbered 3, 6, 9, 12 and 15. The box 2 contains chips numbered 6, 11, 16, 21 and 26. Two chips, one from each box, are drawn at random. The numbers written on these chips are multiplied. The probability for the product to be an even number is

(A) $\frac{6}{25}$ (B) $\frac{2}{5}$
(C) $\frac{3}{5}$ (D) $\frac{19}{25}$

## 2012 IIT Delhi

**9.12** A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$
(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

**9.13** Two independent random variables $X$ and $Y$ are uniformly distributed in the interval $[-1, 1]$. The probability that max $[X, Y]$ is less than $\frac{1}{2}$ is

(A) $\frac{3}{4}$ (B) $\frac{9}{16}$
(C) $\frac{1}{4}$ (D) $\frac{2}{3}$

## 2013 IIT Bombay

**9.14** A continuous random variable $X$ has a probability density function

$$f(x) = e^{-x},\ 0 < x < \infty$$

then $P(X > 1)$ is

(A) 0.368 (B) 0.5
(C) 0.632 (D) 1.0

**9.15** What is the chance that a leap year, selected at random, will contain 53 Sundays?

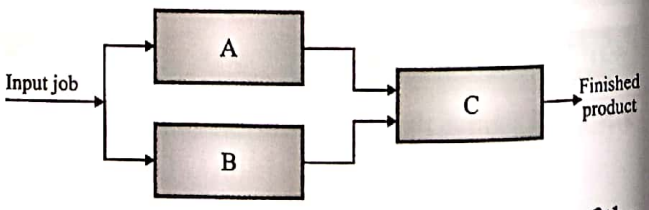
(A) $\frac{2}{7}$ (B) $\frac{3}{7}$
(C) $\frac{1}{7}$ (D) $\frac{5}{7}$

## 2014 IIT Kharagpur

**9.16** Given that $X$ is a random variable in the range $[0, \infty]$ with a probability density function $\frac{e^{\frac{-x}{2}}}{K}$, the value of the constant $K$ is _____.

**9.17** The figure shows the schematic of a production process with machines $A$, $B$ and $C$. An input job needs to be pre-processed either by $A$ or by $B$ before it is fed to $C$, from which the final finished product comes out. The probabilities of failure of the machines are given as,

$P(A) = 0.15$,
$P(B) = 0.05$, $P(C) = 0.1$




Assuming independence of failures of the machines, the probability that a given job is successfully processed (up to the third decimal place) is _____.



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## 2015 IIT Kanpur

9.18 The probability that a thermistor randomly picked up from a production unit is defective is 0.1. The probability that out of 10 thermistors, randomly picked up 3 are defective is

(A) 0.001 (B) 0.057

(C) 0.107 (D) 0.3

9.19 The probability density function of a random variable $X$ is $P_X(x) = e^{-x}$ for $x \geq 0$ and 0 otherwise. The expected value of the function $g_x(x) = e^{3x/4}$ is ____.

## 2016 IISc Bangalore

9.20 An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is

(A) $\frac{65}{156}$ (B) $\frac{67}{156}$

(C) $\frac{79}{156}$ (D) $\frac{89}{156}$

## 2017 IIT Roorkee

9.21 The probability that a communication system will have high fidelity is 0.81. The probability that the system will have both high fidelity and high selectivity is 0.18. The probability that a given system with high fidelity will have high selectivity is

(A) 0.181 (B) 0.191

(C) 0.222 (D) 0.826

## 2018 IIT Guwahati

9.22 $X$ and $Y$ are two independent random variables with variances 1 and 2, respectively. Let $Z = X - Y$. The variance of $Z$ is

(A) 0 (B) 1

(C) 2 (D) 3

9.23 Consider a sequence of tossing of a fair coin where the outcomes of tosses are independent. The probability of getting the head for the third time in the fifth toss is

(A) $\frac{5}{16}$ (B) $\frac{3}{16}$

(C) $\frac{3}{5}$ (D) $\frac{9}{16}$

9.24 Two bags $A$ and $B$ have equal number of balls. Bag $A$ has 20% red balls and 80% green balls. Bag $B$ has 30% red balls, 60% green balls and 10% yellow balls. Contents of Bags $A$ and $B$ are mixed thoroughly and a ball is randomly picked from the mixture. What is the chance that the ball picked is red?

(A) 20% (B) 25%

(C) 30% (D) 40%

## 2019 IIT Madras

9.25 A box has a 8 red balls and 8 green balls. Two balls are drawn randomly in succession from the box without replacement. The probability that the first ball drawn is red and the second ball drawn is green is

(A) $\frac{4}{15}$ (B) $\frac{7}{16}$

(C) $\frac{1}{2}$ (D) $\frac{8}{15}$

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**9.26** The function $p(x)$ is given by $p(x) = A / x^{\mu}$ where $A$ and $\mu$ are constants with $\mu > 1$ and $1 \le x < \infty$ and $p(x) = 0$ for $-\infty < x < 1$. For $p(x)$ to be probability density function, the value of $A$ should be equal to

(A) $\mu - 1$ (B) $\mu + 1$

(C) $1/(\mu - 1)$ (D) $1/(\mu + 1)$

## Chemical Engineering : CH

### 2000 IIT Kharagpur

**9.1** A pair of fair dice is rolled simultaneously.

The probability that the sum of the numbers from the dice equals six is

(A) $\frac{1}{6}$ (B) $\frac{7}{36}$

(C) $\frac{5}{36}$ (D) $\frac{1}{12}$

### 2001 IIT Kanpur

**9.2** A fair die is rolled four times. Find the probability that six shows up twice

(A) $\frac{1}{2}$ (B) $\frac{16}{325}$

(C) $\frac{1}{36}$ (D) $\frac{25}{216}$

### 2003 IIT Madras

**9.3** A box contains 6 red balls and 4 green balls, one ball is randomly picked and then a second ball is picked without replacement of the first ball. The probability that the balls are green is

(A) $\frac{1}{15}$ (B) $\frac{2}{25}$

(C) $\frac{2}{15}$ (D) $\frac{4}{25}$

### 2004 IIT Delhi

**9.4** A box contains three blue balls and four red balls. Another identical box contains two blue balls and five red balls. One ball is picked at random from one of the two boxes and it is red. The probability that it came from the first box is

(A) $\frac{2}{3}$ (B) $\frac{4}{9}$

(C) $\frac{4}{7}$ (D) $\frac{2}{7}$

### 2005 IIT Bombay

**9.5** Two bags contain ten coins each and the coins in each bag are numbered from 1 to 10. One coin is drawn at random from each bag. The probability that one of the coins has value 1, 2, 3, or 4, while the other has value 7, 8, 9 or 10 is

(A) $\frac{2}{5}$ (B) $\frac{4}{25}$

(C) $\frac{8}{25}$ (D) $\frac{16}{25}$

### 2006 IIT Kharagpur

**9.6** A pair of fair dice is rolled three times. What is the probability that 10 (sum of the number on the two faces) will show up exactly once?

(A) $\frac{121}{1728}$ (B) $\frac{363}{1728}$

(C) $\frac{121}{576}$ (D) $\frac{363}{576}$



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9.7 A company purchased components from the firms $P$, $Q$, and $R$ as shown in the table below

| Firm | Total number of components purchased | Number of components likely to be defective |
|---|---|---|
| $P$ | 1000 | 5 |
| $Q$ | 2500 | 5 |
| $R$ | 500 | 2 |

The components are closed together. One of the component is selected at random and found to be defective. What is the probability that it was supplied by firm $R$?

(A) $\frac{1}{250}$ (B) $\frac{1}{12}$

(C) $\frac{1}{8}$ (D) $\frac{1}{6}$

## 2008 IISc Bangalore

9.8 The normal distribution is given by

$$f(x)=\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),\ -\infty<x<\infty$$

The point of inflection to the normal curve are,

(A) $x=-\sigma,+\sigma$

(B) $x=\mu+\sigma,\mu-\sigma$

(C) $x=\mu+2\sigma,\mu-2\sigma$

(D) $x=\mu+3\sigma,\mu-3\sigma$

9.9 The Poisson distribution is given by $P(r)=\frac{m^r}{r!}\exp(-m)$. The first moment about the origin for the distribution is

(A) 0 (B) $m$

(C) $\frac{1}{m}$ (D) $m^2$

## 2009 IIT Roorkee

9.10 A fair die is rolled. Let $R$ denote the event of obtaining a number less than or equal to 5 and $S$ denote the event of obtaining an odd number. Then, which one of the following about the probability ($P$) is TRUE?

(A) $P\left(\frac{R}{S}\right)=1$ (B) $P\left(\frac{R}{S}\right)=0$

(C) $P\left(\frac{S}{R}\right)=1$ (D) $P\left(\frac{S}{R}\right)=0$

## 2010 IIT Guwahati

9.11 $X$ and $Y$ are independent random variables. $X$ follows a binomial distribution, with $n=5$ and $P=\frac{1}{2}$.

$Y$ takes integer values 1 and 2, with equal probability. Then the probability that $X=Y$ is

(A) $\frac{15}{64}$ (B) $\frac{15}{32}$

(C) $\frac{1}{2}$ (D) $\frac{15}{16}$

## 2011 IIT Madras

9.12 A box contains 3 red and 2 black balls. Four balls are removed from the box one by one, without replacement. The probability of the ball remaining in the box being red, is

(A) $\frac{609}{625}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{81}{625}$

## 2012 IIT Delhi

9.13 A box containing 10 identical compartments has 6 red balls and 2 blue balls. If each compartment can hold only one ball, then the number of different possible arrangements are

(A) 1026 (B) 1062

(C) 1260 (D) 1620

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## 2013 IIT Bombay

**9.14** The number of emails received on size consecutive days is 11, 9, 18, 18, 4 and 15 respectively.

What are the median and mode for these data?

(A) 18 and 11 respectively
(B) 13 and 18 respectively
(C) 13 and 12.5 respectively
(D) 12.5 and 18 respectively

**9.15** For two rolls of a fair die, the probability of getting a 4 in the first roll and a number less than 4 in the second roll, up to 3 digits after the decimal point, is _____.

**9.16** In a factory, two machines $M_1$ and $M_2$ manufacture 60% and 40% of the auto components respectively. Out of the total production, 2% of $M_1$ and 3% of $M_2$ are found to be defective. If a randomly drawn auto component from the combined lot is found defective, what is the probability that it was manufactured by $M_2$?

(A) 0.35 (B) 0.45
(C) 0.5 (D) 0.4

## 2014 IIT Kharagpur

**9.17** Consider the following two normal distributions

$$f_1(x) = e^{-\pi x^2}$$

$$f_2(x) = \frac{1}{2\pi} e^{\left\{-\frac{1}{4\pi}(x^2+2x+1)\right\}}$$

If $\mu$ and $\sigma$ denote the mean and standard deviation, respectively, then

(A) $\mu_1 < \mu_2$ and $\sigma_1^2 < \sigma_2^2$
(B) $\mu_1 < \mu_2$ and $\sigma_1^2 > \sigma_2^2$
(C) $\mu_1 > \mu_2$ and $\sigma_1^2 < \sigma_2^2$
(D) $\mu_1 > \mu_2$ and $\sigma_1^2 > \sigma_2^2$

**9.18** In rolling of two fair dice, the outcome of an experiment is considered to be the sum of the numbers appearing on the dice. The probability is highest for the outcome of _____.

## 2016 IISc Bangalore

**9.19** The model $y = mx^2$ is to be fit to the data given below.

| $x$ | 1 | $\sqrt{2}$ | $\sqrt{3}$ |
|---|---|---|---|
| $y$ | 2 | 5 | 8 |

Using linear regression, the value (rounded off to the second decimal place) of $m$ is _____.

## 2017 IIT Roorkee

**9.20** A box has 6 red balls and 4 white balls. A ball is picked at random and replaced in the box, after which a second ball is picked.

The probability of both the balls being red, rounded to 2 decimal places, is ___.

**9.21** The marks obtained by a set of students are :

38, 84, 45, 70, 75, 60, 48

The mean and median marks, respectively are

(A) 45 and 75 (B) 55 and 48
(C) 60 and 60 (D) 60 and 70

## 2018 IIT Guwahati

**9.22** A watch uses two electronic circuits (ECs). Each EC has a failure probability of 0.1 in one year of operation. Both ECs are required for functioning of the watch. The probability of the watch functioning for one year without failure is

(A) 0.99 (B) 0.90
(C) 0.81 (D) 0.80



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### 2019 IIT Madras

9.33 Two unbiased dice are thrown. Each dice can show any number between 1 and 6. The probability that the sum of the outcomes of the two dice is divisible by 4 is _____ (rounded off to two decimal places).

## Production & Industrial : PI

### 2005 IIT Bombay

9.1 Two dice are thrown simultaneously. The probability that the sum of numbers on both exceeds 8 is,

(A) $\frac{4}{36}$ (B) $\frac{7}{36}$

(C) $\frac{9}{36}$ (D) $\frac{10}{36}$

9.2 The life of a bulb (in hours) is a random variable with an exponential distribution $f(t)=\alpha e^{-\alpha t}, 0 \le t \le \infty$. The probability that its value lies between 100 and 200 hours is,

(A) $e^{-100\alpha}-e^{-200\alpha}$ (B) $e^{-100}-e^{-200}$

(C) $e^{-100\alpha}+e^{-200\alpha}$ (D) $e^{-200\alpha}-e^{-100\alpha}$

### 2007 IIT Kanpur

9.3 If $X$ is a continuous random variable whose probability density function is given by

$$f(x)=\begin{cases} K(5x-2x^2), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Then $P(X>1)$ is

(A) $\frac{3}{14}$ (B) $\frac{4}{5}$

(C) $\frac{14}{17}$ (D) $\frac{17}{28}$

9.4 Two cards are drawn at random in succession, with replacement, from a deck of 52 well shuffled cards. Probability of getting both 'Aces' is

(A) $\frac{1}{169}$ (B) $\frac{2}{169}$

(C) $\frac{1}{13}$ (D) $\frac{2}{13}$

9.5 The random variable $X$ takes on the values 1, 2, or 3 with probabilities

$$\frac{(2+5P)}{5}, \frac{(1+3P)}{5} \text{ and } \frac{(1.5+2P)}{5}$$

respectively. The values of $P$ and $E[X]$ are respectively

(A) 0.05, 1.87 (B) 1.90, 5.87

(C) 0.05, 1.10 (D) 0.25, 1.40

9.6 The average number of accidents occurring monthly on an assembly shop floor is 2. The probability that there will be at least one accident in this month is estimated to be

(A) 0.055 (B) 0.456

(C) 0.865 (D) 0.950

9.7 $X_1,.....X_{100}$ are Bernoulli random variables with a probability of success equal to 0.6. By the Central Limit Theorem, the random variable $Y=\sum_{i=1,.....100} X_i$ is approximately normally distributed. Then $Y$ has mean and variance respectively equal to

(A) 40 and 24 (B) 60 and 24

(C) 40 and 12 (D) 60 and 12

### 2008 IISc Bangalore

9.8 In a game, two players $X$ and $Y$ toss a coin alternately. Whosoever gets a 'head' first, wins the game and the game is terminated. Assuming that player $X$ starts the game, the probability of player $X$ winning the game is

(A) 1/3 (B) 1/2

(C) 2/3 (D) 3/4



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**9.9** For a random variable $X(-\infty < x < \infty)$ following normal distribution, the mean is $\mu = 100$. If the probability is $P = \alpha$ for $x \geq 110$, then the probability of $X$ lying between 90 and 110, i.e., $P(90 \leq x \leq 110)$ will be equal to

(A) $1-2\alpha$ (B) $1-\alpha$

(C) $1-\frac{\alpha}{2}$ (D) $2\alpha$

## 2009 IIT Roorkee

**9.10** Consider the joint probability mass function of random variables $X$ and $Y$ as shown in table below for instance, $P\{X=1, Y=2\} = 0.3$

| | $X=1$ | $X=2$ |
|---|---|---|
| $Y=1$ | 0.2 | 0.3 |
| $Y=2$ | 0.3 | 0.1 |
| $Y=3$ | 0.1 | |

The value of $P\{X=2 | Y=2\}$ is

(A) 0.10 (B) 0.25

(C) 0.40 (D) 0.75

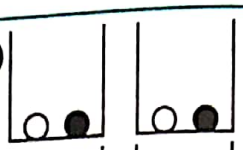
## 2010 IIT Guwahati

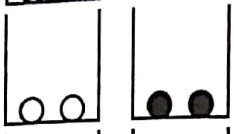
**9.11** If a random variable $X$ satisfies the Poisson's distribution with a mean value of 2, then the probability that $X \geq 2$ is

(A) $2e^{-2}$ (B) $1-2e^{-2}$

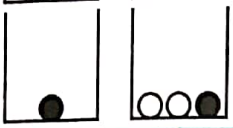
(C) $3e^{-2}$ (D) $1-3e^{-2}$

**9.12** Two white and two black balls, kept in two bins, are arranged in four ways as shown below. In each arrangement, a bin has to be chosen randomly and only one ball needs to be picked randomly from the chosen bin. Which one of the following arrangements has the highest probability for getting a white ball picked?

(A) 

(B) 

(C)

(D) 

## 2011 IIT Madras

**9.13** It is estimated that the average number of events during a year is three. What is the probability of occurrence of not more than two events over a two-year duration? Assume that the number of events follow a poisson distribution.

(A) 0.052 (B) 0.062

(C) 0.072 (D) 0.082

## 2012 IIT Delhi

**9.14** An automobile plant contracted to buy shock absorbers from two suppliers $X$ and $Y$. $X$ supplies 60% and $Y$ supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of $X$'s shock absorbers, 96% are reliable. Of $Y$'s shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by $Y$ is

(A) 0.288 (B) 0.334

(C) 0.667 (D) 0.720

**9.15** A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is



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(A) $\frac{1}{20}$ (B) $\frac{1}{12}$

(C) $\frac{3}{10}$ (D) $\frac{1}{2}$

## 2013 IIT Bombay

9.16 The probability that a student knows the correct answer to a multiple choice question is 2/3. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is 1/4. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$

(C) $\frac{5}{6}$ (D) $\frac{8}{9}$

9.17 Let $X$ be a normal random variable with mean 1 and variance 4. The probability $P\{X<0\}$ is

(A) 0.5

(B) greater than zero and less than 0.5

(C) greater than 0.5 and less than 1.0

(D) 1.0

9.18 A company manufactures 1000 toys every day. On an average, 10% of the toys are defective and 40% of the defective toys can be reworked into defect-free ones. The average number of defect-free toys manufactured daily is

(A) 900 (B) 920

(C) 940 (D) 960

## 2014 IIT Kharagpur

9.19 A simple random sample of 100 observations was taken from a large population. The sample mean and the standard deviation were determined to be 80 and 12, respectively. The standard error of mean is _____.

9.20 Marks obtained by 100 students in an examination are given in the table

| S. No. | Marks obtained | Number of students |
|---|---|---|
| 1. | 25 | 20 |
| 2. | 30 | 20 |
| 3. | 35 | 40 |
| 4. | 40 | 20 |

What would be the mean, median, and mode of the marks obtained by the students?

(A) Mean 33; Median 35; Mode 40

(B) Mean 35; Median 32.5; Mode 40

(C) Mean 33; Median 35; Mode 35

(D) Mean 35; Median 32.5; Mode 35

9.21 In a given day in the rainy season, it may rain 70% of the time. If it rains, chance that a village fair will make a loss on that day is 80%. However, if it does not rain, chance that the fair will make a loss on that day is only 10%. If the fair has not made a loss on a given day in the rainy season, what is the probability that it has not rained on that day?

(A) $\frac{3}{10}$ (B) $\frac{9}{11}$

(C) $\frac{14}{17}$ (D) $\frac{27}{41}$

## 2015 IIT Kanpur

9.22 A product is an assembly of 5 different components. The product can be sequentially assembled in two possible ways. If the 5 components are placed in a box and these are drawn at random from the box, then the probability of getting a correct sequence is

(A) $\frac{2}{5!}$ (B) $\frac{2}{5}$

(C) $\frac{2}{(5-2)!}$ (D) $\frac{2}{(5-3)!}$



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## 2016 IISc Bangalore

**9.23** A fair coin is tossed $N$ times. The probability that head does not turn up in any of the tosses is

(A) $\left(\frac{1}{2}\right)^{N-1}$ (B) $1-\left(\frac{1}{2}\right)^{N-1}$

(C) $\left(\frac{1}{2}\right)^{N}$ (D) $1-\left(\frac{1}{2}\right)^{N}$

**9.24** A normal random variable $X$ has the following probability density function

$$f_x(x)=\frac{1}{\sqrt{8\pi}}e^{-\left\{\frac{(x-1)^2}{8}\right\}}, \quad -\infty<x<\infty$$

then $\int_{1}^{\infty} f_x(x)\,dx =$

(A) 0 (B) $\frac{1}{2}$

(C) $1-\frac{1}{e}$ (D) 1

## 2017 IIT Roorkee

**9.25** Two machines are defective in a lot of 10. A combination of four machines is to be picked at a time from the lot. The maximum number of combinations that can be obtained without any defective machine is _____.

## 2018 IIT Guwahati

**9.26** Weights (in kg) of six products are 3, 7, 6, 2, 3 and 4. The median weight (in kg, up to one decimal place) is _____.

**9.27** The probabilities of occurrence of events $F$ and $G$ are $P(F)=0.3$ and $P(G)=0.4$, respectively. The probability that both events occur simultaneously is $P(F\cap G)=0.2$. The probability of occurrence of at least one event $P(F\cup G)$ is _____.

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## Electronics & Communication : EC

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | A | 9.2 | 0.125 | 9.3 | D | 9.4 | C | 9.5 | D |
| 9.6 | A | 9.7 | C | 9.8 | A | 9.9 | C | 9.10 | D |
| 9.11 | C | 9.12 | C | 9.13 | B | 9.14 | 0.66 | 9.15 | C |
| 9.16 | 3 | 9.17 | 0.7978 | 9.18 | A | 9.19 | 0.444 | 9.20 | C |
| 9.21 | 6 | 9.22 | 1 | 9.23 | 0.072 | 9.24 | 0.027 | 9.25 | 0.25 |
| 9.26 | 11 | | | | | | | | |

## Electrical Engineering : EE

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | B | 9.2 | D | 9.3 | C | 9.4 | C | 9.5 | C |
| 9.6 | B | 9.7 | C | 9.8 | A | 9.9 | C | 9.10 | B |
| 9.11 | A | 9.12 | B | 9.13 | 0.142 | 9.14 | 0.4 | 9.15 | 0.42 |
| 9.16 | 0.25 | 9.17 | D | 9.18 | D | 9.19 | 0.01818 | 9.20 | A |
| 9.21 | 0.9 | 9.22 | A | 9.23 | C | 9.24 | B | 9.25 | 0.265 |

## Mechanical Engineering : ME

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | A | 9.2 | D | 9.3 | C | 9.4 | B | 9.5 | C |
| 9.6 | D | 9.7 | A | 9.8 | D | 9.9 | B | 9.10 | A |
| 9.11 | C | 9.12 | C | 9.13 | D | 9.14 | D | 9.15 | C |
| 9.16 | D | 9.17 | B | 9.18 | D | 9.19 | D | 9.20 | B |
| 9.21 | D | 9.22 | A | 9.23 | D | 9.24 | A | 9.25 | C |
| 9.26 | D | 9.27 | D | 9.28 | B | 9.29 | D | 9.30 | B |
| 9.31 | D | 9.32 | D | 9.33 | A | 9.34 | 0.259 | 9.35 | 0.65 |
| 9.36 | A | 9.37 | 0.8145 | 9.38 | 50 | 9.39 | B | 9.40 | 74.55 |
| 9.41 | D | 9.42 | C | 9.43 | A | 9.44 | 0.4095 | 9.45 | 99.7 |
| 9.46 | A | 9.47 | C | 9.48 | B | 9.49 | C | 9.50 | B |
| 9.51 | A | 9.52 | A | 9.53 | 3 | 9.54 | 0.171 | | |

## Civil Engineering : CE

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | C | 9.2 | C | 9.3 | D | 9.4 | D | 9.5 | B |
| 9.6 | C | 9.7 | B | 9.8 | D | 9.9 | A | 9.10 | D |
| 9.11 | C | 9.12 | C | 9.13 | D | 9.14 | A | 9.15 | 6 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.16 | D | 9.17 | 0.4 | 9.18 | 0.265 | 9.19 | B | 9.20 | B |
| 9.21 | 0.27 | 9.22 | 0.24 | 9.23 | 1.066 | 9.24 | A | 9.25 | 54.5 |
| 9.26 | A | 9.27 | B | 9.28 | 0.5 | 9.29 | B | 9.30 | B |
| 9.31 | 0 | 9.32 | A | 9.33 | 0.5 | 9.34 | C | 9.35 | 17.9 |
| 9.36 | B | | | | | | | | |

## Instrumentation Engineering : IN

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | D | 9.2 | C | 9.3 | A | 9.4 | A | 9.5 | A |
| 9.6 | A | 9.7 | B | 9.8 | A | 9.9 | C | 9.10 | C |
| 9.11 | D | 9.12 | C | 9.13 | B | 9.14 | A | 9.15 | A |
| 9.16 | 2 | 9.17 | 0.893 | 9.18 | B | 9.19 | 4 | 9.20 | A |
| 9.21 | C | 9.22 | D | 9.23 | B | 9.24 | B | 9.25 | A |
| 9.26 | A | | | | | | | | |

## Chemical Engineering : CH

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | C | 9.2 | D | 9.3 | C | 9.4 | B | 9.5 | C |
| 9.6 | B & C | 9.7 | D | 9.8 | B | 9.9 | B | 9.10 | A |
| 9.11 | B | 9.12 | B | 9.13 | C | 9.14 | B | 9.15 | 0.083 |
| 9.16 | C | 9.17 | C | 9.18 | 7 | 9.19 | 4 | 9.20 | 0.36 |
| 9.21 | C | 9.22 | C | 9.23 | 0.25 | | | | |

## Production & Industrial : PI

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9.1 | D | 9.2 | A | 9.3 | D | 9.4 | A | 9.5 | A |
| 9.6 | C | 9.7 | B | 9.8 | C | 9.9 | A | 9.10 | B |
| 9.11 | D | 9.12 | C | 9.13 | B | 9.14 | B | 9.15 | D |
| 9.16 | D | 9.17 | B | 9.18 | C | 9.19 | 1.2 | 9.20 | C |
| 9.21 | D | 9.22 | A | 9.23 | C | 9.24 | B | 9.25 | 70 |
| 9.26 | 3.5 | 9.27 | 0.5 | | | | | | |

## Explanations Probability & Statistics

### Electronics & Communication : EC

**9.1 (A)**

**Given :** A and B are two mutually exclusive (disjoint) events (they cannot occur simultaneously).

$$P(A \cap B) = 0 \qquad \ldots(i)$$

So, option (C) is false.

The union of any two event is given by,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

From equation (i),

$$P(A \cup B) = P(A) + P(B) - 0$$

$$P(A \cup B) = P(A) + P(B)$$

Hence, the correct option is (A).



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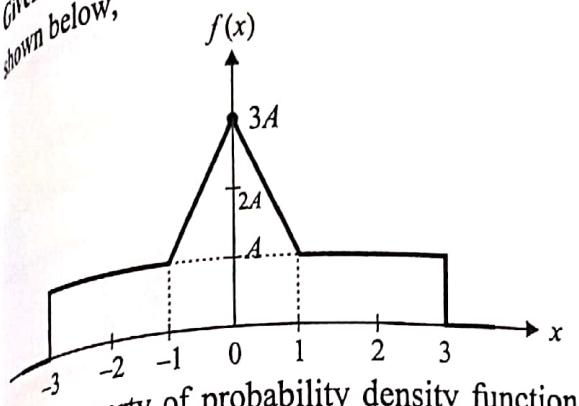
**9.2** **0.125**

Given : A probability density function is as shown below,




By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad \ldots (i)$$

$$\text{Area} = (6\times A) + \left(\frac{1}{2}\times 2\times 2A\right)$$

$$\text{Area} = 6A + 2A = 8A$$

From equation (i),

$$8A = 1$$

$$A = \frac{1}{8} = 0.125$$

Hence, the value of $A$ is **0.125.**

**9.3** **(D)**

Given : A fair dice is rolled twice.

The total number of possible outcomes are 36. The figure given below shows the sample space of given event. The shaded items shows the desired outcomes in which an odd number follow and even number.

$$S = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

So, $n(S) = 36$

$n(A) = 9$

The required probability is given by,

$$P(A) = \frac{n(A)}{n(S)} = \frac{9}{36} = \frac{1}{4}$$

Hence, the correct option is (D).

**9.4** **(C)**

**Given :** A probability density function,

$$p(x) = Ke^{-a|x|},\ x \in (-\infty, \infty)$$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} p(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} Ke^{-a|x|}dx = 1$$

Using properties of even functions,

$$2\int_{0}^{\infty} Ke^{-ax}dx = 1$$

$$\frac{-2K}{a}\left[e^{-\infty} - e^{0}\right] = 1$$

$$\frac{2K}{a} = 1$$

$$K = \frac{a}{2} = 0.5a$$

Hence, the correct option is (C).

**Key Point**

(i) A function $f(x)$ is said to be even if

$$f(-x) = f(x)$$

(ii) For any even function $f(x)$,

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

**9.5** **(D)**

**Given :** The percentage of computers supplied by three companies $X$, $Y$, $Z$ and the probability of those being defective are given in the table below,



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| Company | % of computers supplied | Probability of being defective |
|---|---|---|
| $X$ | 60 % | 0.01 |
| $Y$ | 30 % | 0.02 |
| $Z$ | 10 % | 0.03 |

By the concept of conditional probability, the probability that a computer which is found defective is supplied by $Y$ is given by,

$$P\left(\frac{Y}{D}\right)=\frac{P(Y\cap D)}{P(D)}$$

Where,

$P(D)$ = Probability of defective computer.

$P(Y)$ = Probability of a computer being supplied by company $Y$.

$$P\left(\frac{Y}{D}\right)=\frac{0.3\times0.02}{(0.6\times0.01)+(0.3\times0.02)+(0.1\times0.03)}$$

$$P\left(\frac{Y}{D}\right)=\frac{0.006}{0.015}=0.4$$

Hence, the correct option is (D).

**9.6 (A)**

Variance of a random variable $X$ is given by,

$$\text{Var}(X)=E[X^2]-(E[X])^2$$

Here, $E[X^2]$ = Mean square value

$E[X]$ = Mean or expected value

Hence, the correct option is (A).

**9.7 (C)**

**Given :** The probabilities of failing in two papers in an examination are,

$$P(A)=0.3,\ P(B)=0.2$$

By the concept of conditional probability, the probability of failing in paper 1 when student has failed in paper 2 is given by,

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$$

According to question,

$$P\left(\frac{A}{B}\right)=0.6$$

So, $$0.6=\frac{P(A\cap B)}{0.2}$$

$$P(A\cap B)=0.12$$

Hence, the correct option is (C).

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**9.8 (A)**

**Given :** A probability density function is,

$$P_x(x)=Me^{-2|x|}+Ne^{-3|x|}$$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty}P_x(x)\,dx=1$$

$$\int_{-\infty}^{\infty}(Me^{-2|x|}+Ne^{-3|x|})\,dx=1$$

$P_x(x)$ is even function as $|x|$ is even function.

So, by the property of even function,

$$2\int_{0}^{\infty}(Me^{-2x}+Ne^{-3x})\,dx=1$$

$$2M\left(\frac{e^{-2x}}{-2}\right)_0^{\infty}+2N\left(\frac{e^{-3x}}{-3}\right)_0^{\infty}=1$$

$$-M(e^{-\infty}-e^{0})-\frac{2}{3}N(e^{-\infty}-e^{0})=1$$

$$-M(0-1)-\frac{2}{3}N(0-1)=1$$

$$M+\frac{2}{3}N=1$$

Hence, the correct option is (A).

**9.9 (C)**

**Given :**

The coin is tossed 10 times.

Total number of trials $n=10$

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Probability of getting head in any trial $= \frac{1}{2}$

Probability of getting tail in any trial $= \frac{1}{2}$

Desired condition that only first two tosses will yield heads is

$$A = \{\boxed{H\ H}\ T\ T\ T\ T\ T\ T\ T\ T\}$$

So, required probability $= \left(\frac{1}{2}\right)^{10}$

Hence, the correct option is (C).

**9.10 (D)**

Given : A fair coin is tossed 4 times.

**Method 1**

The sample space for four tosses of a coin is as follows,

| Tosses 1st | 2nd | 3rd | 4th | Favorable combinations |
|---|---|---|---|---|
| H | H | H | H | (4 Heads) |
| H | H | H | T | (3 Heads, 1 Tail) |
| H | H | T | H | (3 Heads, 1 Tail) |
| H | H | T | T | |
| H | T | H | H | (3 Heads, 1 Tail) |
| H | T | H | T | |
| H | T | T | H | |
| H | T | T | T | |
| T | H | H | H | (3 Heads, 1 Tail) |
| T | H | H | T | |
| T | H | T | H | |
| T | H | T | T | |
| T | T | H | H | |
| T | T | H | T | |
| T | T | T | H | |
| T | T | T | T | |

**Fig. Samples space of 4 tosses**

Here in above figure, the favourable combination are in shading that is equal to 5.

Thus, required probability $= \frac{5}{16}$.

Hence, the correct option is (D).

**Method 2**

Required probability is given by,

$P$ = Getting exactly 3 Heads and 1 tail + Getting exactly 4 Heads and 0 tail

According to binomial distribution, the probability of $r$ trails is given by,

$$P(r) = {}^{n}C_{r}\, p^{r} (1-p)^{n-r}$$

where, $P$ is probability of occurrence of the event.

$$P(H > T) = {}^{4}C_{4}\left(\frac{1}{2}\right)^{4}\left(\frac{1}{2}\right)^{0} + {}^{4}C_{3}\left(\frac{1}{2}\right)^{3}\left(\frac{1}{2}\right)^{1}$$

$$P(H > T) = \frac{5}{16}$$

Hence, the correct option is (D).

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**9.11 (C)**

**Given :** A fair dice is tossed 2 times.

**Method 1**

The total number of outcomes for 2 tosses of a fair dice are given by,

$$n(S) = 6^2 = 36$$

If the second toss results in a value higher than the first toss, the possible set of values can be given by the table below :

| Dice 1 | Dice 2 | Total combinations |
|---|---|---|
| 1 | 2, 3, 4, 5, 6 | 5 |
| 2 | 3, 4, 5, 6 | 4 |
| 3 | 4, 5, 6 | 3 |
| 4 | 5, 6 | 2 |
| 5 | 6 | 1 |

Favorable events,

$$n(A) = 5 + 4 + 3 + 2 + 1 = 15$$



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The required probability is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{15}{36}=\frac{5}{12}$$

Hence, the correct option is (C).

**Method 2**

The sample space for tossing of a dice two times is given by,

$$S=\left\{\begin{array}{cccccc}(1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6)\\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6)\\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6)\\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6)\\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6)\\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6)\end{array}\right\}$$

Total number of favorable cases in which second toss results in a value that is higher than the first toss are shaded in above figure = 5 + 4 + 3 + 2 + 1 = 15

Then probability $=\frac{15}{36}=\frac{5}{12}$

Hence, the correct option is (C).

## 9.12 (C)

**Given :** A coin is tossed till a head appears for first time.

If required tosses is odd, then the possible sequence of heads and tails will be,

*H, TTH, TTTTH, TTTTTTH, .....*

Since, these events are mutually exclusive, we can add the probability of each event.

Thus, the required probability is given by,

$$P=\frac{1}{2}+\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)+\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)+.....\infty$$

$$P=\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}+\frac{1}{2^7}+.....\infty$$

This is an infinite geometric series with the first term $a=\frac{1}{2}$ and common ratio $r=\frac{1}{4}$.

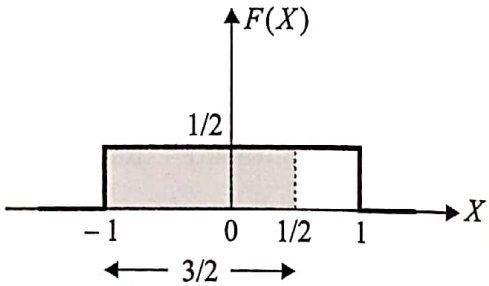
So, $$P=\frac{a}{1-r}$$

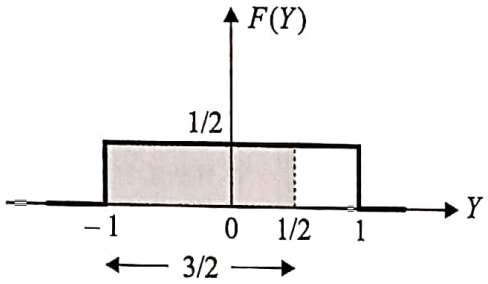
$$P=\frac{\frac{1}{2}}{1-\frac{1}{4}}=\frac{\frac{1}{2}}{\frac{3}{4}}=\frac{2}{3}$$

Hence, the correct option is (C).

## 9.13 (B)

**Given :** Two independent random variables $X$ and $Y$ uniformly distributed in the interval (–1, 1) as follows :




The area of uniformly distributed function = 1

Area of rectangle = Base × Height = 1

$$2\times h=1 \quad\Rightarrow\quad h=\frac{1}{2}$$

To calculate the probability that max[$X$, $Y$] is less than $\frac{1}{2}$, it is needed to calculate individual probability of both random variables as they are independent.

For independent random variable,

$$P[XY]=P[X]P[Y] \qquad \ldots (i)$$

Here the point is to be noted that probability is the area of the distributed function.

$$P\left[X<\frac{1}{2}\right]=\frac{1}{2}\times\frac{3}{2}=\frac{3}{4}=\text{Area}$$



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Similarly, $P\left[Y<\frac{1}{2}\right]=\frac{3}{4}=$ Area

From equation (i),

$$P[XY]=P\left[X<\frac{1}{2}\right]P\left[Y<\frac{1}{2}\right]$$

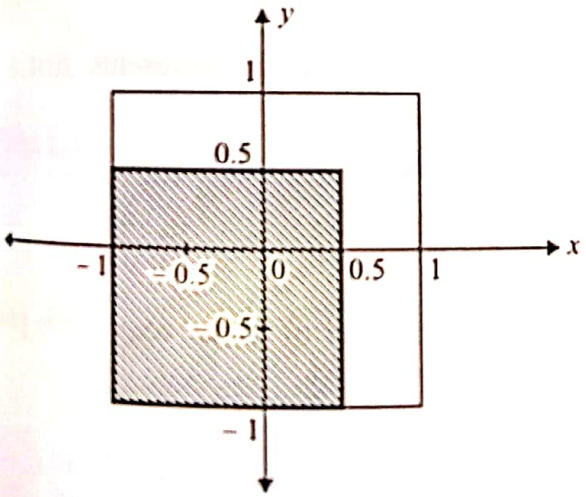
$$P[XY]=\frac{3}{4}\times\frac{3}{4}=\frac{9}{16}$$

Hence, the correct option is (B).

**Method 2**

Let, $-1\le x\le 1$ and $-1\le y\le 1$ represents an entire square.

The region in which maximum of $\{x, y\}$ is less than $\frac{1}{2}$ is shown below as shaded region inside the square.




$$P\left[\{X, Y\}<\frac{1}{2}\right]=\frac{\text{Shaded area of square}}{\text{Entire area of square}}$$

$$P\left[\{X, Y\}<\frac{1}{2}\right]=\frac{1.5\times 1.5}{2\times 2}$$

$$P\left[\{X, Y\}<\frac{1}{2}\right]=\frac{2.25}{4}=\frac{9}{16}$$

Hence, the correct option is (B).

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## 9.14 0.66

**Given :** In a housing society, half of the families have a single child, and remaining half of the families have 2 children.

**Method 1**

Let there are total $N$ families.

So, total number of children are given by,

$$=\left(2\times\frac{N}{2}\right)+\left(1\times\frac{N}{2}\right)=\frac{3N}{2}$$

A sibling can occur only in those families in which there are 2 children.

Thus, the required probability is given by,

$$P=\frac{\left(2\times\frac{N}{2}\right)}{\frac{3N}{2}}=\frac{2}{3}=0.66$$

Hence, the probability is **0.66.**

**Method 2**

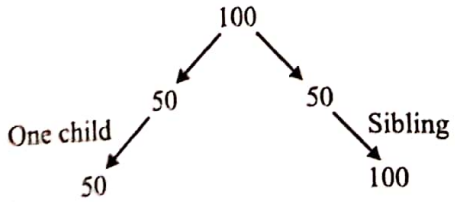
Assume there are 100 families in that society.

According to question, 50 families have one child and remaining 50 families have two children.

So, total number of children

$$=50+(50\times 2)=150$$

The below figure shows the families and their children.




Thus, probability that the child has a sibling is given by,

$$P=\frac{100}{150}=\frac{2}{3}=0.66$$

Hence, the probability is **0.66.**



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## 9.15 (C)

**Given :** An unbiased coin is tossed infinite times.

Probability of getting a head, $p = \frac{1}{2}$

Probability of getting a tail, $q = \frac{1}{2}$

To calculate probability that the fourth head appears at the tenth toss, our approach is to check the 3 heads appears in first 9 trials.

According to binomial distribution,

$$P_1 = {}^{n}C_r \times p^r \times q^{n-r}$$

$r = 3,\ n = 9$

Probability of getting exactly 3 heads in 9 trials is given by,

$$P_1 = {}^{9}C_3 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^6 = {}^{9}C_3 \times \left(\frac{1}{2}\right)^9$$

and in $10^{th}$ trial head must appear.

Probability of getting head in $10^{th}$ trial is given by,

$$P_2 = \frac{1}{2}$$

So, the required probability is,

$$P = P_1 \times P_2$$

$$P = {}^{9}C_3 \left(\frac{1}{2}\right)^9 \times \frac{1}{2} = \frac{9 \times 8 \times 7}{3!} \times \left(\frac{1}{2}\right)^{10}$$

$$P = \frac{84}{1024} = 0.082$$

Hence, the correct option is (C).

## 9.16 3

**Given :** A fair coin is tossed till both head and tail appear at least once.

### Method 1

At least two tosses are required.

The random variable $X$ represents first toss starting with head.

| $X$ | 2 | 3 | 4 | 5 | ..... |
|---|---|---|---|---|---|
| $P(X)$ | $\frac{1}{2^2}$ | $\frac{1}{2^3}$ | $\frac{1}{2^4}$ | $\frac{1}{2^5}$ | ..... |

Expected value of $X$ is given by,

$$E[X] = \Sigma X P(X)$$

$$E[X] = 2 \times \frac{1}{2^2} + 3 \times \frac{1}{2^3} + 4 \times \frac{1}{2^4} + .....$$

$$E[X] = \frac{1}{2}\left(2 \times \frac{1}{2} + 3 \times \frac{1}{2^2} + 4 \times \frac{1}{2^3} + ....\right)$$

$$E[X] = \frac{1}{2}\left(1 + 2 \times \frac{1}{2} + 3 \times \frac{1}{2^2} + 4 \times \frac{1}{2^3} + ..... - 1\right)$$

Now, by Binomial Expansion,

$$E[X] = \frac{1}{2}\left[\left(1 - \frac{1}{2}\right)^{-2} - 1\right] = \frac{1}{2}\left[\left(\frac{1}{2}\right)^{-2} - 1\right]$$

$$E[X] = \frac{1}{2}(4 - 1) = \frac{3}{2}$$

The random variable $Y$ represents first toss starting with tail.

Similarly as above

$$E[Y] = \frac{3}{2}$$

The average number of tosses required is given by,

$$E = E[X] + E[Y] = \frac{3}{2} + \frac{3}{2} = 3$$

Hence, the average number of tosses required is **3.**

**Key Point**

Binomial Expansion

$$(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + .....\infty$$

### Method 2

In first toss, either a head or a tail occurs.

Let us assume that a tail occurs then the experiment is repeatedly performed till, a head comes.



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Probability of getting head in second trial $= \frac{1}{2}$

Probability of getting head in third trial $= \left(\frac{1}{2}\right)^2$

Thus, average number of tosses required,

$$N = 2\times\frac{1}{2} + 3\times\left(\frac{1}{2}\right)^2 + 4\times\left(\frac{1}{2}\right)^3 \ldots\ldots$$

$$\frac{N}{2} = 2\times\left(\frac{1}{2}\right)^2 + 3\times\left(\frac{1}{2}\right)^3 + 4\times\left(\frac{1}{2}\right)^4 \ldots\ldots$$

Now, $$\left(N - \frac{N}{2}\right) = 2\times\frac{1}{2} + (3-2)\times\left(\frac{1}{2}\right)^2 + (4-3)\times\left(\frac{1}{2}\right)^3 \ldots\ldots$$

$$\frac{N}{2} = 1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \ldots\ldots$$

$$\frac{N}{2} = 1 + \left(\frac{1}{2}\right)^2\left[1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \ldots\ldots\right]$$

Formula for infinite GP series,

$$a + ar + ar^2 + \ldots. = \frac{a}{1-r}$$

$$\frac{N}{2} = 1 + \left(\frac{1}{2}\right)^2\left(\frac{1}{1-\frac{1}{2}}\right)$$

$$\frac{N}{2} = \frac{3}{2}$$

$$N = 3$$

Hence, the average number of tosses required is 3.

## 9.17 0.7978

**Given :** A zero mean unit variance Gaussian random variable $X$.

For a Gaussian random variable,

$$f_x(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

For mean $(\mu) = 0$ and variance $(\sigma^2) = 1$

$$f_x(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$

$$E[|x|] = \int_{-\infty}^{\infty} |x| f_x(x)dx$$

$$E[|x|] = \int_{-\infty}^{\infty} |x| \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dx$$

$$E[|x|] = 2\int_0^{\infty} x \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dx$$

$$E[|x|] = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} xe^{\frac{-x^2}{2}} dx$$

Let, $\frac{x^2}{2} = t$

$$\frac{2x\,dx}{2} = dt$$

Thus, $$E[|x|] = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} dt$$

$$E[|x|] = \frac{2}{\sqrt{2\pi}} \left[\frac{e^{-t}}{-1}\right]_0^{\infty}$$

$$E[|x|] = \frac{-2}{\sqrt{2\pi}} (e^{-\infty} - e^0)$$

$$E[|x|] = 0.7978$$

Hence, the value of $E[|x|]$ is **0.7978.**

## 9.18 (A)

**Given :** The total number of calls arriving at a telephone exchange in a fixed time interval is represented by a probability density function.

The point is to be noted that the probability distribution function of total number of calls in fixed time interval will be a **discrete probability distribution.**

Here Gaussian, exponential and Gamma function distribution all are continuous probability distributions.



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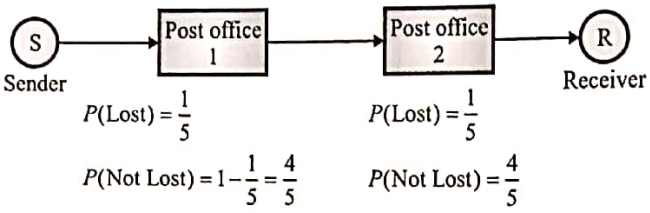
From all the options, only **Poisson distribution** is discrete distribution.

Hence, the correction option is (A).

## 9.19 0.444




Let, $P(A)$ = Probability to lost at post office 1

$$P(A)=\frac{1}{5}$$

$P(L\cap B)$ = Probability to lost at post office 2 i.e. the parcel is not lost by post office 1

$$P(L\cap B)=\frac{4}{5}\times\frac{1}{5}=\frac{4}{25}$$

Total probability to lost the parcel is,

$$P(L)=\frac{1}{5}+\frac{4}{25}=\frac{9}{25}$$

Required probability is given by,

$$P\left(\frac{B}{L}\right)=\frac{P(L\cap B)}{P(L)}=\frac{\frac{4}{25}}{\frac{9}{25}}=\frac{4}{9}=0.444$$

Hence, the probability is **0.444.**

## 9.20 (C)

**Given :** For two independent events $A$ and $B$,

$$P(A\cap B)=P(A)\times P(B)$$

From additive property,

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$

$$P(A\cup B)=P(A)+P(B)-P(A)P(B)$$

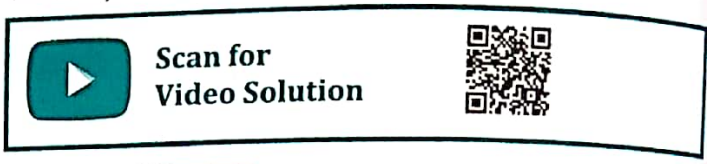
So, option (C) is clearly false.

$$P(A/B)=\frac{P(A\cap B)}{P(B)}$$

$$P(A/B)=\frac{P(A)P(B)}{P(B)}=P(A)$$

Thus, all other options are true, except (C).

Hence, the correct option is (C).




## 9.21 6

**Given :** A fair dice is thrown repeatedly till '3' is observed.

For a fair die,

The probability of getting 3 in each trial $=\frac{1}{6}$

The probability of getting other than 3 in each trial $=1-\frac{1}{6}=\frac{5}{6}$

$x$ is the number of times the die is thrown till '3' is observed first time.

The probability of getting 3 in $(x=1)$ one trial

$$=\frac{1}{6}$$

The probability of getting 3 in $(x=2)$ two trial

$$=\frac{5}{6}\times\frac{1}{6}$$

For $(x=3)$ i.e. third trial $=\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}$

So, probability is

$$p(x)=\begin{cases}\frac{1}{6} & , \quad x=1\\ \frac{5}{6}\times\frac{1}{6} & , \quad x=2\\ \frac{5}{6}\times\frac{5}{6}\times\frac{1}{6} & , \quad x=3\\ \vdots & \vdots\\ \vdots & \vdots\end{cases}$$

The expected value of $X$ is given by,

$$E(X)=\sum_{i=1}^{n}x_i p(x_i)$$

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$$E[X]=1\times\frac{1}{6}+2\times\left(\frac{5}{6}\times\frac{1}{6}\right)$$
$$+3\times\left(\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}\right)+.....$$
$$E[X]=\frac{1}{6}\left[1+2\times\left(\frac{5}{6}\right)+3\times\left(\frac{5}{6}\right)^2\right.$$
$$\left.+4\left(\frac{5}{6}\right)^3+.....\right]$$
$$E[X]=\frac{1}{6}\left[1-\frac{5}{6}\right]^{-2}$$
$$\left[1+2x+3x^2+.....=(1-x)^{-2}\right]$$
$$E[X]=\frac{1}{6}\times\left(\frac{1}{6}\right)^{-2}$$
$$E[X]=\frac{1}{6}\times 36=6$$

Hence, the expected value of $X$ is **6.**

**9.22 1**

Given : Second moment of Poisson-distributed random variable = 2

In Poisson distribution,

Mean = First moment $=\lambda$

Second moment $=\lambda^2+\lambda$

Thus, $\lambda^2+\lambda=2$

$\lambda^2+\lambda-2=0$

$(\lambda+2)(\lambda-1)=0$

$\lambda_1=1,\ \lambda_2=-2$

Since, $\lambda$ cannot be negative, $\lambda=1$.

Hence, the mean of the random variable is **1.**

**9.23 0.072**

Given : Probability of getting head, $P(H)=0.3$

Probability of getting tail, $P(T)=0.7$

$[P(H)+P(T)=1]$

Since, all tosses are independent.

So, probability of getting head for the first time in $5^{th}$ toss

$$=P(T)\,P(T)\,P(T)\,P(T)\,P(H)$$
$$=0.7\times0.7\times0.7\times0.7\times0.3=0.072$$

Hence, the probability is **0.072.**

**9.24 0.027**

**Given :**

Three fair dice are thrown simultaneously.

So, number of sample space is $n(S)=6^3=216$.

Favourable outcome is that where all dice should get same number.

$$\{(1,1,1),(2,2,2),(3,3,3),$$
$$(4,4,4),(5,5,5),(6,6,6)\}$$

So, number of favourable outcomes is,

$n(A)=6$

Required probability is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{6}{216}=\frac{1}{36}=0.027$$

Hence, the probability is **0.027.**

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**9.25 0.25**

**Given :** $X_1, X_2, X_3, X_4$ are four independent random variable with zero mean and unit variance. For four variable total number of combination is $=4!=24$.

All possible conditions are,

(i) Favourable cases for $X_4$ to be smallest :

$X_4<X_1<X_2<X_3$

$X_4<X_1<X_3<X_2$

$X_4<X_2<X_3<X_1$

$X_4<X_2<X_1<X_3$

$X_4<X_3<X_1<X_2$

$X_4<X_3<X_2<X_1$


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(ii) Unfavourable cases for $X_4$ to be smallest

$$X_1 < X_2 < X_3 < X_4$$
$$X_1 < X_2 < X_4 < X_3$$
$$X_1 < X_3 < X_2 < X_4$$
$$X_1 < X_3 < X_4 < X_2$$
$$X_1 < X_4 < X_3 < X_2$$
$$X_1 < X_4 < X_2 < X_3$$

Similarly we can find cases for smallest $X_2$ and $X_3$.

So, total possible cases are 24 out of which 6 are favorable.

$$\therefore \quad P[X_1 \text{ is smallest}] = \frac{6}{24} = 0.25$$

**Note :** As $X_1, X_2, X_3, X_4$ are independent, hence probability of being smallest is same for each of them.

Hence, probability of $X_4$ being smallest

$$= \frac{1}{4} = 0.25$$

## 9.26 11

**Given :** $E[2X+Y]=0$

$$2E[X]+E[Y]=0 \qquad \ldots(i)$$

Also, $E[X+2Y]=33$

$$E[X]+2E[Y]=33 \qquad \ldots(ii)$$

Solving the above equations

$$\begin{array}{r} 4E[X]+2E[Y]=0 \\ E[X]+2E[Y]=33 \\ \hline 3E[X] \qquad\quad =-33 \end{array}$$

$$\therefore \quad E[X]=-11$$

Substituting value of $E[X]$ in equation (i),

$$E[Y]=-2E[X]=22$$

$$\therefore \quad E[X]+E[Y]=-11+22=11$$

## Electrical Engineering : EE

## 9.1 (B)

**Given :** A fair coin is tossed three times in succession.

All possible outcomes this event is given in below figure :

Tosses

| 1st | 2nd | 3rd | Favorable combinations |
|---|---|---|---|
| H | H | H | (3 Heads) |
| H | H | T | (2 Heads, 1 Tail)✓ |
| H | T | H | (2 Heads, 1 Tail)✓ |
| H | T | T | (1 Heads, 2 Tail) |
| T | H | H | Not necessary |
| T | H | T | |
| T | T | H | |
| T | T | T | |

$n(S)=4=$ Number of combinations with a Head produced in the first toss

$n(A)=2=$ Number of combinations with exactly two Heads produced with a Head in the first toss

So, the required probability is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{2}{4}=\frac{1}{2}$$

Hence the correct option is (B).

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## 9.2 (D)

**Given :** Two random events $P$ and $Q$.

By the concept of conditional probability,

$$P\left(\frac{Q}{P}\right)=\frac{P(P\cap Q)}{P(P)} \qquad \ldots(i)$$

The total probability cannot exceed 1.

So, $$P\left(\frac{Q}{P}\right)\le 1$$

From equation (i),

$$\frac{P(P\cap Q)}{P(P)}\le 1$$



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$$P(P \cap Q) \leq P(P)$$

Hence, the correct option is (D).

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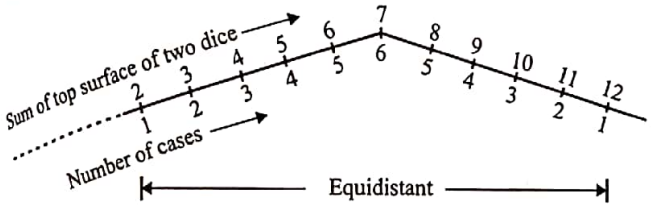
**9.3 (C)**

**Given :**

(i) Two fair dice are rolled.

(ii) $r$ is the sum of the numbers which turned up.

From the Triangle theory for two dice,




Total number of combinations are, $6 \times 6 = 36$

Checking from the options,

**Option (A) :**

$$Pr(r > 6) = \frac{6+5+4+3+2+1}{36}$$

$$Pr(r > 6) = \frac{7}{12} \neq \frac{1}{6}$$

So, this option is False.

**Option (B) :**

$$Pr\left(\frac{r}{3} \text{is an integer}\right) = \frac{n(3)+n(6)+n(9)+n(12)}{36}$$

$$Pr\left(\frac{r}{3} \text{is an integer}\right) = \frac{2+5+4+1}{36} = \frac{1}{3} \neq \frac{5}{6}$$

So, this option is False.

**Option (C) :**

$$Pr(r = 8/(r/4) \text{ is an integer}) = \frac{5}{9} = P(\text{Let})$$

By the concept of conditional probability,

$$Pr\left(\frac{Q}{P}\right) = \frac{Pr(P \cap Q)}{Pr(P)}$$

So,
$$P = \frac{Pr\left[(r=8) \cap \left(\frac{r}{4} \text{ is an integer}\right)\right]}{Pr\left(\frac{r}{4} \text{is an integer}\right)} \quad \ldots(i)$$

Cases when $r = 8$,

$(2, 6), (3,5), (4, 4), (5, 3), (6, 2) = 5$

Cases when $\frac{r}{4}$ is an integer,

(i) $r = 4$

$(1, 3), (2, 2), (3, 1)$

(ii) $r = 8$

$(2, 6), (3, 5), (4, 4), (5, 3), (6, 2),$

(iii) $r = 12$

$(6, 6)$

Cases when $(r = 8) \cap \left(\frac{r}{4} \text{is an integer}\right)$,

$(2, 6), (3,5), (4, 4), (5, 3), (6, 2) = 5$

From equation (i),

$$P = \frac{\frac{5}{36}}{Pr(4)+Pr(8)+Pr(12)}$$

$$P = \frac{\frac{5}{36}}{\frac{3}{36}+\frac{5}{36}+\frac{1}{36}} = \frac{5}{36} \times \frac{36}{9} = \frac{5}{9}$$

Hence, the correct option is (C).

**9.4 (C)**

**Given :** Three identical dice have following probabilities,

$$P(1) = \frac{1}{4} \qquad P(2) = \frac{1}{8} \qquad P(3) = \frac{1}{8}$$

$$P(4) = \frac{1}{8} \qquad P(5) = \frac{1}{8} \qquad P(6) = \frac{1}{4}$$

Occurrence of a value in a throw of dice represent independent events.

So, the probability of occurrence of 1, 5 and 6 is,

$$P(1 \cap 5 \cap 6) = P(1) \times P(5) \times P(6)$$

$$P(1 \cap 5 \cap 6) = \frac{1}{4} \times \frac{1}{8} \times \frac{1}{4}$$

$$P(1 \cap 5 \cap 6) = \frac{1}{128}$$

Hence, the correct option is (C).



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**Key Point**

The probability of independent events to occur simultaneously is given by,

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

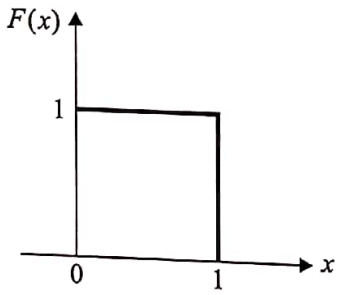
## 9.5 (C)

**Given :** $X$ is uniformly distributed random variable in the interval $X \in [0, 1]$.

So, $$F(X) = \begin{cases} 1, & 0 < X < 1 \\ 0, & \text{otherwise} \end{cases}$$

The mean of $X^3$ is given by,

$$E(X^3) = \int_{-\infty}^{\infty} X^3 F(X)\, dX$$




$$E(X^3) = \int_0^1 X^3 dX = \left[\frac{X^4}{4}\right]_0^1 = \frac{1}{4} - 0 = \frac{1}{4}$$

Hence, the correct option is (C).

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## 9.6 (B)

**Given :** $N$ people, all born in April (a month of 30 days) irrespective of their year of birth. Probability of a person to have born on a particular date in April $= \frac{1}{30}$

**Method 1**

Let, $N$ persons are named as,

$$G_1, G_2, G_3, \ldots\ldots G_N.$$

For two people to have born on April 1st, the possible combinations are :

Either, $[(G_1, G_2)\ (G_1, G_3)\ (G_1, G_4) \cdots\cdots (G_1, G_N)]$

Or $[(G_2, G_3)\ (G_2, G_4) \cdots\cdots (G_2, G_N)]$

Or $[(G_3, G_4) \cdots\cdots (G_3, G_N)]$

$\vdots$

Or $[(G_{(N-1)}, G_N)]$

So, the probability for two people born on April 1st is given by,

$$P = \left[\frac{1}{30} \times \frac{1}{30} + \frac{1}{30} \times \frac{1}{30} + \ldots..(N-1)\text{ times}\right] + \left[\frac{1}{30} \times \frac{1}{30} + \ldots..(N-2)\text{times}\right] + \ldots\ldots + \left[\frac{1}{30} \times \frac{1}{30}\right]$$

$$P = \frac{1}{(30)^2}[(N-1) + (N-2) + \ldots.. + 3 + 2 + 1]$$

$$P = \frac{1}{900} \times \frac{N(N-1)}{2} \quad \text{[Sum of AP]}$$

$$P = \frac{N(N-1)}{1800}$$

This probability will be same for any date in April.

So for all the dates, total probability is given by,

$$P_{total} = P + P + \ldots..(30 \text{ times}) = 30P$$

$$P_{total} = 30 \times \frac{N(N-1)}{1800} = \frac{N(N-1)}{60}$$

According to question,

$$P_{total} > 0.5 \text{ or } \frac{1}{2}$$

$$\frac{N(N-1)}{60} > \frac{1}{2}$$

$$N(N-1) > 30$$

$$N^2 - N - 30 > 0$$

$$(N-6)(N+5) > 0$$

$$N > 6 \quad [N \text{ cannot be negative}]$$

So, the minimum value $N$ can possess is 7.

Hence, the correct option is (B).

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**Method 2**

Let, the probability for any two people to have born at any one day of April is $P$.

So for all the dates, total probability is given by,

$$P_{total} = P + P + \ldots\ldots(30 \text{ times})$$

$$P_{total} = 30P \qquad \ldots(i)$$

Then, $P = {}^{N}C_2 \times \frac{1}{30} \times \frac{1}{30}$

Here ${}^{N}C_2$ represents the number of ways two persons out of $N$ can be chosen.

From equation (i),

$$P_{total} = \left[ {}^{N}C_2 \times \frac{1}{30} \times \frac{1}{30} \right] \times 30$$

$$P_{total} = \left[ \frac{N!}{2!(N-2)!} \times \frac{1}{30} \times \frac{1}{30} \right] \times 30$$

$$P_{total} = \left[ \frac{N(N-1)}{2} \times \frac{1}{30} \times \frac{1}{30} \right] \times 30$$

$$P_{total} = \left[ \frac{N(N-1)}{2} \times \frac{1}{30} \right] = \frac{N(N-1)}{60}$$

According to question,

$$P_{total} > 0.5 \text{ or } \frac{1}{2}$$

$$\frac{N(N-1)}{60} > \frac{1}{2}$$

$$N(N-1) > 30$$

$$N^2 - N - 30 > 0$$

$$(N-6)(N+5) > 0$$

$N > 6$ [$N$ cannot be negative]

So, the minimum value $N$ can possess is 7.

Hence, the correct option is (B).

**9.7 (C)**

**Given :** A box contains 4 white and 3 red balls.

First ball removed is white, so the remaining balls in box are :

3 white balls and 3 red balls.

Now, the probability of removing a red ball is given by,

$$P = \frac{n(\text{Red})}{n(\text{Total})} = \frac{3}{6} = \frac{1}{2}$$

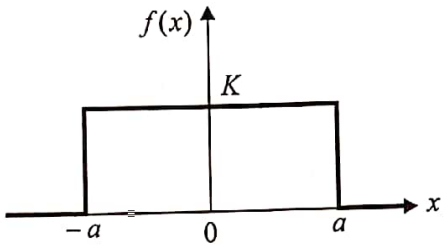
Hence, the correct option is (C).

**9.8 (A)**

**Given :** A uniformly distributed random signal within limits $(-a, a)$.

By the property of probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad \ldots(i)$$




Area for above figure is given by,

$$K \times 2a = 1$$

$$K = \frac{1}{2a}$$

**Note :** For even functions, mean or expected value or average value is equal to 0.

The variance is given by,

$$Var(x) = MSV - (\text{Mean})^2$$

From figure, Mean = 0.

Thus, $Var(x) = MSV$

$$Var(x) = \int_{-a}^{a} x^2 f(x)\,dx = \int_{-a}^{a} x^2 \frac{1}{2a}\,dx$$

$$Var(x) = \frac{1}{2a}\left(\frac{x^3}{3}\right)_{-a}^{a} = \frac{2a^3}{6a} = \frac{a^2}{3}$$

The r.m.s. value is given by,

$$\text{r.m.s.} = \sqrt{Var(x)} = \frac{a}{\sqrt{3}}$$

Hence, the correct option is (A).

**9.9 (C)**

**Refer Soluion 9.12 [Electronics]**

**9.10 (B)**

**Refer Soluion 9.13 [Electronics]**



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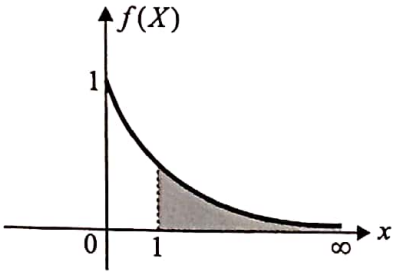
**9.11 (A)**

**Given :** A continuous random variable $X$ with probability density function,

$$f(x)=e^{-x}\ (0<x<\infty)$$




By the property of probability density function,

$$P[X>x]=\int_x^{\infty} f(x)\,dx$$

Thus, $$P\{X>1\}=\int_1^{\infty} e^{-x}dx=\left[-e^{-x}\right]_1^{\infty}$$

$$P\{X>1\}=0-(-e^{-1})=0.368$$

Hence, the correct option is (A).

**9.12 (B)**

**Given :** A fair coin is tossed $n$ times.

Let, $P=$ Number of heads

$Q=$ Number of tails

Then, $P+Q=n$

$$Q=n-P$$

According to question,

$$|P-Q|=n-3$$

$$|P-(n-P)|=n-3$$

$$|2P-n|=n-3$$

Taking positive sign,

$$2P-n=n-3$$

$$P=n-\frac{3}{2}$$

[Not possible to toss in fraction]

Taking negative sign,

$$-(2P-n)=n-3$$

$$P=\frac{3}{2}$$ [which is also not possible]

For any **impossible event,** the probability is always equal to **zero.**

Hence, the correct option is (B).

**9.13 0.142**

**Given :** For a particular dice, the probability of a face with $n$ dots showing up is proportional to $n$.

$$P(\text{'}n\text{' dots}) \propto n$$

Let, the probability of occurrence of one dot is $P$.

Then, the probabilities of face with 2, 3, 4, 5 and 6 dots are respectively $2P$, $3P$, $4P$, $5P$ and $6P$

These are mutually exhaustive events, so the sum of all probabilities is unity.

So, $$P+2P+3P+4P+5P+6P=1$$

$$21P=1$$

$$P=\frac{1}{21}$$

Thus, probability of occurrence of 3 dots is given by,

$$3P=\frac{3}{21}=\frac{1}{7}=0.142$$

Hence, the probability of the face with three dots showing up is **0.142.**

**9.14 0.4**

**Given :** $X$ be a random variable with probability density function,

$$f(x)=\begin{cases}0.2, & \text{for} \quad |x|\le 1\\ 0.1, & \text{for} \quad 1<|x|\le 4\\ 0, & \text{otherwise}\end{cases}$$

**Method 1**

$$P(a<X<b)=\int_a^b f(x)\,dx$$

$$P(0.5<X<5)=\int_{0.5}^{5} f(x)\,dx$$

$$P(0.5<X<5)=\int_{0.5}^{1}0.2\,dx+\int_{1}^{4}0.1\,dx+\int_{4}^{5}0\,dx$$

$$P(0.5<X<5)=0.2[1-0.5]+0.1\times 3$$

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$$P(0.5<X<5)=0.2\times0.5+0.1\times3$$

$$P(0.5<X<5)=0.4$$

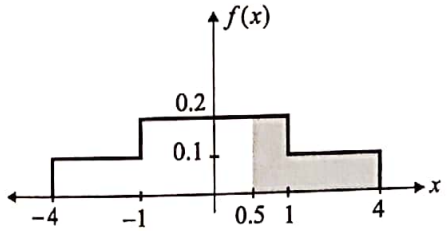
Hence, the probability $P(0.5<X<5)$ is **0.4.**

**Method 2**

A function is defined as,

$$f(x)=\begin{cases}0.2, & \text{for } -1<x<1\\ 0.1, & \text{for } (1<x\le 4)\\ & \text{and } 1<-x<4\\ & \Rightarrow (-4\le x<-1)\\ 0, & \text{Otherwise}\end{cases}$$

$$f(x)=\begin{cases}0.2, & \text{for}(-1<x<1)\\ 0.1, & \text{for } (1<x\le 4) \text{ and}(-4\le x<-1)\\ 0, & \text{Otherwise}\end{cases}$$




Now, $P(0.5<X<5)$ will be under shaded area.

Thus,

$$P(0.5<X<5)=(0.5\times0.2)+[(4-1)\times0.1]$$

$$P(0.5<X<5)=0.1+0.3=0.4$$

Hence, the probability $P(0.5<X<5)$ is **0.4.**

**9.15 0.42**

**Given :** Life time of an electric bulb with density,

$$f(x)=kx^2$$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} f(x)\,dx=1 \qquad \ldots(i)$$

If minimum and maximum lifetime of bulb are 1 and 2 year respectively.

Then from equation (i),

$$\int_{1}^{2} kx^2\,dx=1$$

$$k\left[\frac{x^3}{3}\right]_1^2=1$$

$$k\left(\frac{8}{3}-\frac{1}{3}\right)=1$$

$$\frac{7k}{3}=1\Rightarrow k=\frac{3}{7}=0.42$$

Hence, the value of $k$ is **0.42.**

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**9.16 0.25**

**Given :** $f(x)=\begin{cases}a+bx & ; \text{ for } 0<x<1\\ 0 & ; \text{ otherwise}\end{cases}$

and $E(X)=\frac{2}{3}$

Expected value $E(X)$ is given by,

$$E(X)=\int_{-\infty}^{\infty} xf(x)\,dx$$

Thus, $\int_{-\infty}^{\infty} xf(x)dx=\frac{2}{3}$

$$\int_0^1 x(a+bx)dx=\frac{2}{3}$$

$$a\left(\frac{x^2}{2}\right)_0^1+b\left(\frac{x^3}{3}\right)_0^1=\frac{2}{3}$$

$$a\left(\frac{1}{2}\right)+b\left(\frac{1}{3}\right)=\frac{2}{3}$$

$$3a+2b=4 \qquad \ldots(i)$$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} f(x)\,dx=1$$

$$\int_0^1 f(x)dx=1$$



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$$\int_0^1 (a+bx)dx = 1 \Rightarrow \left[ax + \frac{bx^2}{2}\right]_0^1 = 1$$

$$a + \frac{b}{2} = 1$$

$$2a + b = 2 \qquad \ldots(ii)$$

From equations (i) and (ii),

$$a = 0, \quad b = 2$$

So, $$f(x) = \begin{cases} 2x, & \text{for } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Required probability is given by,

$$P(x < a) = \int_{-\infty}^{a} f(x)dx$$

$$P(x < 0.5) = \int_0^{0.5} 2x\,dx = 2\left[\frac{x^2}{2}\right]_0^{0.5}$$

$$P(x < 0.5) = 0.5^2 - 0^2 = 0.25$$

Hence, $P[X < 0.5]$ is **0.25**.

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## 9.17 (D)

**Given :** Two players $A$ and $B$ alternately keep rolling a fair dice until a six is appeared.

and the person to get a six first wins.

So, the probability of winning a person is same as that of getting a six in a fair dice.

Let probability that $A$ wins is given by,

$$(A) = \frac{1}{6}$$

So, the probability that $A$ loses is given by,

$$(\bar{A}) = 1 - \frac{1}{6} = \frac{5}{6}$$

Similarly for person $B$,

$$B = \frac{1}{6}, \quad \bar{B} = \frac{5}{6}$$

Since $A$ starts first, the probability for him/her to win is given by,

$$= A + \bar{A}\bar{B}A + \bar{A}\bar{B}\,\bar{A}\bar{B}A + \ldots\ldots (\infty \text{ series})$$

$$= \frac{1}{6} + \left(\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}\right) + \left(\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6}\right) + \ldots\ldots(\infty \text{ series})$$

$$= \frac{1}{6}\left[1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \ldots\ldots(\infty \text{series})\right]$$

The above series is infinite GP.

$$= \frac{1}{6}\left[\frac{1}{1-\left(\frac{5}{6}\right)^2}\right] = \frac{1}{6}\left[\frac{36}{36-25}\right] = \frac{6}{11}$$

Hence, the correct option is (D).

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## 9.18 (D)

**Given :** Two coins $R$ and $S$ and the probabilities for joint events are following :

$$H_R H_S = 0.28, \quad H_R T_S = 0.30$$

$$T_R T_S = 0.18, \quad T_R H_S = 0.24$$

Checking from the options,

**Option (A) :**

For coins to be independent,

$$H_R H_S = \text{P(Head in } R) \times \text{P(Head in } S)$$

$$H_R H_S = \frac{1}{2}\times\frac{1}{2} = \frac{1}{4} = 0.25$$

and similarly, $T_R T_S = 0.25$,

$$H_R T_S = 0.25 \text{ and } T_R H_S = 0.25$$

These values do not match with the provided values.

So, this option is False.

**Option (B) :**

If coin $R$ is fair and $S$ is not,

$$H_R = T_R = \frac{1}{2}$$

and also, $H_R H_S = T_R H_S$

But, $0.28 \neq 0.24$

So, this option is False.



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Option (C) :

If coin $S$ is fair and $R$ is not,

$$H_S = T_S = \frac{1}{2}$$

and also, $H_R H_S = H_R T_S$

But, $0.28 \neq 0.30$

So, this option is False.

Thus, it is concluded that the coin tosses are biased and dependent.

Hence, the correct option is (D).

**9.19 0.01818**

Given : Black pens = 3, Blue pens = 3,

Green pens = 3 and Red pens = 3.

So, total number of pens $= 12$

According to question,

3 pens are to be selected among these 12 pens.

And all 3 pens must be of the same colour.

Probability of bringing 3 pens is given by,

$$P = \frac{{}^3C_3 + {}^3C_3 + {}^3C_3 + {}^3C_3}{{}^{12}C_3}$$

$$P = \frac{4}{220} = 0.01818$$

Hence, the probability that a candidate comes with all 3 pens having the same colour is 0.01818.

**9.20 (A)**

Given : $f_X(x) = \frac{3}{2}e^{-3x}u(x) + ae^{4x}u(-x) \ldots (i)$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{\infty}\left[\frac{3}{2}e^{-3x}u(x) + ae^{4x}u(-x)\right]dx = 1$$

$$\int_{-\infty}^{\infty}\frac{3}{2}e^{-3x}u(x)\,dx + \int_{-\infty}^{\infty} ae^{4x}u(-x)\,dx = 1$$

$$\int_{0}^{\infty}\frac{3}{2}e^{-3x}dx + \int_{-\infty}^{0} ae^{4x}dx = 1$$

$$\frac{3}{2}\left[\frac{e^{-3x}}{-3}\right]_0^{\infty} + \left[\frac{ae^{4x}}{4}\right]_{-\infty}^{0} = 1$$

$$\frac{a}{4}(1-0) + \frac{3}{2\times(-3)}(0-1) = 1$$

$$\frac{a}{4} + \frac{1}{2} = 1$$

$$a = 2$$

Probability is given by,

$$P[X \le 0] = \int_{-\infty}^{0} f(x)\,dx$$

So, $$P[X \le 0] = a\int_{-\infty}^{0} e^{4x}dx = 2\int_{-\infty}^{0} e^{4x}dx$$

$$P[X \le 0] = 2\left(\frac{e^{4x}}{4}\right)_{-\infty}^{0} = 2\times\frac{1}{4} = \frac{1}{2}$$

The required value of $a$ and $P[X \le 0] = 2, \frac{1}{2}$

Hence, the correct option is (A).

**Key Point**

A unit step function acts as a limit changer for integration as follows :

(i) $u(x) = \begin{cases} 1 ; & \text{for } 0 < x < \infty \\ 0 ; & \text{otherwise} \end{cases}$

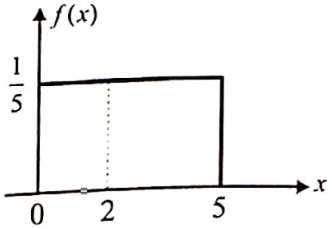
(ii) $u(-x) = \begin{cases} 1 ; & \text{for } -\infty < x < 0 \\ 0 ; & \text{otherwise} \end{cases}$

**9.21 0.9**

**Given :** Cycle of traffic signal lights,

Green = 2 min, Red = 3 min

Probability density function of arrival time of vehicle at the junction is given by,



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If vehicle will arrive at $t = 2$ min then, it will wait for 3 min.

Let, vehicle will arrive at traffic junction at $t = (2 + x)$ min.

So, it will have to wait $5 - x$ min.

Expected waiting time for the vehicle at the junction is given by,

$$E(5-x) = \int_2^5 (5-x) \times \frac{1}{5} dx$$

$$E(5-x) = \frac{1}{5}\left[5x - \frac{x^2}{2}\right]_2^5$$

$$E(5-x) = \frac{1}{5}\left[\left(25 - \frac{25}{2}\right) - \left(10 - \frac{4}{2}\right)\right]$$

$$E(5-x) = 0.9 \text{ min}$$

Hence, the expected waiting time (in minutes) for the vehicle at the junction is **0.9**.

## 9.22 (A)

**Given :** Total numbers of red balls $= 5$

Total numbers of black balls $= 5$

In first draw, 1 ball is picked randomly and discarded without noticing its color.

**Case 1 :** When the discarded ball is red.

Probability for a red ball in second draw is given by,

$$P_1 = P(\text{red in first draw}) \times P(\text{red in second draw})$$

$$P_1 = \frac{5}{10} \times \frac{4}{9} = \frac{2}{9}$$

**Case 2 :** When the discarded ball is black.

Probability for a red ball in second draw is given by,

$$P_2 = P(\text{black in first draw}) \times P(\text{red in second draw})$$

$$P_2 = \frac{5}{10} \times \frac{5}{9} = \frac{5}{18}$$

Then the required probability is,

$$P_1 + P_2 = \frac{2}{9} + \frac{5}{18} = \frac{1}{2}$$

Hence, the correct option is (A).

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## 9.23 (C)

**Given :** A fair coin is tossed repeatedly person. The experiment stops either at 3 tosses or if person gets a head.

All the possible outcomes and their probabilities are :

| Outcome | H | TH | TTH | TTT |
|---|---|---|---|---|
| Number of Heads ($Y$) | 1 | 1 | 1 | 0 |
| Probability $P(Y)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |

This is a general discrete distribution.

The variance of a random variable ($Y$) is given by,

$$\text{Var}(Y) = E(Y)^2 - [E(Y)]^2 \quad \ldots(i)$$

where, $E(Y) = \sum (Y) \times P(Y) = \text{Mean}$

$$E(Y) = \left(1 \times \frac{1}{2} + 1 \times \frac{1}{4} + 1 \times \frac{1}{8}\right) + 0 \times \frac{1}{8}$$

$$E(Y) = \frac{7}{8}$$

Similarly, $E(Y^2) = \sum (Y^2) \times P(Y)$

$$E(Y^2) = \left(1^2 \times \frac{1}{2} + 1^2 \times \frac{1}{4} + 1^2 \times \frac{1}{8}\right) + 0^2 \times \frac{1}{8}$$

$$E(Y^2) = \frac{7}{8}$$

From equation (i),

$$\text{Var}(Y) = \frac{7}{8} - \frac{49}{64} = \frac{7}{64}$$

Hence, the correct option is (C).

## 9.24 (B)

**Given :** Mean Square Value

$$E[x^2] = \frac{kT}{C}$$



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Mean $E[x]=0$

Hence, variance, $\sigma_{x^2}=E[x^2]-\{E[x]\}^2$

$$\sigma_{x^2}=\frac{kT}{C}$$

$\therefore$ Standard derivation,

$$\sigma=\sqrt{\text{Variance}}=\sqrt{\frac{kT}{C}}$$

Hence, the correct option is (B).

**9.25 0.265**

Given : Probability of resistor being defective $p=0.02$

$\therefore$ Probability of resistor being non-defective $q=1-p=0.98$

$\therefore$ Probability of two or more resistors out of 50 to be defective is

$$P(x\geq 2)=1-[P(0)+P(1)] \quad \ldots(i)$$

From binomial distribution, probability of '$r$' successes in '$n$' trials is given as

$$P(x)={}^{n}C_r p^r q^{n-r}$$

$$P(0)={}^{50}C_0 p^0 q^{50-0}=1\times(0.02)^0\times(0.98)^{50}$$

$$=0.364$$

$$P(1)={}^{50}C_1(p)^1.(q)^{49}=50\times 0.02\times(0.98)^{49}$$

$$=0.371$$

Hence from equation (i),

$\therefore$ $P(x\geq 2)=1-[0.364+0.371]=0.265$

## Mechanical Engineering : ME

**9.1 (A)**

Given : A random committee of 5 managers.

Probability of choosing a technocrat

$$=\frac{20}{100}=\frac{1}{5}=p$$

Probability of not choosing a technocrat

$$=1-\frac{1}{5}=\frac{4}{5}=q$$

According to binomial distribution, the probability of $r$ trials is given by,

$$P(r)={}^{n}C_r\times p^r\times q^{n-r}$$

where, $P$ is probability of occurrence of the event.

Here, $n=5$ and $r=2$

$$P(r=2)={}^{5}C_2\times\left(\frac{1}{5}\right)^2\times\left(\frac{4}{5}\right)^{5-2}$$

$$P(r=2)=\frac{5!}{2!3!}\times\frac{1}{25}\times\frac{64}{125}=\frac{128}{625}$$

$$P(r=2)=0.2048$$

Hence, the correct option is (A).

**9.2 (D)**

**Given :** Probability of a defective piece being produced, $p=0.01$

Probability of a non-defective piece being produced, $q=0.99$

According to binomial distribution, the probability of $r$ trials is given by,

$$P(r)={}^{n}C_r\times p^r\times q^{n-r}$$

where, $n=5$ and $r=1$

$$P(r=1)={}^{5}C_1(0.01)^1(0.99)^4$$

$$P(r=1)=5\times(0.99)^4(0.01)^1$$

Hence, the correct option is (D).

**9.3 (C)**

**Given :** A box contains 5 black balls and 3 red balls.

Total number of balls $=8$

The number of ways 3 balls can be randomly picked from 8 balls $={}^{8}C_3$.

Similarly, the number of ways 2 black balls can be selected from 5 black balls $={}^{5}C_2$.

and the number of ways 1 red balls can be picked from 3 red balls $={}^{3}C_1$.



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The total probability of selecting two black balls and one red ball is given by,

$$P = \frac{{}^5C_2 \times {}^3C_1}{{}^8C_3} = \frac{15}{28}$$

Hence, the correct option is (C).

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**9.4 (B)**

The probability of first friend being born in any of the month out of the 12 months,

$$= \frac{12}{12} = 1$$

Then, the probability that second friend was born in the same month is $\frac{1}{12}$.

So, the probability that two friends will share the same birth month is,

$$= 1 \times \frac{1}{12} = \frac{1}{12}$$

Hence, the correct option is (B).

**9.5 (C)**

**Given :** $P(E_1) = \frac{1}{2}$, $P(E_2) = \frac{1}{3}$,

$$P(E_1 \text{ and } E_2) = P(E_1 \cap E_2) = \frac{1}{5}$$

**From option (A) :**

By addition rule,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$P(E_1 \cup E_2) = \frac{1}{2} + \frac{1}{3} - \frac{1}{5}$$

$$P(E_1 \cup E_2) = \frac{5}{6} - \frac{1}{5}$$

$$P(E_1 \cup E_2) = \frac{25-6}{30} = \frac{19}{30}$$

So, option (A) is incorrect.

**From option (B) :**

For two events to be independent

$$P(E_1 \cap E_2) = P(E_1).P(E_2)$$

$$\frac{1}{5} = \frac{1}{2} \times \frac{1}{3}$$

$$\frac{1}{5} \neq \frac{1}{6}$$

Thus, $E_1$ and $E_2$ are not independent events.

**From option (D) :**

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{1}{5}}{\frac{1}{3}} = \frac{3}{5}$$

Thus, option (D) is incorrect.

Hence, the correct option is (C).

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**9.6 (D)**

**Given :** $P(E_1 \cup E_2) = 1$ and $P(E_1) = P(E_2)$

Let $P(E_1) = P(E_2) = P(E)$ ...(i)

By the addition rule,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$1 = P(E) + P(E) - P(E) \times P(E)$$

[Given $E_1$ and $E_2$ are independent then $P(E_1 \cap E_2) = P(E_1) \times P(E_2) = P(E)^2$ ]

$$P(E)^2 - 2P(E) + 1 = 0$$

$$[P(E) - 1]^2 = 0$$

$$P(E) = 1$$

Therefore, from equation (i),

$$P(E_1) = 1$$

Hence, the correct option is (D).

**9.7 (A)**

**Given :** Probability of a defective bolt, $p = 0.1$

Probability of making bolts with no defect,

$$q = 1 - 0.1 = 0.9$$

Total number of bolts $(n) = 900$

Mean, $\mu = np$

$$\mu = 900 \times 0.1 = 90$$



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and standard deviation $= \sqrt{npq}$

$= \sqrt{900 \times 0.1 \times 0.9} = 9$

Hence, the correct option is (A).

**Key Point**

For binomial distribution,

Mean $= np$

Variance $= npq$

where, $n =$ Total number of event

$p =$ Probability of happening

$q =$ Probability of not happening

## 9.8 (D)

The sample space of a coin for three tosses is given in below figure,

| Tosses 1st | 2nd | 3rd | Favorable combinations |
|---|---|---|---|
| H | H | H | |
| H | H | T | (2 Heads, 1 Tail) |
| H | T | H | (2 Heads, 1 Tail) |
| H | T | T | |
| T | H | H | (2 Heads, 1 Tail) |
| T | H | T | |
| T | T | H | |
| T | T | T | |

Total number of cases when unbiased coin is tossed three times $= 8$ cases

Head turns up in exactly two tosses $= 3$ cases

So, the required probability, $P(E) = \frac{3}{8}$

Hence, the correct option is (D).

## 9.9 (B)

**Given :** Seven car accidents occurred in a week $n = 7$.

Probability of one accident in a week $p = \frac{1}{7}$

$$q = 1 - p = 1 - \frac{1}{7} = \frac{6}{7}$$

Probability of all 7 accidents occurred on a particular day of week (by binominal distribution) is given by,

Probability $= {}^nC_r\, p^r q^{n-r}$

where, $n = 7, r = 7$

$$\text{Probability} = {}^7C_7 \left(\frac{1}{7}\right)^7 \left(\frac{6}{7}\right)^0 = \left(\frac{1}{7}\right)^7$$

So, for seven car accidents, the required probability $= 7 \times \left(\frac{1}{7}\right)^7 = \frac{1}{7^6}$

Hence, the correct option is (B).

## 9.10 (A)

**Given :** Two dice are thrown.

**Method 1**

The sample space for tossing of a dice two times is given by,

$$S = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

$n(S) = 36$

These shaded items shows the desired outcomes in which sum of top surface is 8.

i.e. $[(2,6),(6,2),(3,5),(5,3),(4,4)]$

$n(A) = 5$

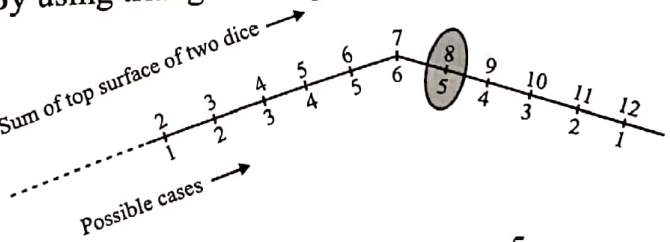
The required probability is given by,

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36}$$

Hence, the correct option is (A).

**Method 2**

By using triangle theory,




Probability of getting total sum $8 = \frac{5}{36}$.

Hence, the correct option is (A).



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## 9.11 (C)

**Given** : Four fair coins are tossed simultaneously.

So, the total number of cases $n(S)=16$

Number of cases in which at least one head and at least one tail is present in below figure is represented in shaded form.

Total favorable combinations, $n(A)=14$

| Tosses 1st | 2nd | 3rd | 4th | Favorable combinations |
|---|---|---|---|---|
| H | H | H | H | |
| H | H | H | T | Atleast one head & atleast one tail |
| H | H | T | H | |
| H | H | T | T | |
| H | T | H | H | |
| H | T | H | T | |
| H | T | T | H | |
| H | T | T | T | |
| T | H | H | H | |
| T | H | H | T | |
| T | H | T | H | |
| T | H | T | T | |
| T | T | H | H | |
| T | T | H | T | |
| T | T | T | H | |
| T | T | T | T | |

**Fig. Samples space of 4 tosses**

The required probability is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{14}{16}=\frac{7}{8}$$

Hence, the correct option is (C).

## 9.12 (C)

The line of regression of $Y$ on $X$ is given by,

$$(Y-\overline{Y})=b_{YX}(X-\overline{X})$$

and also the linear relationship as,

$$Y=b_{YX}(X)+c$$

So, value of $X$ exactly determine value of $Y$.

Hence, the correct option is (C).

## 9.13 (D)

**Given :** A box contains 5 black and 5 red balls.

**Method 1**

Probability of picking out 1 red ball out of 10 balls $=\frac{5}{10}$

Here the point is to be noted that there is no replacement of balls.

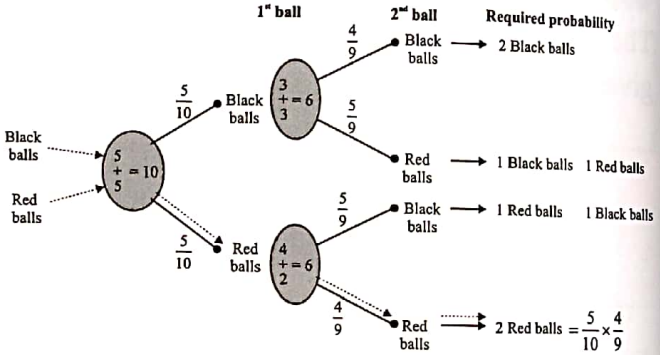
So, probability of drawing two red balls

$$=\frac{5}{10}\times\frac{4}{9}=\frac{2}{9}$$

Hence, the correct option is (D).

**Method 2**

By using tree diagram,




So, probability of drawing two red balls

$$=\frac{5}{10}\times\frac{4}{9}=\frac{2}{9}$$

Hence, the correct option is (D).

## 9.14 (D)

**Given :** $P(A)=1, P(B)=1/2$

Assume independent events,

$$P(A\cap B)=P(A)\times P(B)=1\times\frac{1}{2}=\frac{1}{2}$$

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{1/2}{1/2}=1$$

or $$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=P(A)=1$$

$$P\left(\frac{B}{A}\right)=\frac{P(A\cap B)}{P(A)}=\frac{1/2}{1}=\frac{1}{2}$$

or $$P\left(\frac{B}{A}\right)=\frac{P(A\cap B)}{P(A)}=P(B)=\frac{1}{2}$$

Hence, the correct option is (D).



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**9.15 (C)**

Here the flow rate is given in range, so taking the mid value as shown in below table :

| Mid value ($x_i$) : | 7.6 | 7.8 | 8.0 | 8.2 | 8.4 | 8.6 |
|---|---|---|---|---|---|---|
| Frequency ($f_i$) : | 1 | 5 | 35 | 17 | 12 | 10 |

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{652.8}{80} = 8.16$$

Hence, the correct option is (C).

**9.16 (D)**

Given : A pack of regular playing cards .

Probability of drawing a king is given by,

$$P(\text{King}) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{1}{13}.$$

The point is to be noted that the drawn card is not replaced. So there are remaining 3 cards that can be a king and total number of cards are 51.

Probability of drawing two king cards without replacement is,

$$P(2\,\text{Kings}) = \frac{1}{13} \times \frac{3}{51} = \frac{1}{221}$$

Hence, the correct option is (D).

**9.17 (B)**

Given : A lot has 10 % defective items.

So, the probability of an item to be defective is,

$$p = 0.1$$

Then, the probability of a non-defective item is given by,

$$q = 1 - p = 1 - 0.1 = 0.9$$

By binominal distribution, the probability that exactly 2 of the chosen items are defective is,

$$P = {}^{10}C_2 (p)^2 (q)^8$$

$$P = {}^{10}C_2 (0.1)^2 (0.9)^8 = 0.1937$$

Hence, the correct option is (B).

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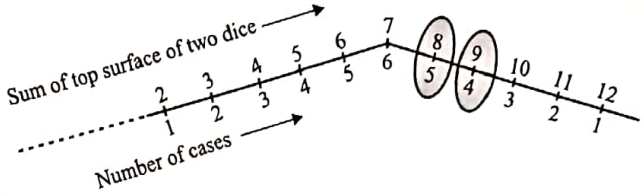
**9.18 (D)**

**Method 1**

**Given :** A single die is thrown twice.

From the Triangle theory for two dice,




Total number of combinations are,

$$n(S) = 36$$

The number of cases when the sum is neither 8 nor 9 are,

$$n(A) = 1+2+3+4+5+6+3+2+1$$

$$n(A) = 27$$

Required probability is given by,

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{27}{36} = \frac{3}{4}$$

Hence, the correct option is (D).

**Method 2**

The number of cases when the sum is either 8 or 9 are,

$$n(A) = 5+4 = 9$$

$$P(A) = \frac{9}{36}$$

Required probability is given by,

$$P(B) = 1 - P(A) = 1 - \frac{9}{36} = \frac{27}{36} = \frac{3}{4}$$

Hence, the correct option is (D).

**9.19 (D)**

**Given :** A box contains 20 defective items and 80 non-defective items.

Probability of first item being defective is $\frac{20}{100}$.

The point is to be noted that the items are selected without replacement, so after selecting 1 item the remaining items are 19 out of 99.



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So, the probability that both are defective is,

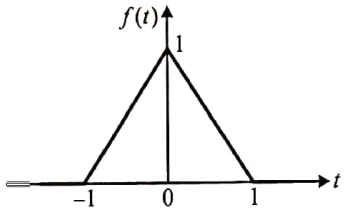
$$P=\frac{20}{100}\times\frac{19}{99}=\frac{19}{495}$$

Hence, the correct option is (D).

**9.20 (B)**

**Given :** $f(t)=\begin{cases}1+t\ ; & -1\le t\le 0\\ 1-t\ ; & 0\le t\le 1\end{cases}$

The function is represented as follows,




The given function is an even function, and for even function the mean value is equal to 0.

Variance = Mean square value – (Mean)$^2$

Now, variance $=\int_{-\infty}^{\infty} t^2 f(t)dt$

$$\sigma^2=\int_{-1}^{0} t^2(1+t)\,dt+\int_{0}^{1} t^2(1-t)\,dt$$

$$\sigma^2=\int_{-1}^{0}(t^2+t^3)\,dt+\int_{0}^{1}(t^2-t^3)\,dt$$

$$\sigma^2=\left[\frac{t^3}{3}+\frac{t^4}{4}\right]_{-1}^{0}+\left[\frac{t^3}{3}-\frac{t^4}{4}\right]_{0}^{1}$$

$$\sigma^2=\frac{1}{12}+\frac{1}{12}=\frac{1}{6}$$

Since, standard deviation = $\sqrt{\text{Variance}}$

Standard deviation $=\frac{1}{\sqrt{6}}$

Hence, the correct option is (B).

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**9.21 (D)**

For $X$ and $Y$ be two independent random variables.

(i) $E(XY)=E(X)E(Y)$ ...(i)

(ii) $\text{Cov}(X,Y)=E(XY)-E(X)E(Y)$

$=E(X)E(Y)-E(X)E(Y)$ [From equation (i)]

$=0$

(iii) $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$

(iv) $E(X^2Y^2)=E(X^2)E(Y^2)$

Therefore, relation in option (D) is false.

Hence, the correct option is (D).

**9.22 (A)**

**Given :** A coin is tossed 4 times.

**Method 1**

Probability of getting head, $p(H)=0.5$

Probability of not getting head, $q(H)=0.5$

According to binomial distribution, the probability of $r$ trials is given by,

$$P(r)={}^nC_r\times p^r\times q^{n-r}$$

where $P$ is probability of occurrence of the event.

Here, $n=4$ and $r=3$

$$P={}^4C_3(0.5)^3(0.5)=\frac{1}{4}$$

Hence, the correct option is (A).

**Method 2**

Tosses

| 1st | 2nd | 3rd | 4th | Favorable combinations |
|---|---|---|---|---|
| H | H | H | H | |
| H | H | H | T | |
| H | H | T | H | |
| H | H | T | T | |
| H | T | H | H | |
| H | T | H | T | |
| H | T | T | H | |
| H | T | T | T | Shaded part represent exactly 3 heads |
| T | H | H | H | |
| T | H | H | T | |
| T | H | T | H | |
| T | H | T | T | |
| T | T | H | H | |
| T | T | H | T | |
| T | T | T | H | |
| T | T | T | T | |

**Fig. Samples space of 4 tosses**

For 4 tosses the total number of combinations is

$$n(S)=16$$



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The shaded part represent the combination of exactly 3 heads that is equal to 4.

$$n(E) = 4$$

So, the required probability is given by,

$$P(E) = \frac{n(E)}{n(S)} = \frac{4}{16} = \frac{1}{4}$$

Hence, the correct option is (A).

**9.23 (D)**

Given : 3 coins are tossed simultaneously.

**Method 1**

The total number of combinations $= 2^3 = 8$.

Number of ways to get all tails = 1.

Probability to get no heads (all tails) is given by,

$$q = \frac{\text{Number of ways to get tails}}{\text{Total number of ways}} = \frac{1}{8}$$

Therefore, probability to get at least one head is,

$$p = 1 - q = 1 - \frac{1}{8} = \frac{7}{8}$$

Hence, the correct option is (D).

**Method 2**

Three coins are tossed simultaneously, the total number of combinations, $n(S) = 2^3 = 8$.

Coins

| 1st | 2nd | 3rd | Favorable combinations |
|---|---|---|---|
| H | H | H | Shaded part represents At least 1 head |
| H | H | T | |
| H | T | H | |
| H | T | T | |
| T | H | H | |
| T | H | T | |
| T | T | H | |
| T | T | T | |

The shaded part represent at least 1 head which means total number of favorable combinations $n(E) = 7$.

Therefore, probability to get at least one head is

$$P(E) = \frac{n(E)}{n(S)} = \frac{7}{8}.$$

Hence, the correct option is (D).

**Method 3**

This can also be solved by binomial distribution as follows :

Probability of at least one head is given by,

$$P(x \geq 1) = 1 - P(x = 0)$$

$$P(x \geq 1) = 1 - {}^3C_0 (0.5)^0 (0.5)^3$$

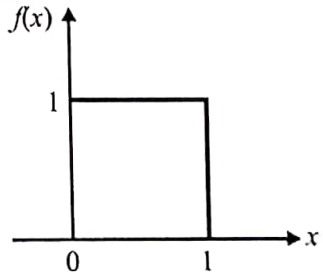
$$P(x \geq 1) = 1 - \left(\frac{1}{2}\right)^3$$

$$P(x \geq 1) = 1 - \frac{1}{8}$$

$$P(x \geq 1) = \frac{7}{8}$$

Hence, the correct option is (D).

**9.24 (A)**




For a random variable, the mean is given by,

$$\mu = \int_0^1 x f(x)dx = \int_0^1 x dx = \frac{1}{2}$$

Mean square value is given by,

$$E[X^2] = \int_0^1 x^2 f(x)dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

Variance is given by,

$$\text{Var}(x) = \sigma^2 = \text{M.S.V.} - (\mu)^2$$

$$\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$\sigma = \frac{1}{\sqrt{12}}$$

Hence, the correct option is (A).

**Key Point**

For a uniformly distributed random variable between $a$ to $b$,



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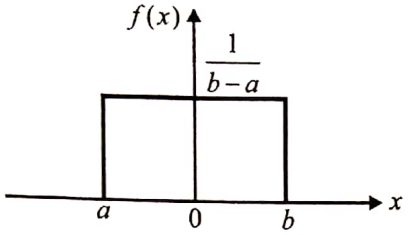



$$\text{Mean} = \frac{a+b}{2}$$

$$\text{Variance} = \frac{(b-a)^2}{12}$$

$$\text{M.S.V.} = \frac{a^2+ab+b^2}{3}$$

$$\text{S.D.} = \frac{(b-a)}{\sqrt{12}}$$

## 9.25 (C)

**Given :** Number of nuts = 3

Number of washers = 2

Number of bolts = 4

Total number of items in a box is $n = 9$

The probability of drawing 2 washers first followed by 3 nuts and subsequently the 4 bolts is given by,

$$P = P(\text{2 washers}) \times P(\text{3 nuts}) \times P(\text{4 bolts})$$

$$P = \frac{2}{9}\times\frac{1}{8}\times\frac{3}{7}\times\frac{2}{6}\times\frac{1}{5}\times\frac{4}{4}\times\frac{3}{3}\times\frac{2}{2}\times\frac{1}{1}$$

$$P = \frac{1}{1260}$$

Hence, the correct option is (C).

## 9.26 (D)

**Given :** An unbiased coin is tossed five times.

Therefore, the total number of combination

$$= 2^5 = 32$$

$$\text{Probability of getting head} = \frac{1}{2}$$

$$\text{Probability of getting no head (tail)} = \frac{1}{2}$$

$P$(at least one head) = 1 – $P$ (no heads)

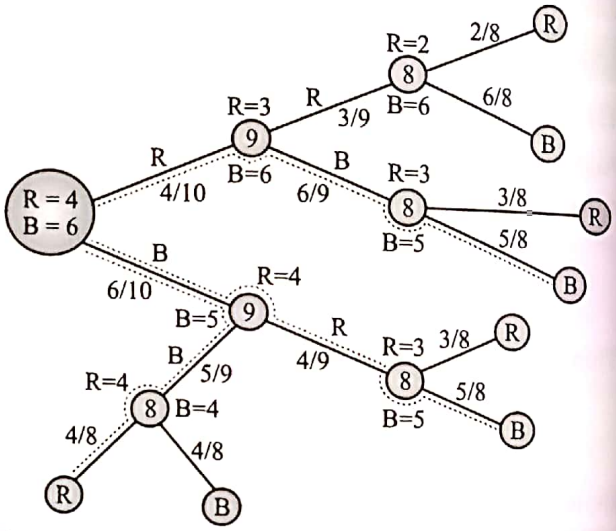
$$= 1 - \left(\frac{1}{2}\right)^5 = \frac{31}{32}$$

Hence, the correct option is (D).

## 9.27 (D)

**Given :** A box contains 4 red balls and 6 black balls.

Let us consider red balls (R) and black balls (B) with proper tree diagram representation as follows :




For 1 red and 2 black balls, the following sequence can occur :

1. (1 red and 2 black, RBB)

$$\frac{4}{10}\times\frac{6}{9}\times\frac{5}{8} = \frac{1}{6}$$

2. (1 black, 1 red and 1 black, BRB)

$$\frac{6}{10}\times\frac{4}{9}\times\frac{5}{8} = \frac{1}{6}$$

3. (2 black and 1 red, BBR)

$$\frac{6}{10}\times\frac{5}{9}\times\frac{4}{8} = \frac{1}{6}$$

Total probability is given by,

$$P = \frac{1}{6}+\frac{1}{6}+\frac{1}{6} = \frac{1}{2}$$

Hence, the correct option is (D).

## 9.28 (B)

**Given :** For an automobile plant, the part of shock absorbers supplied are :

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(i) 60 % by $X$.
(ii) 40 % by $Y$.

So, probability that a product belongs to a particular supplier is given by,

$$P(X)=0.6 \text{ and } P(Y)=0.4$$

Also the corresponding reliabilities are given as;

$$P\left(\frac{R}{X}\right)=0.96 \quad P\left(\frac{R}{Y}\right)=0.72$$

By the concept of conditional probability,

$$P\left(\frac{Y}{R}\right)=\frac{P(Y)P\left(\frac{R}{Y}\right)}{P(X)P\left(\frac{R}{X}\right)+P(Y)P\left(\frac{R}{Y}\right)}$$

$$P\left(\frac{Y}{R}\right)=\frac{(0.4)(0.72)}{(0.6)(0.96)+(0.4)(0.72)}$$

$$P\left(\frac{Y}{R}\right)=\frac{0.288}{0.576+0.288}$$

$$P\left(\frac{Y}{R}\right)=\frac{0.288}{0.864}=0.334$$

Hence, the correct option is (B).

**9.29 (D)**

Let, event $A$ = student knows the answer and event $C$ = student answered the question correctly.

The probability that the student knows the answer and answered the question correctly is given by,

$$P(A\cap C)=\frac{2}{3}$$

The student guess the answer if he does not know the answer.

Then, the probability of answering correctly by guessing $=\frac{1}{3}\times\frac{1}{4}=\frac{1}{12}$

The probability of correct answer is,

$$P(A)=\frac{2}{3}+\frac{1}{3}\times\frac{1}{4}=\frac{3}{4}$$

Hence the required probability is,

$$P\left(\frac{C}{A}\right)=\frac{P(A\cap C)}{P(A)}=\frac{\frac{2}{3}}{\frac{3}{4}}=\frac{8}{9}$$

Hence, the correct option is (D).

**9.30 (B)**

**Refer Solution 9.17 [Production & Industrial]**

**9.31 (D)**

Total 2-digit numbers between 1 and 100 is 90 (10 to 99).

i.e. $n(S)=90$

The numbers between 1 to 100 which are not divisible by 7 are 77.

i.e. $n(E)=77$

The required probability is,

$$P(E)=\frac{n(E)}{n(S)}=\frac{77}{90}$$

Hence, the correct option is (D).

**9.32 (D)**

**Given :** A table showing random variable and probability distribution is as follows :

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | 0.3 | 0.6 | 0.1 |
| $x^2$ | 1 | 4 | 9 |

The mean of discrete random variable is given by,

$$E(x)=\Sigma xP(x)$$
$$E(x)=(1\times0.3)+(2\times0.6)+(3\times0.1)$$
$$E(x)=1.8$$

Mean square value of discrete random variable,

$$\text{M.S.V.}=\Sigma x^2P(x)$$
$$\text{M.S.V.}=(1^2\times0.3)+(2^2\times0.6)+(3^2\times0.1)$$
$$\text{M.S.V.}=3.6$$

Variance is given by,

$$\text{Var}(x)=\Sigma x^2P(x)-[\Sigma xP(x)]^2$$
$$\text{Var}(x)=3.6-1.8^2$$
$$\text{Var}(x)=0.36$$



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Standard deviation is given by,

$$\sigma = \sqrt{\text{Variance}} = \sqrt{0.36}$$

$$\sigma = 0.6$$

Hence, the correct option is (D).

**9.33 (A)**

**Given :** In a box containing 25 parts, the number of :

(i) Defective parts = 10

(ii) Good parts = 25 – 10 = 15

If two parts are being drawn simultaneously in a random manner, the probability of both the parts being good is given by,

$$P = \frac{{}^{15}C_2}{{}^{25}C_2} = \frac{7}{20}$$

Hence, the correct option is (A).

**9.34 0.259**

**Given :** An unbiased cubic dice with opposite faces coloured identically.

**Method 1**

Assume $R$ denotes Red, and similarly $B$ denotes Blue and $G$ for Green.

The dice is thrown thrice $n = 3$.

Probability of obtaining red colour $= \frac{2}{6} = \frac{1}{3}$

Probability of obtaining blue colour $= \frac{2}{6} = \frac{1}{3}$

Probability of obtaining green colour $= \frac{2}{6} = \frac{1}{3}$

In 3 trials, the probability for at least 2 Red is given by,

$$P = P(2R \,\&\, 1B) + P(2R \,\&\, 1G) + P(3R)$$

$$P = {}^3C_2\left(\frac{1}{3}\right)^2\left(\frac{1}{3}\right) + {}^3C_2\left(\frac{1}{3}\right)^2\left(\frac{1}{3}\right) + {}^3C_3\left(\frac{1}{3}\right)^3$$

$$P = \frac{1}{9} + \frac{1}{9} + \frac{1}{27} = 0.259$$

Hence, the probability of obtaining red colour on top face of the dice at least twice is **0.259.**

**Method 2**

The dice is thrown thrice $n = 3$.

Probability of obtaining red colour $= \frac{2}{6} = \frac{1}{3}$

Probability of not obtaining red colour

$$= 1 - \frac{1}{3} = \frac{2}{3}$$

The probability of red colour at least twice is

$$P = P(r = 2) + P(r = 3)$$

$$P = {}^3C_2\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right) + {}^3C_3\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^0$$

$$P = \frac{2}{9} + \frac{1}{27} = \frac{7}{27} = 0.259$$

Hence, the probability of obtaining red colour on top face of the dice at least twice is **0.259.**

**9.35 0.65**

**Given :** In a group of people (with equal number of men and women),

20% of the men and 50% of the women are unemployed.

Let $P(M)$ = Probability of selecting a man $= \frac{1}{2}$

$P(W)$ = Probability of selecting a woman $= \frac{1}{2}$

Probability of selecting an **employed** man is given by,

$$P\left(\frac{E}{M}\right) = 80\% = \frac{80}{100} = \frac{4}{5}$$

Similarly, the probability of selecting an **employed** woman is given by.

$$P\left(\frac{E}{W}\right) = 50\% = \frac{50}{100} = \frac{1}{2}$$

Therefore, the required probability is,

$$P = P(M)P\left(\frac{E}{M}\right) + P(W)P\left(\frac{E}{W}\right)$$

$$P = \left(\frac{1}{2} \times \frac{4}{5}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{2}\left(\frac{4}{5} + \frac{1}{2}\right)$$

$$P = \frac{13}{20} = 0.65$$



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Hence, the probability of the selected person being employed is **0.65.**

**9.36 (A)**

The random variable and probability distribution are as follows,

| No. of defective pieces $x$ | 0 | 1 | 2 |
|---|---|---|---|
| Probability $P(x)$ | 1/6 | 2/3 | 1/6 |
| $x^2$ | 0 | 1 | 4 |

By the concept of discrete random variables,

Mean of random variable,

$$E(x)=\Sigma xP(x)$$

$$E(x)=\left(0\times\frac{1}{6}\right)+\left(1\times\frac{2}{3}\right)+\left(2\times\frac{1}{6}\right)=1$$

Mean square value of random variable,

$$E\left[x^2\right]=\Sigma x^2P(x)$$

$$E\left[x^2\right]=0\times\frac{1}{6}+1\times\frac{2}{3}+4\times\frac{1}{6}$$

$$E\left[x^2\right]=0+\frac{2}{3}+\frac{4}{6}=\frac{4}{3}$$

Variance of random variable,

$$\sigma^2=E(x^2)-\left[E(x)\right]^2=\frac{4}{3}-(1)^2$$

$$\text{Variance}=\frac{1}{3}$$

Hence, the correct option is (A).

**9.37 0.8145**

Given : In a batch of 100 bulbs,

Number of defective bulbs = 5

So, the number of non-defective bulbs is 95.

The probability for one bulb to be non-defective is,

$$p=\frac{95}{100} \qquad \ldots(i)$$

According to binomial distribution, the probability of $r$ trials is given by,

$$P(r)={}^nC_r\times p^r\times q^{n-r}$$

where $P$ is probability of occurrence of the event.

Here, $n=4$ and $r=0$

Probability that none of the bulb is defective is given by,

$$P={}^4C_0\left(\frac{5}{100}\right)^0\left(\frac{95}{100}\right)^4=\left(\frac{95}{100}\right)^4$$

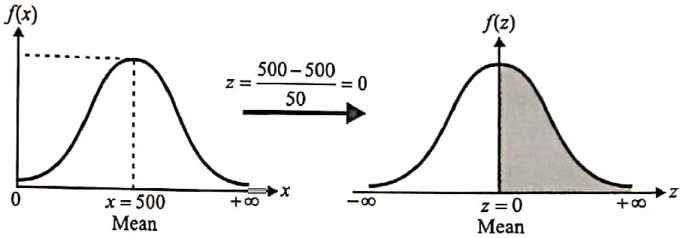
$$P=0.8145$$

Hence, the probability that the current batch accepted is **0.8145.**

**9.38 50**

**Given :** Mean $\mu=500$,

Standard deviation $\sigma=50$




$$z=\frac{x-\mu}{\sigma}$$

$$z=\frac{500-500}{50}=0$$

So, $P(z>0)=50\%$ (lies between 49 to 51)

Hence, the percentage of savings account holders is **50.**

**9.39 (B)**

**Given :** Mean, $\mu=5.2$

According to Poisson distribution, the probability is given by,

$$P(X=r)=\frac{e^{-\mu}\mu^r}{r!}$$

$$P(X<2)=P(X=0)+P(X=1)$$

$$P(X<2)=e^{-\mu}+\mu e^{-\mu}=e^{-\mu}(\mu+1)$$

$$P(X<2)=e^{-5.2}(5.2+1)$$

$$P(X<2)=0.034$$

Hence, the correct option is (B).

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**9.40 74.55**

**Given :** In below table random variable and its probability distribution are shown :

| $x$ | 50 | 70 | 75 | 80 | 85 |
|---|---|---|---|---|---|
| $P(x)$ | 0.15 | 0.14 | 0.21 | 0.20 | 0.30 |

By the concept of discrete random variable,

Expected value is given by,

$$E(x) = \Sigma x P(x)$$

$$E(x) = 50\times0.15 + 70\times0.14 + 75\times0.21 + 80\times0.20 + 85\times0.30$$

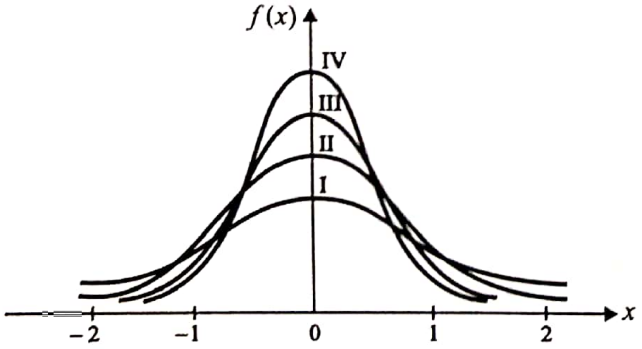
$$E(x) = 7.5 + 9.8 + 15.75 + 16 + 25.5$$

$$E(x) = 74.55$$

Hence, the expected demand during lead time is **74.55**.

**9.41 (D)**

**Given :** Four normal distribution curves are shown below,




The trace of a normal distribution curve is given by the following equation :

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(\frac{x-\mu}{\sigma}\right)^2}{2}} \quad \ldots(i)$$

where, $\sigma$ = Standard deviation.

$\mu$ = Mean

Let the variance be $V$.

Then, $\sigma = \sqrt{V}$

From equation (i),

$$f(x) \propto \frac{1}{\sigma} \propto \frac{1}{\sqrt{V}}$$

Curve (IV) has maximum peak hence minimum variance.

Hence, the correct option is (D).

**9.42 (C)**

**Given :** $P(X) = \frac{1}{4}$, $P(Y) = \frac{1}{3}$

and $P(X \cap Y) = \frac{1}{12}$

By the concept of conditional probability, (Bayes' Theorem)

$$P\left(\frac{Y}{X}\right) = \frac{P(Y \cap X)}{P(X)}$$

$$P\left(\frac{Y}{X}\right) = \frac{1/12}{1/4} = \frac{1}{3}$$

Hence, the correct option is (C).

**9.43 (A)**

**Given :** The mean for distribution $= \mu$

For Poisson distribution, Mean = Variance

Therefore, Variance $= \mu$

Standard deviation $= \sqrt{\text{Variance}}$

So, standard deviation $= \sqrt{\mu}$

Hence, the correct option is (A).

**9.44 0.4095**

**Given :**

The probability of defective screw $p = 0.1$

Probability of a screw being not defective is given by,

$$q = 1 - 0.1 = 0.9$$

Each packet contains 5 screws and would be replaced if one or more screws will be defective.

So, probability of replacement is given by,

$P = 1 -$ (Probability that no screw is defective)

$$P = 1 - \left[{}^5C_0\, p^0 q^5\right]$$

$$P = 1 - \left[{}^5C_0 \times (0.1)^0 \times (0.9)^5\right]$$

$$P = 0.4095$$

Hence, the probability that a packet would have to be replaced is **0.4095**.



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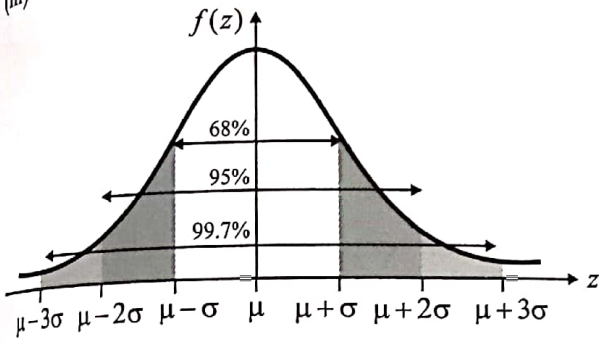
**9.45** **99.7**

By a rule of statistics for normal distribution curves, the area (in percentage) under the curve within :

(i) $\pm 1$ of the mean is 68.27%.

(ii) $\pm 2$ of the mean is 95.45%.

(iii) $\pm 3$ of the mean is 99.73%.




This rule is generally referred to as "68-95-99.7" rule.

Hence, the area (in percentage) under standard normal distribution curve is **99.7**.

**9.46** **(A)**

**Given :** Three cards were drawn from 52 cards.

The number of ways in which three cards can be picked is,

$$n(S) = {}^{52}C_3$$

The number of ways a king, a queen and a jack can be picked from a pack of cards is given by,

$$n(A) = {}^{4}C_1 \times {}^{4}C_1 \times {}^{4}C_1$$

So required probability is given by,

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{{}^{4}C_1 \times {}^{4}C_1 \times {}^{4}C_1}{{}^{52}C_3} = \frac{16}{5525}$$

Hence, the correct option is (A).

**9.47** **(C)**

**Given :** An ungrouped data of 15 numbers is as follows :

17, 18, 17, 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3.

Mode refers to value that appears most frequently in a set of data.

The number 17 has occurred for the maximum (four) number of times.

So, the mode of this sample is 17.

Hence, the correct option is (C).

**9.48** **(B)**

**Given :** Four red balls, four green balls and four blue balls are put in a box.

**Method 1**

Total number of balls in a box = 12

The number of ways 3 balls selected from 12 balls $= {}^{12}C_3$

The number of ways 3 red balls selected from 4 red balls $= {}^{4}C_3$

Probability that all the three balls are red is

$$= \frac{{}^{4}C_3}{{}^{12}C_3} = \frac{4}{220} = \frac{1}{55}$$

Hence, the correct option is (B).

**Method 2**

Total number of balls in a box

$$= 4R + 4G + 4B = 12$$

Probability that all the 3 balls are red without replacement

$$= \frac{4}{12} \times \frac{3}{11} \times \frac{2}{10} = \frac{1}{55}$$

Hence, the correct option is (B).

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**9.49** **(C)**

**Given :** A dice is rolled five times,

$$n = 5$$

A dice has 6 faces therefore probability of getting $1 = p = \frac{1}{6}$.

Probability of not getting $1 = q = 1 - \frac{1}{6} = \frac{5}{6}$.



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**According to binomial distribution :**

Probability of getting 1 at least four times is

$$p(x \geq 4) = p(x=4) + p(x=5)$$

$$= {}^5C_4\, p^4 q^{5-4} + {}^5C_5\, p^5 q^{5-5}$$

$$= \frac{5!}{1!4!}\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right)^{5-4} + \frac{5!}{0!5!}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^{5-5}$$

$$= \frac{25}{6^5} + \frac{1}{6^5} = 0.0033 = 0.33\%.$$

Hence, the correct option is (C).

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**9.50 (B)**

**Given :** $\mu_1 = 1,\ \mu_2 = 1,\ \sigma_1 = 1,\ \sigma_2 = 2$

If $x_1$ and $x_2$ are two independent random variables.

and $Y = X_1 - X_2$

Mean or variance of random variable $Y$ is given by,

$$\mu(Y) = \mu(X_1 - X_2)$$

$$\mu(Y) = \mu(X_1) - \mu(X_2)$$

$$\mu(Y) = \mu_1 - \mu_2 = 1 - 1 = 0$$

$$\text{Var}(Y) = \text{Var}(X_1 - X_2)$$

$$\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2) - \text{Cov}(X_1, X_2)$$

Let $X_1$ and $X_2$ are independent variables therefore $\text{Cov}(X_1, X_2) = 0$

$$\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2)$$

$$\text{Var}(Y) = \sigma_1^2 + \sigma_2^2 = 1 + 4$$

$$\text{Var}(Y) = 5$$

Hence, the correct option is (B).

**9.51 (A)**

Probability distribution function for exponential random variable defined as

$$f(x) = \lambda e^{-\lambda x};\ x \geq 0$$

where $\lambda$ is parameter and mean of exponential random variable.

Mean $(X_1) = 0.5$

$$\frac{1}{\lambda_1} = 0.5 \Rightarrow \lambda_1 = \frac{1}{0.5} = 2$$

Mean $(X_2) = 0.25$

$$\frac{1}{\lambda_2} = 0.25 \Rightarrow \lambda_2 = \frac{1}{0.25} = 4$$

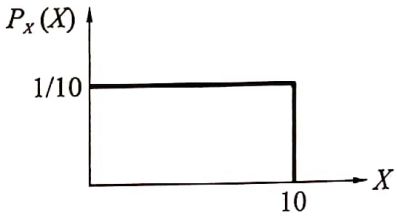
$y =$ Mean $(X_1, X_2)$

$$\text{Mean}\ (Y) = \frac{1}{\lambda_1 + \lambda_2} = \frac{1}{2+4} = \frac{1}{6}$$

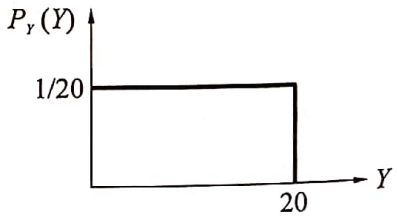
Hence, the correct option is (A).

**9.52 (A)**

**Given :** $X$ uniformly distributed between 0 and 10.




$Y$ uniformly distributed between 0 and 20.




**Method 1**

**Given :** $x + y > 20 \Rightarrow y > 20 - x$

$$P(x+y>20) = \int_0^{10}\int_{20-x}^{20} P(x, y)\, dy\, dx$$

$$= \int_0^{10}\int_{20-x}^{20}\left(\frac{1}{10}\times\frac{1}{20}\right) dy\, dx$$

[As $X$ and $Y$ are independent

$$P(x, y) = P_X(x) P_Y(y)]$$

$$P(x+y>20) = \frac{1}{200}\int_0^{10} [y]_{20-x}^{20}\, dx$$

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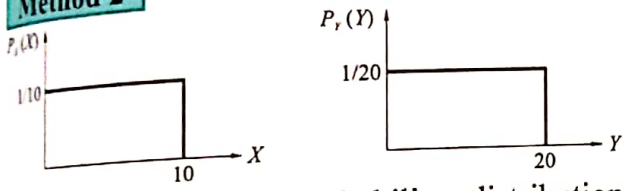
$$P(x+y>20)=\frac{1}{200}\int_0^{10}[20-(20-x)]dx$$

$$P(x+y>20)=\frac{1}{200}\int_0^{10}x\,dx=\frac{1}{200}\times\left[\frac{x^2}{2}\right]_0^{10}$$

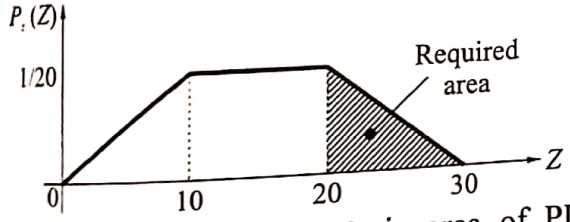
$$P(x+y>20)=\frac{1}{400}[100-0]=\frac{1}{4}=0.25$$

Hence, the correct option is (A).

**Method 2**




Let $Z=X+Y$, then probability distribution function of $Z$ is convolution of PDF's of $X$ and $Y$.




$\therefore$ Probability of $Z>20$ is area of PDF between 20 and 30,

$$=\frac{1}{2}\times10\times\frac{1}{20}=\frac{1}{4}=0.25$$

Hence, the correct option is (A).

**9.53 3**

Given : Mean of the data $[3,x,2,4]$ is $x$.

$\therefore$ $$\frac{3+x+2+4}{4}=x$$

$$9+x=4x$$

$$3x=9$$

$$x=3$$

Hence, the given data is $[3,3,2,4]$.

Arranging the data in proper form $2,3,3,4$.

Mode of the data is given as the number with highest occurrence. As '3' has occurred maximum two times, so the mode of the given data is '3'.

**9.54 0.171**

**Given :** Probability of a part to be defective $p=0.05$

$\therefore$ Probability of the part to be non-defective

$$q=1-p=1-0.05=0.95$$

Probability of atleast two defective parts

= 1 – {(Probability of no part defective) + (Probability of one defective part)}

Using binomial distribution, probability of '$r$' success in '$n$' trials is given as

$$P(r)={}^nC_r\,p^r\cdot q^{n-r}$$

Here, $n=15$

$$P(0)={}^{15}C_0\,(0.05)^0\times(0.95)^{15}=0.4633$$

$$P(1)={}^{15}C_1\,(0.05)^1\times(0.95)^{14}=0.3658$$

$\therefore$ $$P(r\geq2)=1-[P(0)+P(1)]=0.171$$

Hence, the probability that at least two parts will be defective is **0.171**.

## Civil Engineering : CE

**9.1 (C)**

**Given :** Total number of screws = 10

Number of defective screws = 3

$$P\text{ (defective screws)}=\frac{3}{10}=0.3$$

$$P\text{ (non-defective screws)}=1-0.3=0.7$$

According to binomial distribution, the probability of $r$ trails is given by,

$$P(r)={}^nC_r\times p^r\times q^{n-r}$$

Here, $p=0.3$, $q=0.7$, $n=2$, $r=0$

Probability of none of the drawn two screws are defective is given by,

$$P(r=0)={}^2C_0p^0\,q^2$$

$$P(r=0)=1\times0.3^0\times0.7^2=0.49$$



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Hence the percentage probability

$$= 0.49 \times 100\% = 49\%$$

Hence, the correct option is (C).

**9.2 (C)**

**Given :** $P$ (failure of each gate) $= 0.2$

From the given data it is clear that all gates operate independently,

For independent events,

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

According to Bayes' theorem,

Probability that gate 2 and gate 3 will fail when gate 1 has failed, is given by,

$$P\left(\frac{G_2 \cap G_3}{G_1}\right) = \frac{P(G_3) \cap P(G_2) \cap P(G_1)}{P(G_1)}$$

$$P\left(\frac{G_2 \cap G_3}{G_1}\right) = \frac{0.2 \times 0.2 \times 0.2}{0.2} = 0.04$$

Hence, the correct option is (C).

**Key Point**

**Independent events :**

If $E_1$ and $E_2$ are independent event i.e. one does not affects the others outcome.

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

**9.3 (D)**

**From option (A) :**

Skewness tells about the direction of the variation or the departure from symmetry. Therefore, the measure of skewness is dependent upon the amount of dispersion.

Hence, option (A) is true.

**From option (B) :**

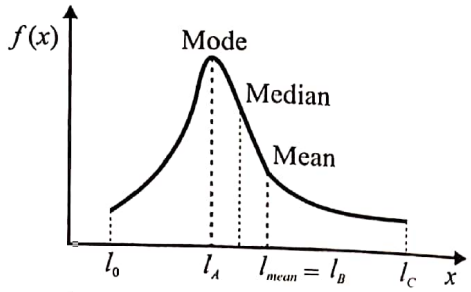
Symmetrical distribution is a distribution in which the values of variables occurs at regular frequencies and the mean, median and mode occurs at same point.

Hence, option (B) is true.

**From option (C) :**

**In a positively or right skewed distribution :**

mean > median > mode.

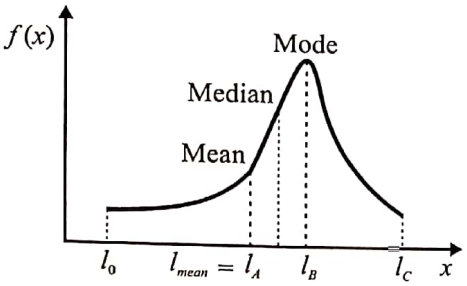



Hence, option (C) is true.

**From option (D) :**

**In negatively or left skewed distribution :**

mode > median > mean




So, option (D) is not true.

Hence, the correct option is (D).

**9.4 (D)**

Let the mean and standard deviation of the students of batch $C$ be $\mu_C$ and $\sigma_C$ respectively, and the mean and standard deviation of entire class of first year students be $\mu$ and $\sigma$ respectively.

$\mu_C = 6.6$, $\sigma_C = 2.3$ and $\mu = 5.5$, $\sigma = 4.2$

In order to normalize batch $C$ to entire class, the normalized score must be equated.

Since, $Z = \frac{x - \mu}{\sigma}$

$$Z_C = \frac{x_C - \mu_C}{\sigma_C} = \frac{8.5 - 6.6}{2.3} \quad \ldots(i)$$

Now, $Z = \frac{x - \mu}{\sigma} = \frac{x - 5.5}{4.2}$ $\ldots(ii)$



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Equating equation (i) and (ii),

$$\frac{8.5-6.6}{2.3}=\frac{x-5.5}{4.2}$$

$$x=8.969\simeq 9.0$$

Hence, the correct option is (D).

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**9.5 (B)**

Given : Total number of calculators = 25

Number of defective calculators = 2

Number of non-defective calculators = 23

Given that out of 25 calculators 2 are defective that means 23 are non-defective.

Probability of being defective, $p=\frac{2}{25}$

Probability of being non-defective,

$$q=1-\frac{2}{25}=\frac{23}{25}$$

Hence, the probability of one defective calculators in the set of 5 calculators is given by,

$p$(1 defective in 5 calculators)

$$=\frac{{}^{2}C_1\times{}^{23}C_4}{{}^{25}C_5}=\frac{\frac{2!}{1!\,1!}\times\frac{23!}{19!\,4!}}{\frac{25!}{20!\,5!}}=\frac{1}{3}$$

Hence, the correct option is (B).

**9.6 (C)**

Given : Standard deviation, $\sigma=8.8$ kmph

and mean, $\mu=33$ kmph

Hence, by the formula of coefficient variation

$$C_v=\frac{\sigma}{\mu}=\frac{8.8}{33}=0.2666$$

Hence, the correct option is (C).

**9.7 (B)**

Given : $f(x)=\begin{cases}x^2, & -1\le x\le 1\\ 0, & \text{elsewhere}\end{cases}$

The probability is given by,

$$P(-\infty\le x\le\infty)=\int_{-\infty}^{\infty}f(x)\,dx$$

Hence, $P\left(\frac{-1}{3}\le x\le\frac{1}{3}\right)=\int_{\frac{-1}{3}}^{\frac{1}{3}}f(x)dx=\int_{\frac{-1}{3}}^{\frac{1}{3}}x^2dx$

$$P\left(\frac{-1}{3}\le x\le\frac{1}{3}\right)=\left[\frac{x^3}{3}\right]_{\frac{-1}{3}}^{\frac{1}{3}}$$

$$P\left(\frac{-1}{3}\le x\le\frac{1}{3}\right)=\frac{1}{3}\left[\left(\frac{1}{3}\right)^3+\left(\frac{1}{3}\right)^3\right]=\frac{2}{81}$$

The probability expressed in percentage

$$P=\frac{2}{81}\times 100\%=2.469\%=2.47\%$$

Hence, the correct option is (B).

**9.8 (D)**

Given : $y=a+bx$

$n=3,\Sigma x=6,\Sigma y=21,\Sigma x^2=14$ and $\Sigma xy=46$

Since, $y=a+bx$

$$\sum_{n=1}^{3}y=\sum_{n=1}^{3}a+\sum_{n=1}^{3}bx$$

$$21=3a+b\sum_{n=1}^{3}x$$

$$21=3a+6b \qquad \ldots(i)$$

Multiplying equation (i) by $x$ on both side,

$$xy=xa+bx^2$$

$$\sum_{n=1}^{3}xy=\sum_{n=1}^{3}xa+\sum_{n=1}^{3}bx^2$$

$$46=a\sum_{n=1}^{3}x+b\sum_{n=1}^{3}x^2$$



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$$46 = a6 + b \times 14$$

$$6a + 14b = 46 \quad \ldots(ii)$$

Solving equation (i) and equation (ii),

$$a = 3,\ b = 2$$

Hence, the correct option is (D).

**9.9 (A)**

**Given :** $P$ (private car) $= 0.45$

$$P\left(\frac{\text{bus}}{\text{public transport}}\right) = 0.55$$

Since, a person has a choice between private car and public transport,

$$P \text{ (public transport)} = 1 - P(\text{private car})$$
$$= 1 - 0.45 = 0.55$$

$$P(\text{bus}) = P(\text{Bus} \cap \text{Public transport})$$

$$P(\text{bus}) = P\left(\frac{\text{bus}}{\text{public transport}}\right) \times P(\text{public transport})$$

$$P \text{ (bus)} = 0.55 \times 0.55$$

$$P \text{ (bus)} = 0.3025 \simeq 0.30$$

$$P(\text{metro}) = 1 - [P(\text{private car}) + P(\text{bus})]$$
$$= 1 - (0.45 + 0.30) = 0.25$$

$$P(\text{private car}) = 0.45$$

$P$(bus) = 0.30 and P(metro) = 0.25

Hence, the correct option is (A).

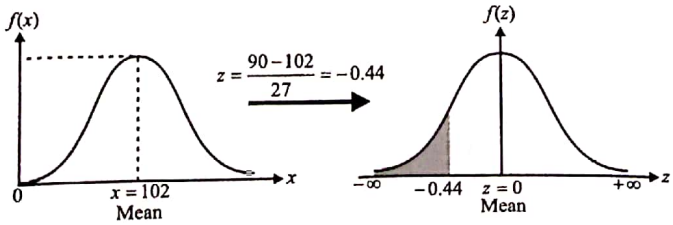
**9.10 (D)**

**Given :** $\mu = 102\,\text{cm};\ \sigma = 27\,\text{cm}$

The required probability calculated as

$$p(90 \le x \le 102) = p\left(\frac{90-102}{27} \le x \le \frac{102-102}{27}\right)$$
$$= p(-0.44 \le x \le 0)$$

This area is shown below :




The shades area in above figure is given by

$$F(0) - F(-0.44) = \frac{1}{1+\exp(0)} - \frac{1}{1+\exp[(-1.7255(-0.44)(0.44)^{0.12}]}$$

$$F(0) - F(-0.44) = 0.5 - 0.3345$$

$$F(0) - F(-0.44) = 0.1655 \simeq 16.55\%$$

Hence, the correct option is (D).

**9.11 (C)**

The probability of getting one head $= \frac{1}{2}$

The probability of getting two heads are independent events hence by product rule

$$P(A \cap B) = P(A) \times P(B)$$

Hence, the required probability $P$(2 heads) is,

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Hence, the correct option is (C).

**9.12 (C)**

**Given :** Container 1 has 4 red balls and 3 green balls = total 7 balls
Container 2 has 3 blue balls and 4 green balls = total 7 balls.

**Method 1**

This is the case of independent event.

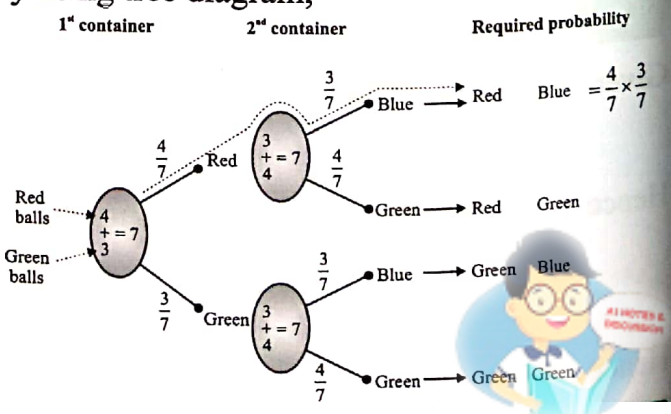
Probability that one is red and other is blue

$$= \frac{{}^4C_1}{{}^7C_1} \times \frac{{}^3C_1}{{}^7C_1} = \frac{12}{49}$$

Hence, the correct option is (C).

**Method 2**

By using tree diagram,

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Total probability $= \frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$

Hence, the correct option is (C).

**9.13 (D)**

Since negative and positive are equally likely, Probability of occurrence of negative number is,

$$p = \frac{1}{2}$$

So, probability of occurrence of non-negative number is,

$$q = 1 - \frac{1}{2} = \frac{1}{2}$$

Let $x$ represent number of negative values.

So, $n = 5$

and $p$ (at most 1 negative value)

$$= p(x \le 1)$$

$$= p(x = 0) + p(x = 1)$$

By binomial distribution required probability is given by,

$$P[x \le 1] = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 + {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4$$

$$P[x \le 1] = \frac{6}{32}$$

Hence, the correct option is (D).

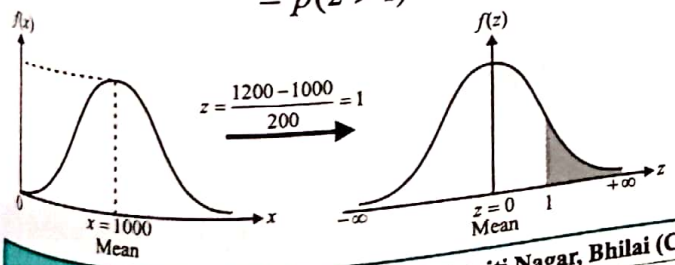
**9.14 (A)**

Given : The annual precipitation is normally distributed with $\mu = 1000$ mm and $\sigma = 200$ mm

For simplification of normal distribution let us consider $z = \frac{x - \mu}{\sigma}$.

$$p(x > 1200) = p\left(z > \frac{1200 - 1000}{200}\right)$$

$$= p(z > 1)$$




The probability for above distribution from figure is given by,

$$P(x > 1200) = P(z > 1)$$

According to property of normal distribution, the normal distribution curve is symmetric with mean $z = 0$ which means distribution is divided into two parts with equal probability of 0.5.

Hence, $P(z > 1) = 0.5 - P(0 < z < 1)$

That means the required probability is less than 0.5 that means less than 50%.

Hence, the correct option is (A).

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**9.15 6**

**Given :** $f(x) = \begin{cases} \lambda(-x^2 + 3x - 2); & 1 \le x \le 2 \\ 0 & ; \text{ otherwise} \end{cases}$

By the property of probability density function (P.D.F.),

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_1^2 \lambda(-x^2 + 3x - 2)\,dx = 1$$

$$\lambda\left[-\frac{x^3}{3} + 3\frac{x^2}{2} - 2x\right]_1^2 = 1$$

$$\lambda\left[-\left(\frac{8}{3} - \frac{1}{3}\right) + \frac{3}{2}(4-1) - 2(2-1)\right] = 1$$

$$\lambda\left[-\frac{7}{3} + \frac{9}{2} - 2\right] = 1$$

$$\lambda\left[\frac{-14 + 27 - 12}{6}\right] = 1$$

$$\lambda\left[\frac{1}{6}\right] = 1$$

$$\lambda = 6$$

Hence, the value of $\lambda$ is **6**.

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**Key Point**

**Probability distribution or probability density function of $X$ :**

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(iii) $P(a < X < b) = \int_{b}^{a} f(x)\,dx$

**9.16 (D)**

Return period of rainfall, $T = 50$ years.

Probability of occurrence once in 50 years,

$$P = \frac{1}{50} = 0.02$$

Probability of occurrence in each of 2 successive years,

$$P = 0.02 \times 0.02 = 0.0004$$

Hence, the correct option is (D).

**9.17 0.4**

**Given :** $f(E) = \begin{cases} \frac{1}{5}; & 0 \leq E \leq 5 \text{ mm/day} \\ 0; & \text{otherwise} \end{cases}$

Hence, the required probability is,

$$P(2 \leq E \leq 4) = \int_{2}^{4} f(E)\,dE$$

$$= \int_{2}^{4} \frac{1}{5}\,dx = \frac{1}{5}[x]_2^4$$

$$= \frac{1}{5}(4-2)$$

$$P(2 \leq E \leq 4) = \frac{2}{5} = 0.4$$

Hence, the probability that $E$ lies in between 2 and 4 mm/day in the watershed is **0.4**.

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**9.18 0.265**

**Given :**

The mean of poisson distribution is $m = 5$

By poisson distribution probability is given by,

$$P(r) = \frac{e^{-m} m^r}{r!}$$

Let $X$ = Number of penalties per day

$$P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$P(X < 4) = e^{-m} + me^{-m} + \frac{m^2}{2!}e^{-m} + \frac{m^3}{3!}e^{-m}$$

$$P(X < 4) = e^{-m}\left(1 + m + \frac{m^2}{2} + \frac{m^3}{6}\right)$$

$$= e^{-5}\left(1 + 5 + \frac{25}{2} + \frac{125}{6}\right)$$

$$P(X < 4) = \frac{e^{-5}}{6}(6 + 30 + 75 + 125)$$

$$P(X < 4) = 0.265$$

Hence, the probability that there will be less than 4 penalties in a day is **0.265**.

**9.19 (B)**

$P$ (4 successively head in 4 tosses) $= \frac{1}{2^4}$

$P$ (4 successively head and 1 tail in last toss during 5 toss) $= P(H\,H\,H\,H\,T) = \frac{1}{2^5}$

$P$ (4 successively head and head in last toss during 5 toss) $= \frac{1}{2^5}$

$$\text{Required probability} = \frac{P(4\text{H, T})}{P(4\text{H, H}) + P(4\text{H, T})}$$

$$\text{Required probability} = \frac{\left(\frac{1}{2}\right)^4 \times \frac{1}{2}}{\left(\frac{1}{2}\right)^4 \times \frac{1}{2} + \left(\frac{1}{2}\right)^4 \times \frac{1}{2}}$$

Hence, the correct option is (B).

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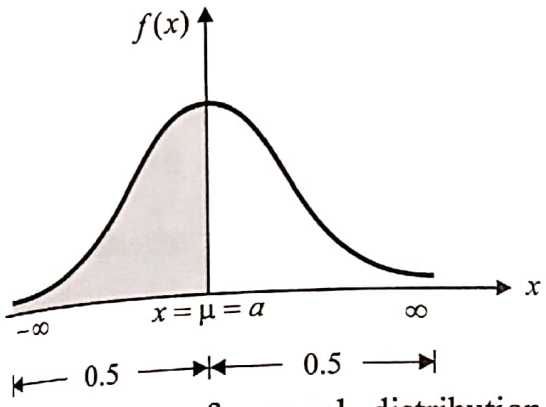
**9.20 (B)**

Given : $f(x)=\frac{1}{\sqrt{2\pi}\times b}e^{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2}$




By the property of normal distribution, the distribution curve is divided into two parts which are equally probable.

So,

$$\int_{-\infty}^{a}\frac{1}{\sqrt{2\pi}\times b}e^{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2}=\int_{a}^{\infty}\frac{1}{\sqrt{2\pi}\times b}e^{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2}=0.5$$

Hence, the correct option is (B).

**9.21 0.27**

Given : $m=240$ veh/h

$$m=\frac{240}{60}\text{veh/min}=4\text{ veh/min}$$

$$m=\frac{4}{60}\text{veh/sec}=2\text{ veh/30sec}$$

On average, two vehicle arrive in 30 sec time interval.

So, $m=2$

Hence, according to poisson distribution, the probability of having one vehicle arriving over 30 sec. time interval is given by,

$$P(r)=\frac{e^{-m}m^r}{r!}\qquad[\text{where, } r=1]$$

The required probability,

$$=P(X=1)=\frac{e^{-2}2^1}{1!}=2e^{-2}=0.27$$

Hence, the probability of having one vehicle arriving over a 30-second time interval is **0.27**.

**9.22 0.24**

Given : $p(X,q)=\begin{cases}q, & \text{if } X=0\\ 1-q, & \text{if } X=1\\ 0, & \text{otherwise}\end{cases}$

$q=0.4$

| $X$ | 0 | 1 |
|---|---|---|
| $p(X)$ | 0.4 | 0.6 |

Variance of $X$ is given by,

$$Var(X)=E(X^2)-[E(X)]^2\quad\ldots(i)$$

where, $E[X]=\Sigma X P(X)$

$$E[X]=(0\times0.4+1\times0.6)=0.6$$

Similarly, $E[X^2]=\Sigma X^2\times P(X)$

$$E[X^2]=(0^2\times0.4+1^2\times0.6)=0.6$$

From equation (i),

$$Var(X)=E(X^2)-[E(X)]^2$$

$$Var(X)=0.6-(0.6)^2$$

$$Var(X)=0.6-0.36=0.24$$

Hence, the variance of $X$ is **0.24**.

**9.23 1.066**

Given : $f(x)=\begin{cases}\frac{x}{4}(4-x^2); & \text{for } 0\le x\le 2\\ 0 \quad ; & \text{otherwise}\end{cases}$

Mean $\mu_x$ of random variable is given by,

$$\mu_x=\int_{-\infty}^{\infty}xf(x)\,dx=\int_0^2 x\times\frac{x}{4}(4-x^2)\,dx$$

$$\mu_x=\int_0^2\left(x^2-\frac{x^4}{4}\right)dx=\left[\frac{x^3}{3}-\frac{1}{4}\times\frac{x^5}{5}\right]_0^2$$

$$\mu_x=\frac{2^3}{3}-\frac{1}{4}\times\frac{2^5}{5}=\frac{8}{3}-\frac{32}{20}=1.066$$

The mean $\mu_x$ of the random variable is **1.066**.

**9.24 (A)**

Given : $f(x)=\begin{cases}0.25; & \text{if } 1\le x\le 5\\ 0 \quad ; & \text{otherwise}\end{cases}$



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$$P(X \le 4) = \int_{-\infty}^{4} f(x)dx$$

$$P(X \le 4) = \int_{-\infty}^{1} f(x)dx + \int_{1}^{4} f(x)dx$$

$$P(X \le 4) = \int_{1}^{4} \frac{1}{4}dx = \frac{1}{4}(x)_1^4 = \frac{3}{4}$$

Hence, the correct option is (A).

**9.25 54.5**

**Given :** Increasing order of spot speeds are,

32, 45, 49, 51, **53**, **56**, 60, 62, 66, 79

Median = Middle value of observed data

Since, the number of observation are 10 i.e. even number.

Therefore, we have to take average of 5th and 6th observation.

Therefore,

$$\text{Median} = \frac{53+56}{2} = 54.5 \text{ kmph}$$

Hence, the median speed is **54.5**.

**9.26 (A)**

**Given :** $P(X) = 0.40$

$$P(X \cup Y^C) = 0.7$$

Hence,

$$P(X \cup Y^C) = P(X) + P(Y^C) - P(X \cap Y^C)$$

Since, $X$ and $Y$ are two independent events,

Therefore, $P(X \cap Y^C) = P(X) \cdot P(Y^C)$

Hence,

$$P(X \cup Y^C) = P(X) + P(Y^C) - P(X) \cdot P(Y^C)$$

$$0.7 = 0.4 + P(Y^C) - 0.4 \times P(Y^C)$$

$$0.3 = 0.6 \times P(Y^C)$$

$$P(Y^C) = 0.5$$

$$1 - P(Y) = 0.5$$

$$P(Y) = 0.5$$

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

$$P(X \cup Y) = 0.4 + 0.5 - 0.4 \times 0.5 = 0.7$$

Hence, the correct option is (A).

**9.27 (B)**

**Given :**

$f(x)$ and $g(x)$ are two probability density functions,

$$f(x) = \begin{cases} \frac{x}{a} + 1 & ; \quad -a \le x < 0 \\ -\frac{x}{a} + 1 & ; \quad 0 \le x \le a \\ 0 & ; \quad \text{otherwise} \end{cases}$$

and

$$g(x) = \begin{cases} -\frac{x}{a} & ; \quad -a \le x < 0 \\ \frac{x}{a} & ; \quad 0 \le x \le a \\ 0 & ; \quad \text{otherwise} \end{cases}$$

Mean of $f(x)$ is given by,

$$E[f(x)] = \int_{-\infty}^{\infty} xf(x)\,dx$$

$$E[f(x)] = \int_{-a}^{0} \left(\frac{x^2}{a} + x\right)dx + \int_{0}^{a} \left(\frac{-x^2}{a} + x\right)dx$$

$$E[f(x)] = \left(\frac{x^3}{3a} + \frac{x^2}{2}\right)_{-a}^{0} + \left(\frac{-x^3}{3a} + \frac{x^2}{2}\right)_{0}^{a}$$

$$E[f(x)] = -\frac{(-a)^3}{3a} - \frac{(-a)^2}{2} + \frac{(-a^3)}{3a} + \frac{a^2}{2}$$

$$E[f(x)] = \frac{a^2}{3} - \frac{a^2}{2} - \frac{a^2}{3} + \frac{a^2}{2}$$

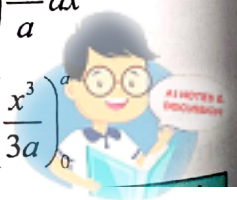
$$E[f(x)] = 0$$

Mean of $g(x)$ is given by,

$$E[g(x)] = \int_{-\infty}^{\infty} xg(x)\,dx$$

$$E[g(x)] = \int_{-a}^{0} \frac{-x^2}{a}dx + \int_{0}^{a} \frac{x^2}{a}dx$$

$$E[g(x)] = \left(-\frac{x^3}{3a}\right)_{-a}^{0} + \left(\frac{x^3}{3a}\right)_{0}^{a}$$



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$$E[g(x)] = -\left(\frac{a^2}{3}\right) + \frac{a^2}{3} = 0$$

Variance of $f(x)$ is given by,

$$\text{Var}[f(x)] = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$E(X^2) = \int_{-a}^{0}\left(\frac{x^3}{a} + x^2\right)dx + \int_{0}^{a}\left(\frac{-x^3}{a} + x^2\right)dx$$

$$E(X^2) = \left(\frac{x^4}{4a} + \frac{x^3}{3}\right)_{-a}^{0} + \left(\frac{-x^4}{4a} + \frac{x^3}{3}\right)_{0}^{a}$$

$$E(X^2) = -\left(\frac{a^3}{4} - \frac{a^3}{3}\right) + \left(\frac{-a^3}{4} + \frac{a^3}{3}\right)$$

$$E(X^2) = \frac{-2a^3}{4} + \frac{2a^3}{3}$$

$$E(X^2) = \frac{-6a^3 + 8a^3}{12} = \frac{2a^3}{12} = \frac{a^3}{6}$$

$$\text{Var}[f(x)] = \frac{a^3}{6} - 0 = \frac{a^3}{6}$$

Variance of $g(x)$ is given by,

$$\text{Var}[g(x)] = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 g(x)\,dx$$

$$E(X^2) = \int_{-a}^{0} \frac{-x^3}{a}\,dx + \int_{0}^{a} \frac{x^3}{a}\,dx$$

$$E(X^2) = \left(-\frac{x^4}{4a}\right)_{-a}^{0} + \left(\frac{x^4}{4a}\right)_{0}^{a}$$

$$E(X^2) = (0) - \left(-\frac{a^3}{4}\right) + \frac{a^3}{4}$$

$$E(X^2) = \frac{2a^3}{4} = \frac{a^3}{2}$$

$$Var[g(x)] = \frac{a^3}{2} - 0 = \frac{a^3}{2}$$

Hence, the correct option is (B).

**9.28 0.5**

Probability of getting $[H, H, H] = \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$

and the probability of getting $[H, H, H, H]$

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{16}$$

According to given question, $(H, H, H)$ has already come in toss.

Condition is that to toss a coin one more time to obtain $H$ again,

Hence, probability of getting '$H$' again after $(H, H, H)$

$$= \frac{\text{Probability of getting } (H,H,H,H)}{\text{Probability of getting } (H,H,H)}$$

$$= \frac{1/16}{1/8} = \frac{8}{16} = \frac{1}{2} = 0.5$$

Hence, the conditional probability according to the question is **0.5**.

**9.29 (B)**

For the univariate exponential function,

Probability density function is given by,

$$f(x) = \begin{cases} a e^{-ax} & ; \quad x \geq 0 \\ 0 & ; \quad x < 0 \end{cases}$$

where, $a$ is a parameter

For Gaussian distribution, probability density function is given by,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where, $\sigma$ and $\mu$ are parameters.

Hence, the number of parameters in univariate exponential distribution is 1and for Gaussian distribution the parameters are 2.

Hence, the correct option is (B).

**9.30 (B)**

**Given :** $f(x) = a + bx,\ 0 \leq x \leq 1$

By probability density function,



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$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Hence, $\int_0^1 (a+bx)\,dx = 1$

$$\left[ax + \frac{bx^2}{2}\right]_0^1 = 1$$

$$a + \frac{b}{2} = 1 \qquad \ldots(i)$$

Only option (B) i.e. $a = 0.5$ and $b = 1$ satisfies the above equation

Hence, the correct option is (B).

**9.31 0**

**Given :** $n = 20$

| $f$ (MPa) | Number of specimens with compressive strength equal to $f$ |
|---|---|
| 23 | 4 |
| 28 | 2 |
| 22.5 | 5 |
| 31 | 5 |
| 29 | 4 |

$n = 20$

Mean strength is given by,

$$\mu = \frac{(23\times4)+(28\times2)+(22.5\times5)+(31\times5)+(29\times4)}{20}$$

$\mu = 26.575$ MPa

Standard deviation is given by,

$$\sigma = \sqrt{\frac{\Sigma(f-\mu)^2}{n-1}}$$

$$\sigma = \sqrt{\frac{\begin{array}{l}4(23-26.575)^2+2(28-26.575)^2\\+5(22.5-26.575)^2+5(31-26.575)^2\\+4(29-26.575)^2\end{array}}{19}}$$

$$\sigma = \sqrt{\frac{\begin{array}{l}51.1225+4.06125+83.028\\+97.903+23.5225\end{array}}{19}}$$

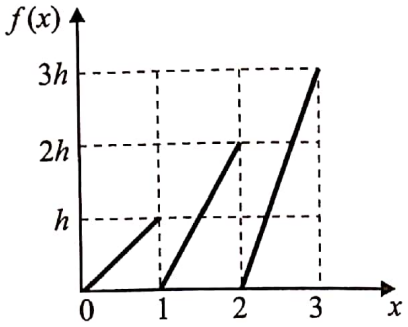
$\sigma = 3.697$ MPa

Thus, $\mu - 3\sigma = 26.575 - 3(3.697) = 15.485$ MPa

Hence, the number of specimen with compressive strength less than $\mu - 3\sigma$ is **0**.

**9.32 (A)**

**Given :**




According to the given graph, $h$ lies between 0 and 1, $2h$ lies between 1 and 2 and $3h$ between 2 and 3.

$$\underbrace{\int_0^1 f(x)\,dx}_{1^{st}\text{ interval}} + \underbrace{\int_1^2 f(x)\,dx}_{2^{nd}\text{ interval}} + \underbrace{\int_2^3 f(x)\,dx}_{3^{rd}\text{ interval}} = 1$$

$$\frac{1}{2}\times h\times 1 + \frac{1}{2}\times 2h\times(2-1) + \frac{1}{2}\times 3h\times(3-2) = 1$$

$$\frac{h}{2} + \frac{2h}{2} + \frac{3h}{2} = 1$$

$$0.5h + 1h + 1.5h = 1$$

$$3h = 1$$

$$h = \frac{1}{3}$$

Hence, the correct option is (A).

**9.33 0.5**

**Given :** Number of red balls in the box = 5

Number of white balls in the box = 1

Probability of first ball to be red $= \frac{5}{6}$

Probability of second ball to be red (out of remaining 5 balls) $= \frac{4}{5}$

Probability of third ball to be red (out of remaining 4 balls) $= \frac{3}{4}$



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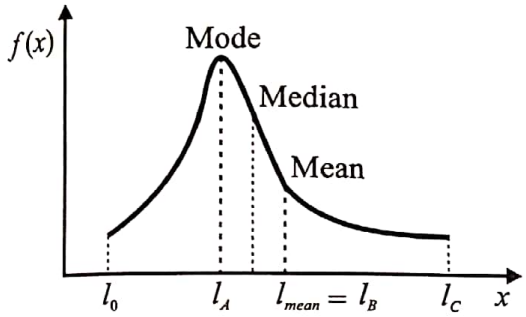
Hence, the probability of consecutively picking 3 red balls without replacement

$$=\frac{5}{6}\times\frac{4}{5}\times\frac{3}{4}=\frac{1}{2}=0.5$$

Hence, the probability is **0.5.**

**9.34 (C)**

In a right skew distribution curve,




From the above curve, mean is always right of the peak. Mode comes at the peak and median comes in between mean and mode.

Therefore, for a right skew curve (from the graph)

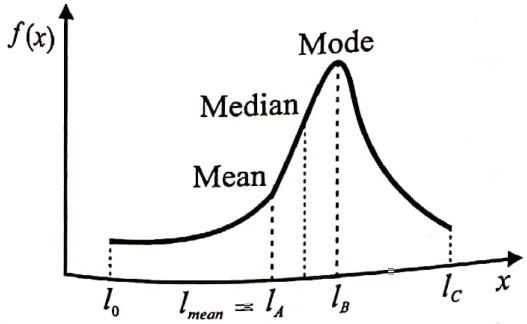
$$l_B > l_A$$

So, Mean > Mode

Hence, the correct option is (C).

**Key Point**

Similarly, in a left skew distribution curve, mean is always left of the peak. Mode comes at the peak and median comes in between mean and mode.




For a left skew curve (from the graph) $l_A < l_B$

Mean < Mode

Mean is less than mode and median.

**9.35 17.9**

The probability is of binominal distribution.

$n$ = number of times of occurrence of event = 5

$r$ = number of times of favorable case

Probability of flood discharge

$$= P(r \geq 1) = 1 - P(r = 0)$$

$$= 1 - {}^5C_0 p^0 q^5$$

$$= 0.25$$

So, $q = 0.944 = 1 - p$

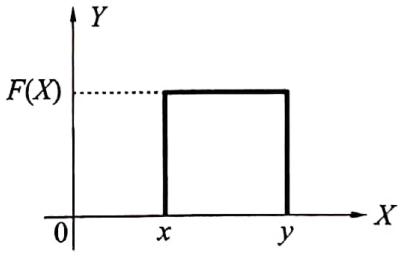
So, $p = 1 - 0.944 = 0.056$

Hence, return period

$$T = \frac{1}{p} = \frac{1}{0.056} = 17.85 \text{ years}$$

**9.36 B**

The probability density function of a continuous random variable distributed uniformly between $x$ and $y$ (for $y > x$) is $\frac{1}{y-x}$.




Total area of P.D.F. is 1

$$F(X) \times (y - x) = 1$$

$$F(X) = \frac{1}{y-x}$$

Hence, the correct option is (B).

## Instrumentation Engineering : IN

**9.1 (D)**

**Given :** A leap year which has 366 days.

One week contains only one Sunday.

So, the minimum number of Sundays in a leap year is equal to the number of weeks.

$$\text{Weeks} = \frac{366}{7} = 52\frac{2}{7}$$



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This mixed fraction shows that a leap year contains 52 weeks and 2 days.

These 2 days are consecutive and have following possibilities :

| Sunday | Monday |
|---|---|
| Monday | Tuesday |
| Tuesday | Wednesday |
| Wednesday | Thursday |
| Thursday | Friday |
| Friday | Saturday |
| **Saturday** | **Sunday** |

Let $A$ be the event of occurrence of 53 Sundays.

$$n(S)=7,\ n(A)=2$$

Required probability is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{2}{7}$$

Hence, the correct option is (D).

## 9.2 (C)

**Given :** Two persons have gone to a cyber café that has only three terminals.

Let $A$ be the event of selecting the same terminal.

### Method 1

Let $X$, $Y$ and $Z$ be the three terminals.

Since, each person can select only one of the three terminals, the following are the possibilities :

$$\begin{Bmatrix} X,X & X,Y & X,Z \\ Y,X & Y,Y & Y,Z \\ Z,X & Z,Y & Z,Z \end{Bmatrix}$$

Number of sample space, $n(S)=3\times3=9$

Number of events, when both friends do not select same terminals, is given by,

$$n(A)=6$$

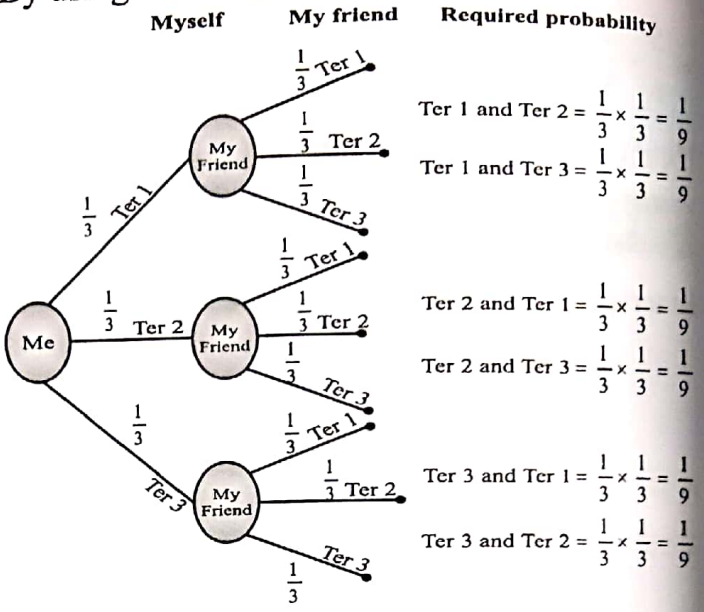
So, the require probability that both person do not select the same terminal is given by,

$$P(A)=\frac{n(A)}{n(S)}=\frac{6}{9}=\frac{2}{3}$$

Hence, the correct option is (C).

### Method 2

By using tree diagram



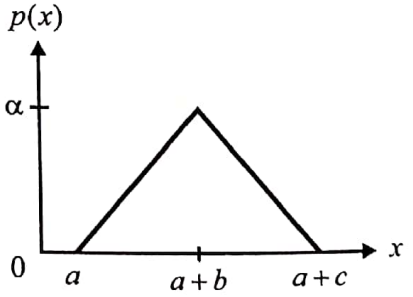

Probability that both friends will not select the same terminal,

$$=\frac{1}{9}+\frac{1}{9}+\frac{1}{9}+\frac{1}{9}+\frac{1}{9}+\frac{1}{9}=\frac{6}{9}=\frac{2}{3}$$

Hence, the correct option is (C).

## 9.3 (A)

**Given :** Probability density function (PDF) of $x$ is as shown below,




By the property of probability density function, area under the PDF curve is 1.

So, $$\frac{1}{2}\times(a+c-a)\times\alpha=1$$

$$\alpha=\frac{2}{c}$$

Hence, the correct option is (A).

## 9.4 (A)

**Given :** Two dices are simultaneously tossed.

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**Method 1**

Let $A$ be the event of having the sum of digits on top of two dices as an even number.

The sample space for two dices is given as follows :

$$S=\left\{\begin{array}{cccccc}(1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6)\\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6)\\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6)\\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6)\\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6)\\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6)\end{array}\right\}$$

So, $n(S)=36$

$n(A)=18$

Required probability is given by,

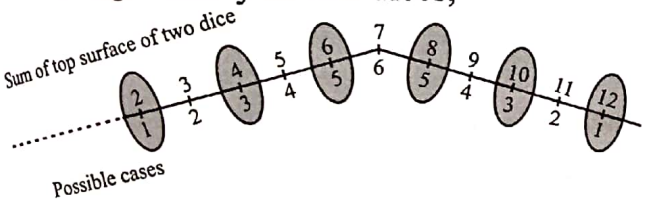
$$P(A)=\frac{n(A)}{n(S)}$$

$$P(A)=\frac{18}{36}=0.5$$

Hence, the correct option is (A).

**Method 2**

From triangle theory of two dices,




The table below shows the probabilities when the sum of top surfaces of two dice is an even number :

| Sum of top surface digit | Probability |
|---|---|
| 2 | $\frac{1}{36}$ |
| 4 | $\frac{3}{36}$ |
| 6 | $\frac{5}{36}$ |
| 8 | $\frac{5}{36}$ |
| 10 | $\frac{3}{36}$ |
| 12 | $\frac{1}{36}$ |

Required probability is given by,

$$P=\frac{1}{36}+\frac{3}{36}+\frac{5}{36}+\frac{5}{36}+\frac{3}{36}+\frac{1}{36}$$

$$P=\frac{18}{36}=0.5$$

Hence, the correct option is (A).

**Method 3**

There are total 36 outcomes possible.

The sum of digits on top of each die can either be even or odd.

Both of these are equiprobable events.

Therefore, $P(\text{sum is even})=P(\text{sum is odd})$

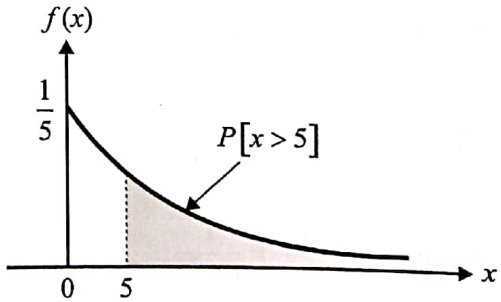
$$=\frac{1}{2}=0.5$$

Hence, the correct option is (A).

**9.5 (A)**

**Given :** An exponential distribution of a telephone conversation is given by,

$$f(x)=\frac{1}{5}e^{\frac{-x}{5}};$$




The probability for conversation to exceed 5 minutes (shown by the shaded portion) is given by,

$$P=\int_5^{\infty} f(x)\,dx=\int_5^{\infty}\frac{1}{5}e^{\frac{-x}{5}}\,dx$$

$$P=\frac{1}{5}\times(-5)\left[e^{\frac{-x}{5}}\right]_5^{\infty}=-1\left[e^{-\infty}-e^{-1}\right]$$



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$$P = -1\left[\frac{-1}{e}\right] = \frac{1}{e}$$

Hence, the correct option is (A).

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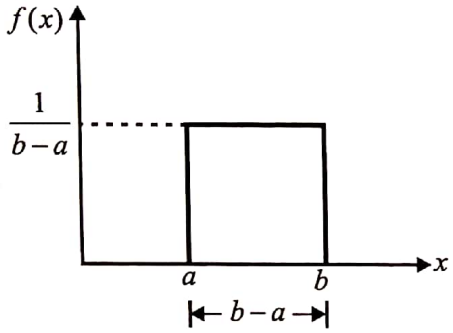
**9.6 (A)**

**Given :** A random variable is uniformly distributed in the interval (2, 10).

Consider a uniform distribution function $f(x)$ over the interval $(a, b)$.




By the property of probability density function,

$$f(x)\times(b-a)=1$$

$$f(x)=\frac{1}{b-a} \qquad (a<x<b)$$

Mean of $f(x)$ is given by,

$$\text{Mean} = \int_a^b x f(x)dx$$

$$\text{Mean} = \left[\frac{x^2}{2}\right]_a^b \times \frac{1}{b-a}$$

$$\text{Mean} = \frac{b^2-a^2}{2}\times\frac{1}{b-a}$$

$$\text{Mean} = \frac{b+a}{2}$$

Mean square value is given by,

$$\text{M.S.V.} = \int_a^b x^2 f(x)dx$$

$$\text{M.S.V.} = \left[\frac{x^3}{3}\right]_a^b \times \frac{1}{b-a}$$

$$\text{M.S.V.} = \frac{b^3-a^3}{3}\times\frac{1}{b-a}$$

$$\text{M.S.V.} = \frac{(b-a)(b^2+ab+a^2)}{3}\times\frac{1}{b-a}$$

$$\text{M.S.V.} = \frac{b^2+ab+a^2}{3}$$

Variance is given by,

$$\sigma^2 = \text{M.S.V} - (\text{Mean})^2$$

$$\sigma^2 = \frac{b^2+ab+a^2}{3} - \left(\frac{b+a}{2}\right)^2$$

$$\sigma^2 = \frac{4(b^2+ab+a^2)-3(b^2+2ab+a^2)}{12}$$

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Putting $a=2$ and $b=10$,

$$\sigma^2 = \frac{(10-2)^2}{12} = \frac{64}{12} = \frac{16}{3}$$

Hence, the correct option is (A)

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**9.7 (B)**

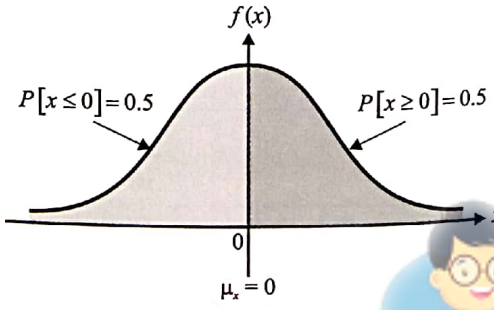
**Given :** A Gaussian distribution random variable with zero mean and standard deviation is $\sigma$.

The probability function for Gaussian distributed random variable is given by,

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{\frac{-(x-\mu_x)^2}{2\sigma_x^2}}$$

where, $\mu_x = \text{Mean} = 0$,

$\sigma_x = \text{Standard deviation}$



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$$f(x)=\frac{1}{\sqrt{2\pi\sigma_x^2}}e^{\frac{-x^2}{2\sigma_x^2}}$$

By the property of probability density function,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_{-\infty}^{0} f(x)dx = \int_{0}^{\infty} f(x)dx = \frac{1}{2}$$

[Mean is zero]

The cumulative distribution function is given by,

$$F(x) = \int_{-\infty}^{x^+} f(x)dx$$

At $x=0$,

$$F(0) = \int_{-\infty}^{0} f(x)dx = \frac{1}{2} = 0.5$$

Hence, the correct option is (B).

**9.8 (A)**

**Refer Soluion 9.8 [Electronics]**

**9.9 (C)**

Given : For an screening test,

(i) 12% of positive reports are incorrect.

(ii) 15% of negative reports are incorrect.

Let, $P$ = Number of positive reports

$N$ = Number of negative reports

Total number of reports $= P+N$

According to question,

Probability of getting a positive report is 0.01.

So, $\frac{P}{P+N}=0.01$

$N = 99P$ ....(i)

The number of incorrect positive reports

$$=\frac{12\times P}{100}=0.12P$$

and the number of incorrect negative reports

$$=\frac{15\times N}{100}=0.15N$$

Total number of incorrect reports

$$=0.12P+0.15N$$

$$=0.12P+0.15(99P)=14.97P$$

[From equation (i)]

Total number of reports

$$=P+N=P+99P=100P$$

The required probability

$$=\frac{\text{Number of incorrect reports}}{\text{Number of total reports}}$$

$$=\frac{14.97P}{100\ P}=0.1497$$

Hence, the correct option is (C).

**9.10 (C)**

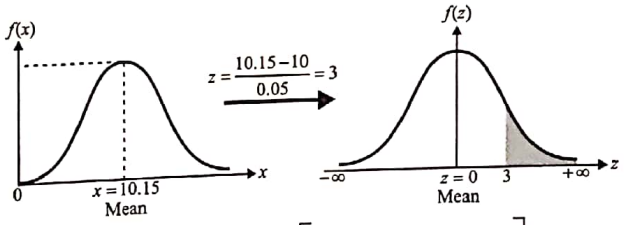
**Given :** $\sigma = 0.05\,\text{mm}$, $\mu_x = 10\,\text{mm}$

Then probability,

$$P(X>10.15)=Q\left(\frac{x-\mu_x}{\sigma_x}\right)$$

where, $x = 10.15\,\text{mm}$

$$P=Q\left(\frac{10.15-10}{0.05}\right)=Q(3)$$




$$P=Q(x)\big|_{x=3}\cong\left[\frac{1}{\sqrt{2\pi(x)^2}}e^{-x^2/2}\right]_{x=3}$$

$$\left[Q(z)\cong\frac{1}{\sqrt{2\pi z^2}}e^{-z^2/2};\ z>>1\right]$$

$$P\cong\frac{1}{\sqrt{2\pi\times 3^2}}e^{-x^2/2}$$

$$P\cong\frac{1}{\sqrt{18\pi}}e^{-4.5}\cong 1.477\times10^{-3}$$

So, number of measurement more than 10.15 mm is given by,

$N = P\times$ Total number of measurement

$N = 1.477\times10^{-3}\times10^4$

$N = 14.77 \simeq 15$

Hence, the correct option is (C).



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**9.11 (D)**

**Given :** 2 boxes each containing 5 numbered chips are as follows :

| Box 1 | | | | |
|---|---|---|---|---|
| 3 | 6 | 9 | 12 | 15 |

| Box 2 | | | | |
|---|---|---|---|---|
| 6 | 11 | 16 | 21 | 26 |

One chip from each box are drawn at random.

**Method 1**

Total number of possible combinations of chips is given by,

$$n(S) = {}^5C_1 \times {}^5C_1 = 25$$

For the product of numbers on two chips to be even, the possible combinations are,

(3,6) (3,16) (3,26) (6,6) (6,11)
(6,16) (6,21) (6,26) (9,6) (9,16)
(9,26) (12,6) (12,11) (12,16)
(12,21) (12,26) (15,6) (15,16)
(15,26)

Total number of favorable cases,

$$n(A) = 19$$

Required probability is given by,

$$P(A) = \frac{n(A)}{n(S)} = \frac{19}{25}$$

Hence, the correct option is (D).

**Method 2**

Let box 1 = $B_1$ and box 2 = $B_2$

Then, Even $(B_1) = 2$ (6, 12)

Odd $(B_1) = 3$ (3, 9 and 15)

Even $(B_2) = 3$ (6, 16 and 26)

Odd $(B_2) = 2$ (11, 21)

Let the cases when the product is even $= n(A)$

$$n(A) = [\text{Even } (B_1) \text{ and Odd } (B_2)]$$

or [Odd $(B_1)$ and Even $(B_2)$]

or [Even $(B_1)$ and Even $(B_2)$]

$$n(A) = (2\times2) + (3\times3) + (2\times3)$$

$$n(A) = 19$$

Total cases are given by,

$$n(S) = 5\times5 = 25$$

Then required probability is,

$$P(A) = \frac{n(A)}{n(S)} = \frac{19}{25}$$

Hence, the correct option is (D).

**9.12 (C)**

**Refer Soluion 9.12 [Electronics]**

**9.13 (B)**

**Refer Soluion 9.13 [Electronics]**

**9.14 (A)**

**Refer Soluion 9.11 [Electrical]**

**9.15 (A)**

**Refer Solution 9.1 [Instrumentation]**

**9.16 2**

**Given :** A probability density function is as given below,

$$f(x) = \frac{e^{-\frac{x}{2}}}{K} \text{ and } x \in [0, \infty]$$

By the property of probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \quad \Rightarrow \quad \int_0^{\infty} \frac{e^{\frac{-x}{2}}}{K}\,dx = 1$$

$$\frac{1}{K}\int_0^{\infty} e^{\frac{-x}{2}}\,dx = 1$$

$$-2\left[e^{\frac{-x}{2}}\right]_0^{\infty} = K$$

$$-2(e^{-\infty} - e^0) = K$$

$$K = 2$$

Hence, the value of constant $K$ is **2**.

**9.17 0.893**

**Given :** $P(A) = 0.15$, $P(B) = 0.05$, $P(C) = 0.1$ where, $A$, $B$ and $C$ shows the failure of respective machines.



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**Method 1**

The probabilities of successes of machines are given by,

$$P(\bar{A}) = 1 - P(A) = 1 - 0.15 = 0.85$$
$$P(\bar{B}) = 1 - P(B) = 1 - 0.05 = 0.95$$
$$P(\bar{C}) = 1 - P(C) = 1 - 0.1 = 0.9$$

Probability for job to complete successfully is,

$$= P[\text{(Machine A or Machine B success) and (Machine C success)}]$$
$$= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$
$$= (0.85)(0.9) + (0.95)(0.9) - (0.85)(0.95)(0.9)$$
$$= 0.89325 \simeq 0.893$$

Hence, the probability that a given job is successfully processed is **0.893**.

**Method 2**

The probability that job fails at both $A$ and $B$ is,

$$= P(A)P(B) = 0.15 \times 0.05 = 0.0075$$

Then, probability that job is fed to $C$ is,

$$= 1 - P(A)P(B) = 1 - 0.0075 = 0.9925$$

Probability that machine $C$ succeeds,

$$= 1 - P(C) = 1 - 0.1 = 0.9$$

The required probability is given by,

$$= 0.9925 \times 0.9 = 0.89325$$

Hence, the probability that a given job is successfully processed is **0.893**.

**9.18 (B)**

**Given :** Probability that a thermistor randomly picked is defective, $p = 0.1$.

Then, probability of being non defective is given by,

$$q = 1 - p = 1 - 0.1 = 0.9$$

From binomial distribution, probability that 3 out of 10 thermistors are defective is given by,

$$P(r) = {}^{n}C_r \times p^r \times q^{n-r}$$

where, $n = 10$, $r = 3$

The required probability is given by,

$$= {}^{10}C_3 (0.1)^3 (0.9)^7 = 0.057$$

Hence, the correct option is (B).

**9.19 4**

**Given :** Probability density function of random variable $X$ is given by,

$$P_X(x) = \begin{cases} e^{-x}; & x \geq 0 \\ 0; & \text{otherwise} \end{cases}$$

and $g_x(x) = e^{3x/4}$

Expected value of a function is given by,

$$E[g_x(x)] = \int_{-\infty}^{\infty} g_x(x) \times P_X(x)\, dx$$
$$E[g_x(x)] = \int_{0}^{\infty} e^{3x/4} \times e^{-x}\, dx$$
$$E[g_x(x)] = \int_{0}^{\infty} e^{-x/4} dx = -4 \left| e^{-x/4} \right|_0^{\infty}$$
$$E[g_x(x)] = -4(0 - 1) = 4$$

Hence, the expected value of function $g_x(x) = e^{3x/4}$ is **4**.

**9.20 (A)**

**Given :** An urn contains,

Total number of red balls = 5

Selected ball is placed back into the urn along with another ball of the same colour.

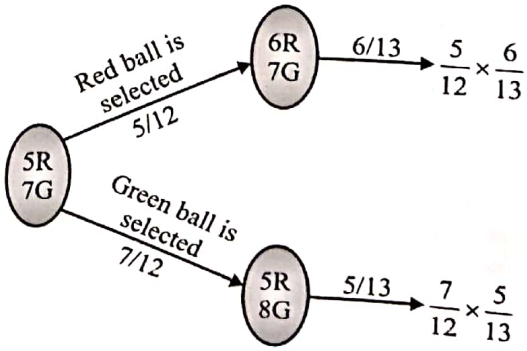
Total number of green balls = 7

Probability of drawing one green ball,

$$P(G) = \frac{7}{12}$$

Probability of drawing one red ball,

$$P(R) = \frac{5}{12}$$



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The required probability is given by,

$$= \frac{5}{12}\times\frac{6}{13}+\frac{7}{12}\times\frac{5}{13}=\frac{65}{156}$$

Hence, the correct option is (A).

**9.21 (C)**

**Given :** Probability of communication system having high fidelity = 0.81

Probability of communication system having both high fidelity and high selectivity = 0.18

According to Bayes' theorem,

Probability of occurrence of event $A$ when event $B$ has already happened, is given by,

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$$

where, $P(A\cap B)$ = Probability of system having both high fidelity and high selectivity = 0.18

$$P(B)=0.81$$

So, probability that a given system with high fidelity will have high selectivity,

$$P\left(\frac{A}{B}\right)=\frac{0.18}{0.81}=0.222$$

Hence, the correct option is (C).

**9.22 (D)**

**Given :** Two independent random variables $X$ and $Y$ have variances 1 and 2.

$$Z=X-Y$$

If two random variables $X$ and $Y$ are independent then,

$$\text{Var}(aX+bY)=a^2\text{Var}(X)+b^2\text{Var}(Y)$$

So, $\text{Var}(Z)=\text{Var}(X-Y)$

$$\text{Var}(X-Y)=1^2\text{Var}(X)+(-1)^2\text{Var}(Y)$$

$$\text{Var}(X-Y)=1^2\times1+(-1)^2\times2=3$$

Hence, the correct option is (D).

**9.23 (B)**

**Given :** In sequence of tossing of a fair coin, the outcomes of tosses are independent.

Probability of getting a head in a toss,

$$P(H)=\frac{1}{2}$$

Probability of getting a tail in a toss,

$$P(T)=\frac{1}{2}$$

To calculate the probability of getting a head for the third time in the fifth toss, first the probability of getting two heads in four tosses has to be calculated.

From binomial distribution,

$$P(r)={}^nC_r p^r q^{n-r}$$

where, $n=4$ and $r=2$

$$p=P(H)=\frac{1}{2},\quad q=P(T)=\frac{1}{2}$$

$$P(r)={}^4C_2\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{4-2}={}^4C_2\times\left(\frac{1}{2}\right)^4$$

Probability of getting a head in the fifth toss is given by,

$$P(F)=\frac{1}{2}$$

The probability of getting the head for the first time in the fifth toss is given by,

$$P=P(r)\times P(F)$$

$$P={}^4C_2\times\left(\frac{1}{2}\right)^4\times\frac{1}{2}=\frac{3}{16}$$

Hence, the correct option is (B).

**9.24 (B)**

**Given :** Bag $A$ contains 20 % red balls and 80% green balls.

Bag $B$ contains 30 % red balls, 60 % green balls and 10 % yellow balls.

Let both bags contain 10 balls each.

Hence, bag $A$ has 2 red balls and 8 green balls.

Bag $B$ has 3 red balls, 6 green balls and 1 yellow ball.

If balls of bags $A$ and $B$ are mixed then,

Total no. of balls = 10 + 10 = 20

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Total no. of red balls = 2 +3 = 5
Total no. of green balls = 8 + 6 = 14
Total no. of yellow balls = 1
Therefore, if one ball is picked at random, then the probability of this ball being red is,

$$P(R)=\frac{\text{No. of red balls}}{\text{Total no. of balls}}\times 100\%$$

$$P(R)=\frac{5}{20}\times 100\%=25\%$$

Hence, the correct option is (B).

**9.25 (A)**

Given, Box contains 8 Red balls and 8 Green balls.
Two balls are drawn randomly in succession without replacement
∴ Probability of first ball red and second ball green is

$$\frac{8}{16}\times\frac{8}{15}=\frac{4}{15}$$

Hence, the correct option is (A).

**9.26 (A)**

Given :

(i) $p(x)=\frac{A}{x^{\mu}}$ $\mu>1$ and $1<x<\infty$

(ii) $p(x)=0$ $-\infty<x<1$

For $p(x)$ to be probability density function, total probability is 1.

So, $\int_1^{\infty} p(x)dx=1$

$$\int_1^{\infty}\frac{A}{x^{\mu}}dx=1$$

$$A\frac{x^{-\mu+1}}{-\mu+1}\Big|_1^{\infty}=1$$

$$\frac{A(0-1)}{(1-\mu)}=1$$

$$A=\mu-1$$

Hence, the correct option is (A).

## Chemical Engineering : CH

**9.1 (C)**

**Given :** A pair of fair dice is rolled simultaneously.
The total number of possible outcomes are $n(S)=36$.
The figure given below shows the sample space of given event. The shaded items shows the desired outcomes in which the sum of number from the dice equals to six.

$$S=\begin{Bmatrix}(1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6)\\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6)\\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6)\\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6)\\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6)\\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6)\end{Bmatrix}$$

$n(S)=36,\ n(A)=5$

$$P(A)=\frac{n(A)}{n(S)}=\frac{5}{36}$$

Hence, the correct option is (C).

**9.2 (D)**

**Given :** A dice is rolled 4 times.

Probability of occurrence of six $=\frac{1}{6}$

Probability of not occurrence of six $=\frac{5}{6}$

According to binomial distribution the probability of $r$ trials is given by,

$$P(r)={}^{n}C_{r}p^{r}q^{n-r}$$

where, $p$ is the probability of success,
$q$ is the probability of failure

Here, $p=\frac{1}{6},\ q=\frac{5}{6},\ n=4,\ r=2$

$$P(r)={}^{4}C_{2}p^{2}q^{4-2}={}^{4}C_{2}\left(\frac{1}{6}\right)^{2}\times\left(\frac{5}{6}\right)^{2}$$

$$P(r)=\frac{4!}{2!2!}\times\frac{1}{36}\times\frac{25}{36}$$



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$$P(r)=\frac{4\times3\times2}{2\times2}\times\frac{1}{36}\times\frac{25}{36}=\frac{25}{6\times36}=\frac{25}{216}$$

Hence, the correct option is (D).

**9.3 (C)**

**Given :** Number of red balls = 6

Number of green balls = 4

Total numbers of balls = 6 + 4 = 10

Probability that one ball is picked randomly is green $=\frac{{}^4C_1}{{}^{10}C_1}=\frac{4}{10}$

Probability that second ball is picked randomly is green without replacement is $=\frac{{}^3C_1}{{}^9C_1}=\frac{3}{9}$

So, required probability $=\frac{4}{10}\times\frac{3}{9}=\frac{2}{15}$

Hence, the correct option is (C).

**9.4 (B)**

Box 1 : 3 blue balls, 4 red balls

Box 2 : 2 blue balls, 5 red balls

Let $B_1$ : The ball drawn from box 1.

$B_2$ : The ball drawn from box 2.

$R$ : The ball is red.

$$P(B_1)=P(B_2)=\frac{1}{2}$$

$$P\left(\frac{B_1}{R}\right)=\frac{P(B_1)\times P\left(\frac{R}{B_1}\right)}{P(B_1)\times P\left(\frac{R}{B_1}\right)+P(B_2)\times P\left(\frac{R}{B_2}\right)}$$

$$P\left(\frac{R}{B_1}\right)=P\text{ (a red ball is drawn from box 1)}=\frac{4}{7}$$

$$P\left(\frac{R}{B_2}\right)=P\text{ (a ball is drawn from box 2)}=\frac{5}{7}$$

Therefore, $P\left(\frac{B_1}{R}\right)=\frac{\frac{1}{2}\times\frac{4}{7}}{\frac{1}{2}\times\frac{4}{7}+\frac{1}{2}\times\frac{5}{7}}=\frac{4}{9}$

Hence, the correct option is (B).

**9.5 (C)**

**Given :** Two bags containing 10 coins each.

Probability of coins having values $(1,2,3,4)=\frac{4}{10}$

Probability of coins having values

$$(7,8,9,10)=\frac{4}{10}$$

So, required probability

$$=\left(\frac{4}{10}\right)\left(\frac{4}{10}\right)+\left(\frac{4}{10}\right)\left(\frac{4}{10}\right)$$

$$=\frac{32}{100}=\frac{8}{25}$$

Hence, the correct option is (C).

**9.6 (B) & (C)**

**Given :** A pair of fair dice is rolled three times.

Sum of the numbers on the two faces in 10 i.e. cases are (6, 4) (4, 6) (5, 5).

$P$(sum of the numbers on the two face is 10)

$$=\frac{3}{36}=\frac{1}{12}$$

$$P\text{(sum of number is not 10)}=1-\frac{1}{12}=\frac{11}{12}$$

By binomial distribution,

Probability of $r$ trials are given by,

$$P(r)={}^nC_r p^r q^{n-r}$$

Here, $n=3,\ r=1,\ p=\frac{1}{12},\ q=\frac{11}{12}$

$$P(r)={}^3C_1\left(\frac{1}{12}\right)^1\left(\frac{11}{12}\right)^{3-1}$$

$$P(r)=3\times\frac{1}{12}\times\left(\frac{11}{12}\right)^2=\frac{363}{1728}=\frac{121}{576}$$

Hence, the correct options are (B) and (C).

**9.7 (D)**

**Given :** Number of components purchased $=1000+2500+500=4000$

Total number of components likely to be defective $=5+5+2=12$



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By concept of conditional probability, the probability that a component was defective is supplied from $R$ is given by,

$$P\left(\frac{R}{D}\right)=\frac{P(R\cap D)}{P(D)}$$

$P(D)$ = Probability of defective component

$$P(D)=\frac{12}{4000}$$

$P(R\cap D)$ = Probability of component likely to be defective and purchase from $R$ $=\frac{2}{4000}$

$$P\left(\frac{R}{D}\right)=\frac{\frac{2}{4000}}{\frac{12}{4000}}=\frac{1}{6}$$

Hence, the correct option is (D).

**9.8 (B)**

Given : $f(x)=\frac{1}{\sqrt{2\pi}\sigma}\ e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty<x<\infty$

Differentiating $f(x)$ with respect to $x$,

$$f'(x)=\frac{1}{\sqrt{2\pi}\sigma}\times\left[-e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right]\times\frac{2(x-\mu)}{2\sigma^2}$$

$$f'(x)=\frac{1}{\sqrt{2\pi}\sigma}\times\left[-e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right]\times\frac{x-\mu}{\sigma^2}$$

$$f'(x)=-\frac{1}{\sqrt{2\pi}\sigma\times\sigma^2}\times\left[-e^{-\frac{(x-\mu)^2}{2\sigma^2}}\times(x-\mu)\right]$$

$$f''(x)=-\frac{1}{\sqrt{2\pi}\sigma\times\sigma^2}\left[e^{-\frac{(x-\mu)^2}{2\sigma^2}}\times1+(x-\mu)e^{-\frac{(x-\mu)^2}{2\sigma^2}}\times\frac{-2(x-\mu)}{2\sigma^2}\right]$$

$$f''(x)=-\frac{1}{\sqrt{2\pi}\sigma^{\frac{5}{2}}}\left\{e^{-\frac{(x-\mu)^2}{2\sigma^2}}\left[\frac{\sigma^2-(x-\mu)^2}{\sigma^2}\right]\right\}$$

$$f''(x)=-\frac{1}{\sqrt{2\pi}\sigma^{\frac{5}{2}}}\left\{e^{-\frac{(x-\mu)^2}{2\sigma^2}}\left[\frac{(\sigma-x+\mu)(\sigma+x-\mu)}{\sigma^2}\right]\right\}$$

Point of inflection is given by,

$f''(x)=0$

$$-\frac{1}{\sqrt{2\pi}\sigma^{\frac{5}{2}}}\left\{e^{-\frac{(x-\mu)^2}{2\sigma^2}}\left[\frac{(\sigma-x+\mu)(\sigma+x-\mu)}{\sigma^2}\right]\right\}=0$$

$(\sigma-x+\mu)=0\Rightarrow x=\sigma+\mu$

$(\sigma+x-\mu)=0\Rightarrow x=\mu-\sigma$

Hence, the correct option is (B).

**9.9 (B)**

**Given :** $P(r)=\frac{m^r}{r!}e^{-m}$

The first moment about the origin = Mean

For a Poisson distribution for $r$ trials,

$$P(r)=\frac{e^{-m}m^r}{r!}$$

So, mean = $m$

Hence, the correct option is (B).

**9.10 (A)**

**Given :** A die is rolled.

So, sample space $S=\{1, 2, 3, 4, 5, 6\}$

Event :

$R\rightarrow$ Obtaining a number less than or equal to 5

$S\rightarrow$ Obtaining an odd number.

$$P(R)=\frac{5}{6}$$

$$P(S)=\frac{3}{6}=\frac{1}{2}$$

Since, Events $R$ and $S$ are dependent events

So, $P(R\cap S)=\frac{3}{6}=\frac{1}{2}$

Now, $P\left(\frac{R}{S}\right)=\frac{P(R\cap S)}{P(S)}=\frac{1/2}{1/2}=1$

$$P\left(\frac{S}{R}\right)=\frac{P(R\cap S)}{P(R)}=\frac{1/2}{5/6}$$

$$=\frac{1}{2}\times\frac{6}{5}=0.6$$

Hence, the correct option is (A).



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**9.11 (B)**

**Given :** $X$ and $Y$ are independent random variables.

For random variable $X$,

$$n = 5,\ P = \frac{1}{2}$$

According to binomial distribution,

$$P(X = Y) = {}^{n}C_{r}\,(p)^{r}\,(q)^{n-r}$$

where, $p$ = Probability of success

$q$ = Probability of failure

For random variable $Y$,

| $Y$ | 1 | 2 |
|---|---|---|
| $P(Y)$ | $\frac{1}{2}$ | $\frac{1}{2}$ |

Since, $Y$ takes values 1 and 2 with equal probability.

Hence, $P(1) = \frac{1}{2}$

$$P(2) = \frac{1}{2}$$

For $X = Y$

$$P(X) = P(Y)$$

$$P(X) = P(Y) = P(1) + P(2)$$

$$P(X) = {}^{5}C_{1}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{4} + {}^{5}C_{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{3}$$

$$P(X) = 5 \times \frac{1}{2^5} + \frac{5!}{3! \times 2!} \times \frac{1}{2^5}$$

$$P(X) = \frac{5}{32} + \frac{5 \times 4 \times 3!}{3! \times 2!} \times \frac{1}{32}$$

$$P(X) = \frac{5}{32} + \frac{10}{32} = \frac{15}{32}$$

Hence, the correct option is (B).

**9.12 (B)**

**Given :** Number of red balls = 3

Number of black balls = 2

Total number of balls = 3 + 2 = 5

Since, 4 balls are removed one by one without replacement therefore the probability of the ball remaining in the box being red is

$$\text{Required probability} = \frac{{}^{3}C_{2} \times {}^{2}C_{2}}{{}^{5}C_{4}}$$

$$= \frac{\frac{3!}{1!2!} \times 1}{\frac{5!}{1!4!}} = \frac{3 \times 1}{5} = \frac{3}{5}$$

Hence, the correct option is (B).

**9.13 (C)**

**Given:** A box having 10 identical compartments

Number of red balls = 6

Number of blue balls = 2

Arrangement of 8 balls in 10 compartment is ${}^{10}P_{8}$.

Arrangement within 6 red ball is not countable.

Similarly, arrangement within 2 blue ball is not countable.

Therefore, number of different possible arrangements are

$$= \frac{{}^{10}P_{8}}{6!2!} = \frac{\frac{10!}{2!}}{6!2!} = \frac{10 \times 9 \times 8 \times 7 \times 6!}{2! \times 6! \times 2!} = 1260$$

Hence, the correct option is (C).

**9.14 (B)**

**Given :**

**For calculation of median :**

Arranging the given observation is ascending/ descending order.

If number of observation, $n$ = Odd

Then, median = Middle term

If number of observation, $n$ = even

Then, median = Mean of two middle terms.

Therefore, arranging the given observation is ascending order :

4, 9, 11, 15, 18, 18

Here, number of terms, $n = 6$ (even)



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So, Median $= \frac{11+15}{2} = \frac{26}{2} = 13$

**For calculation of mode :**

Mode = The term which is repeated maximum number of times in the given observation.

Here, the term 18 is repeated 2 times.

So, mode = 18

Hence, the correct option is (B).

**9.15 0.083**

Probability of getting 4 in first roll $= \frac{1}{6}$

According to the question, $P(\text{getting} < 4)$

$$= P(1) + P(2) + P(3) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

Thus, required probability $= \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

$$P = 0.083$$

Hence, the probability is **0.083**.

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**9.16 (C)**

**Given :** For machine $M_1$, $P(M_1) = 0.6$,

For machine $M_2$, $P(M_2) = 0.4$

$P\left(\frac{D}{M_1}\right)$ = Probability of machine $M_1$ that is defective

$$P\left(\frac{D}{M_1}\right) = \frac{2}{100}$$

$P\left(\frac{D}{M_2}\right)$ = Probability of machine $M_2$ that is defective

$$P\left(\frac{D}{M_2}\right) = \frac{3}{100}$$

By Bayes' theorem,

$P\left(\frac{M_2}{D}\right) = P$ (Component drawn is defective and it was found to be manufactured by $M_2$)

$$P\left(\frac{M_2}{D}\right) = \frac{P(M_2) \times P\left(\frac{D}{M_2}\right)}{P(M_1) \times P\left(\frac{D}{M_1}\right) + P(M_2) \times P\left(\frac{D}{M_2}\right)}$$

$$P\left(\frac{M_2}{D}\right) = \frac{0.4 \times 0.03}{0.6 \times 0.02 + 0.4 \times 0.03}$$

$$P\left(\frac{M_2}{D}\right) = \frac{0.012}{0.012 + 0.012} = \frac{1}{2} = 0.5$$

Hence, the correct option is (C).

**9.17 (C)**

**Given :** The density function of the normal distribution is given by :

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where, $\mu$ = Mean

$\sigma$ = Standard deviation

Given : $f_1(x) = e^{-\pi x^2}$

On comparing with the standard equation,

$$\mu_1 = 0 \text{ and } \frac{1}{\sqrt{2\pi\sigma_1^2}} = 1$$

$$\sigma_1^2 = \frac{1}{2\pi}$$

Also, $f_2(x) = \frac{1}{2\pi} e^{\left\{-\frac{1}{4\pi}(x^2+2x+1)\right\}}$

$$f_2(x) = \frac{1}{2\pi} e^{-\frac{1}{4\pi}(x+1)^2}$$

$$f_2(x) = \frac{1}{2\pi} e^{-\frac{1}{4\pi}(x-(-1))^2}$$

On comparing,

$$\mu_2 = -1 \text{ and } \frac{1}{\sqrt{2\pi\sigma_2^2}} = \frac{1}{2\pi}$$

$$\sigma_2^2 = 2\pi$$

So, $\mu_1 > \mu_2$ and $\sigma_1^2 < \sigma_2^2$

Hence, the correct option is (C).



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**9.18 7**

**Given :** Two fair dice are rolled.

The total number of possible outcomes are 36.

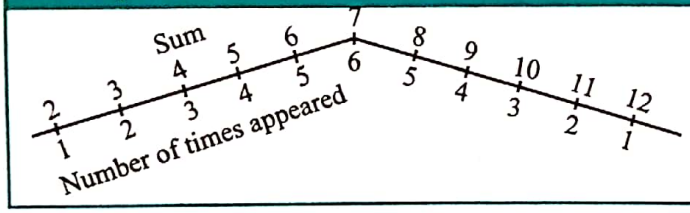
The figure given below shows the sample space of given event. The shaded items shows the desire outcomes in which the sum of the numbers appearing on dice is highest.

$$S = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

When the total sum of top surface of fair dice is 7, then probability $= \frac{6}{36}$, which is highest among them.

Hence, the probability is highest for the outcome of 7.

**Key Point**




**9.19 4**

We can't use linear regression as $y = mx^2$ is non-linear. So, non-linear regression approach needs to be used as under.

**Given :**

| $x$ | 1 | $\sqrt{2}$ | $\sqrt{3}$ |
|---|---|---|---|
| $y$ | 2 | 5 | 8 |

**By Non-linear Regression Method :**

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \Sigma x_i^2 y_i \end{bmatrix}$$

For 2nd order polynomial of the form

$$y = a_0 + a_1 x + a_2 x^2$$

Above matrix is used.

$$\Sigma x_i = 1 + \sqrt{2} + \sqrt{3} = 4.146$$

$$\Sigma x_i^2 = 1 + 2 + 3 = 6$$

$$\Sigma x_i^3 = 1^3 + (\sqrt{2})^3 + (\sqrt{3})^3 = 9.024$$

$$\Sigma x_i^4 = 1 + 4 + 9 = 14$$

$$\Sigma y_i = 2 + 5 + 8 = 15$$

$$\Sigma x_i y_i = 1 \times 2 + \sqrt{2} \times 5 + \sqrt{3} \times 8 = 22.92$$

$$\Sigma x_i^2 y_i = 2 + 10 + 24 = 36$$

$$AX = B$$

$$X = A^{-1}B$$

$$\begin{bmatrix} 3 & 4.146 & 6 \\ 4.146 & 6 & 9.024 \\ 6 & 9.024 & 14 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 15 \\ 22.92 \\ 36 \end{bmatrix}$$

$$3a_0 + 4.146a_1 + 6a_2 = 15 \quad \ldots(i)$$

$$4.146a_0 + 6a_1 + 9.024a_2 = 22.92 \quad \ldots(ii)$$

$$6a_0 + 9.024a_1 + 14a_2 = 36 \quad \ldots(iii)$$

Solving equations (i), (ii) and (iii),

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0.6839 \\ -2.59 \\ 3.9481 \end{bmatrix}$$

$$y = 0.6839 - 2.59x + 3.9481x^2$$

On comparing with $y = mx^2 \Rightarrow$ best possible answer that fits equation is

$$m = 3.94 \approx 4$$

Hence, the value of $m$ is 4.

**9.20 0.36**

**Given :** Number of red balls = 6

Number of white balls = 4

Total number of balls = 6 + 4 = 10



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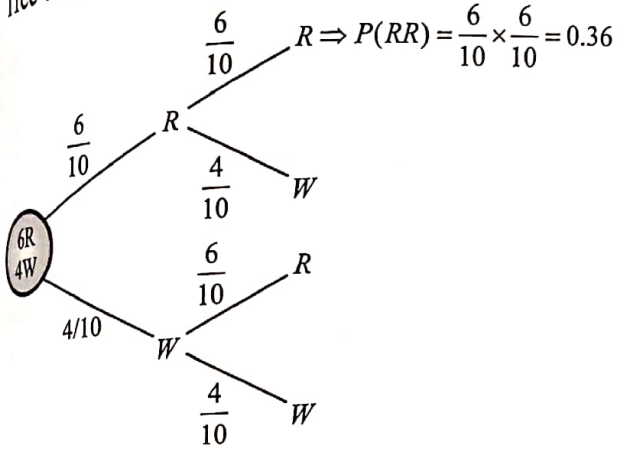
$$P(\text{red balls}) = \frac{6}{10}$$

$$P(\text{white balls}) = \frac{4}{10}$$

Tree diagram,




A ball is picked at random and replaced in the box.

$$P(\text{Both the ball being red}) = \frac{6}{10} \times \frac{6}{10} = 0.36$$

Hence, the probability of both the balls being red is 0.36.

**9.21 (C)**

Given : Marks obtained by set of students,

38, 84, 45, 70, 75, 60, 48

$$\text{Mean} = \frac{\text{Sum of all the observations}}{\text{Total number of observations}}$$

$$= \frac{38+84+45+70+75+60+48}{7}$$

$$= \frac{420}{7} = 60$$

For calculation of median :

Arrange the given observation in ascending/ descending order.

Median = Middle term,

(if number of observations = Odd)

Median = Mean of two middle terms,

(if number of observations = Even)

Arranging the given observations in ascending order,

38, 45, 48, 60, 70, 75, 84

Here, $n = 7$ (odd)

where, $n =$ number of observations

So, median = Middle term = 60

Hence, the correct option is (C).

**9.22 (C)**

**Given :** There are two electronic circuits (ECs).

Probability of failure of each EC = 0.1

So, probability of not failure of each EC $= 1 - 0.1 = 0.9$

Hence, probability of watch functioning for 1 year without failure $= 0.9 \times 0.9 = 0.81$.

Hence, the correct option is (C).

**9.23 0.25**

Two dice are rolled.

$\therefore$ Total number of outcomes,

$$n(s) = 6^2 = 36$$

Let 'A' denote the sum of the outcomes of the two dice is divisible by 4 {4, 8 or 12}

$\therefore$ No. of favorable cases,

$$n(A) = 3+5+1 = 9$$

$[\because (1,3)(2,2)(3,1) | (2,6)(3,5)(4,4)(5,3)(6,2) | (6,6)]$

$\therefore$ Required probability,

$$P(A) = \frac{n(A)}{n(S)} \Rightarrow P(A) = \frac{9}{36} = \frac{1}{4} = 0.25$$



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## Production & Industrial : PI

**9.1 (D)**

The sample space for two dice are as follows,

$$S=\begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

The sum of numbers that exceed 8 is given in shaded form, the favorable outcome is 10.

So, the required probability $=\frac{10}{36}$.

Hence, the correct option is (D).

**9.2 (A)**

Probability is given by,

$$P[a\le t\le b]=\int_a^b f(t)\,dt$$

$$P(100\le t\le 200)=\int_{100}^{200}\alpha e^{-\alpha t}dt=\left[\alpha\frac{e^{-\alpha t}}{-\alpha}\right]_{100}^{200}$$

$$P(100\le t\le 200)=-e^{-200\alpha}+e^{-100\alpha}$$

$$P(100\le t\le 200)=e^{-100\alpha}-e^{-200\alpha}$$

Hence, the correct option is (A).

**9.3 (D)**

By the property of probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx=1 \qquad \ldots(i)$$

So, $\int_0^2 K(5x-2x^2)\,dx=1$

$$K\left[\frac{5x^2}{2}-2\frac{x^3}{3}\right]_0^2=1$$

$$K=\frac{3}{14}$$

From equation (i),

$$P(X>1)=\int_1^{\infty} f(x)\,dx$$

$$P(X>1)=\int_1^2\frac{3}{14}(5x-2x^2)\,dx$$

$$P(X>1)=\frac{3}{14}\left[\frac{5x^2}{2}-\frac{2x^3}{3}\right]_1^2=\frac{17}{28}$$

Hence, the correct option is (D).

**9.4 (A)**

Probability of drawing an ace card from a deck of cards $=\frac{{}^4C_1}{{}^{52}C_1}=\frac{4}{52}=\frac{1}{13}$.

Here the point is to be noted that, the cards are drawn with replacement.

Hence the probability for both cards are same $=\frac{1}{13}$.

So, the required probability is given by,

$$=P(E)=\frac{1}{13}\times\frac{1}{13}=\frac{1}{169}$$

Hence, the correct option is (A).

**9.5 (A)**

Given that the random variable $X$ and probability distribution $P(X)$ are as follows :

| $X$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(X)$ | $\frac{2+5P}{5}$ | $\frac{1+3P}{5}$ | $\frac{1.5+2P}{5}$ |

Sum of probability = 1

$$P(1)+P(2)+P(3)=1$$

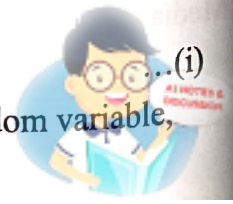
$$\frac{2+5P}{5}+\frac{1+3P}{5}+\frac{1.5+2P}{5}=1$$

$$\frac{4.5+10P}{5}=1$$

$$P=0.05 \qquad \ldots(i)$$

By the concept of discrete random variable,

$$E(X)=\sum X\,P(X)$$



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$$E(X)=1\left(\frac{2+5P}{5}\right)+2\left(\frac{1+3P}{5}\right)+3\left(\frac{1.5+2P}{5}\right)$$

$$E(X)=\frac{8.5+17P}{5}$$

From equation (i),

$$E(X)=\frac{8.5+(17\times 0.05)}{5}$$

$$E(X)=1.87$$

Hence, the correct option is (A).

**9.6 (C)**

Given : Mean $=2$

By Poisson's distribution, probability of $r$ trials is given by,

$$P(X=r)=\frac{e^{-\lambda}\lambda^{r}}{r!}$$

where, $\lambda=$ mean $=2$

Required probability,

$$P(X\geq 1)=1-P(X=0)$$

$$P(X\geq 1)=1-\frac{e^{-\lambda}(\lambda)^{0}}{0!}$$

$$P(X\geq 1)=1-e^{-2}=0.865$$

Hence, the correct option is (C).

**9.7 (B)**

Given :

$X_1,\ldots X_{100}$ are Bernoulli random variables,

Probability of success $(p)=0.6$

So, probability of failure $(q)=1-0.6=0.4$

Random variable $Y=\sum_{i=1,\ldots 100} X_i$

By central limit theorem,

Mean = Sum of mean of individual random variables,

Variance = Sum of variance of individual random variables.

So, Mean $=np=100\times 0.6=60$

Variance $=npq=100\times 0.6\times 0.4=24$

Hence, the correct option is (B).

**9.8 (C)**

Player $X$ wins the game if he tosses a coin and head appears.

The probability of player $X$ getting a head $P(X)$ and getting a tail $P(\bar{X})$ respectively is,

$$P(X)=\frac{1}{2}, P(\bar{X})=\frac{1}{2}$$

Since, player $X$ and $Y$ toss a coin alternatively. So, the probability of player $X$ winning the game is given by,

$$P(X \text{ win})=P(X)+P(\bar{X})P(\bar{Y})P(X)+P(\bar{X})P(\bar{Y})P(\bar{X})P(\bar{Y})P(X).....$$

where, $P(\bar{Y})=$ Probability of player $Y$ getting a tail.

Also, $P(\bar{Y})=\frac{1}{2}$

$$P(X \text{ win})=\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}+.....\infty$$

The above probability represents the geometric progression of infinite terms.

Now, the summation of infinite terms of G.P. is given by,

$$P(X \text{ win})=\frac{a}{1-r}=\frac{\frac{1}{2}}{1-\frac{1}{2^2}}=\frac{2}{3}$$

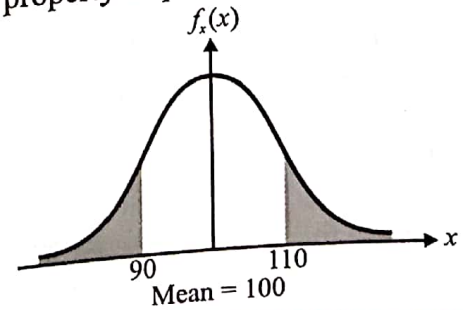
Hence, the correct option is (C).

**9.9 (A)**

**Given :** Normal distribution with mean = 100,

$P(x\geq 110)=\alpha$

By the property of probability density function,



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Above curve is symmetric about its mean $\mu = 100$.

So, $P(x \le 90) = P(x \ge 110) = \alpha$

$$\int_{-\infty}^{\infty} f_x(x)\,dx = 1$$

$$P(x<90) + P(90<x<110) + P(x>110) = 1$$

$$\alpha + P(90<x<110) + \alpha = 1$$

$$P(90<x<110) = 1 - 2\alpha$$

Hence, the correct option is (A).

## 9.10 (B)

**Given :**

| | $X=1$ | $X=2$ |
|---|---|---|
| $Y=1$ | 0.2 | 0.3 |
| $Y=2$ | 0.3 | 0.1 |
| $Y=3$ | 0.1 | |

Random variables $X$ and $Y$,

$$P\{X=1, Y=2\} = 0.3$$

By conditional probability,

$$P\left(\frac{X}{Y}\right) = \frac{P(X,Y)}{P(Y)}$$

So, $$P\left(\frac{X=2}{Y=2}\right) = \frac{P(X=2, Y=2)}{P(Y=2)}$$

$$= \frac{0.1}{0.3+0.1} = \frac{0.1}{0.4}$$

$$P\left(\frac{X=2}{Y=2}\right) = 0.25$$

Hence, the correct option is (B).

## 9.11 (D)

**Given :** The mean of Poisson's distribution $m = 2$.

According to Poisson's distribution, the probability for any event is given by,

$$P(X=r) = \frac{e^{-m}\, m^r}{r!}$$

So, the required probability is given by,

$$P(X \ge 2) = 1 - [P(r=0) + P(r=1)]$$

$$P(X \ge 2) = 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!}\right]$$

$$P(X \ge 2) = 1 - e^{-2}[1+2] = 1 - 3e^{-2}$$

Hence, the correct option is (D).

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## 9.12 (C)

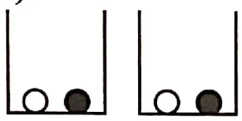
Given that two white and two black balls are kept in two bins.

The probability to choose one bin is $= \frac{1}{2}$.

Let the probability of getting a white ball be $P$.

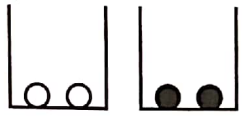
So, we have the following cases for required probability for given arrangements to get white ball,

**From option (A) :**



$$P = \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} = \frac{1}{2}$$

**From option (B) :**



$$P = \frac{1}{2}\times\frac{2}{2} + \frac{1}{2}\times 0 = \frac{1}{2}$$

**From option (C) :**

$$P = \frac{1}{2}\times\frac{1}{1} + \frac{1}{2}\times\frac{1}{3} = \frac{2}{3}$$

**From option (D) :**

$$P = \frac{1}{2}\times 0 + \frac{1}{2}\times\frac{2}{3} = \frac{1}{3}$$

Therefore, the probability calculated in option (C) is the highest.

Hence, the correct option is (C).



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**9.13 (B)**

Given : Average number of event during a year is 3. Therefore for two year average number of event is $m = 6$.

For poisson distribution probability is given by,

$$P = \frac{e^{-m} m^r}{r!}$$

Probability of occurance of event not more than two event (By poisson distribution) is given by,

$$P = P(r=0) + P(r=1) + P(r=2)$$

$$P = \frac{e^{-m} m^0}{0!} + \frac{e^{-m} m^1}{1!} + \frac{e^{-m} m^2}{2!}$$

$$P = \frac{e^{-6} 6^0}{0!} + \frac{e^{-6} 6^1}{1!} + \frac{e^{-6} 6^2}{2!}$$

$$P = e^{-6}\left[1 + 6 + \frac{6^2}{2!}\right]$$

$$P = e^{-6}\left[1 + 6 + \frac{36}{2}\right]$$

$$P = 0.0619 \approx 0.062$$

Hence, the correct option is (B).

**9.14 (B)**

**Refer Solution 9.28 [Mechanical]**

**9.15 (D)**

**Refer Solution 9.27 [Mechanical]**

**9.16 (D)**

**Refer Solution 9.29 [Mechanical]**

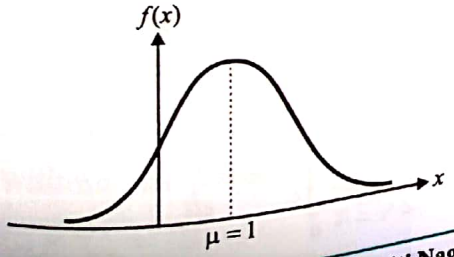
**9.17 (B)**

Given : Mean $(\mu) = 1$, Standard deviation $(\sigma) = 2$

In normal distribution,

$$P(x \geq \mu) = P(x \leq \mu) = 0.5$$

$$P(x \geq 1) = P(x \leq 1) = 0.5$$




So, $P(x \leq 0) = P(x \leq 1) - P(0 \leq x \leq 1)$

$$P(x \leq 0) = 0.5 - P(0 \leq x \leq 1)$$

Therefore, $P(x \leq 0)$ is greater than zero but less than 0.5.

Hence, the correct option is (B).

**9.18 (C)**

**Given :**

Number of toys manufactured everyday = 1000

Number of toys that are defective = 10% of 1000 = 100

Number of defected toys that can be made defect free = 40% of defective toys

$$= \frac{40}{100} \times 100 = 40$$

So, total number of defect free toys

$$= 1000 - 100 + 40 = 940$$

Hence, the correct option is (C).

**9.19 1.2**

**Given :** Mean $(\mu) = 80$, Standard deviation $(\sigma) = 12$, Number of observations $(n) = 100$

By the formula of standard error of mean,

$$= \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{100}} = 1.2$$

Hence, the standard error of mean is **1.2**.

**9.20 (C)**

**Given :**

| S. No. | Marks obtained | Number of students |
|---|---|---|
| 1. | 25 | 20 |
| 2. | 30 | 20 |
| 3. | 35 | 40 |
| 4. | 40 | 20 |

$$\text{Mean} = \frac{\Sigma x_i f_i}{\Sigma f_i}$$

$$\text{Mean} = \frac{(20 \times 25) + (20 \times 30) + (40 \times 35) + (20 \times 40)}{20 + 20 + 40 + 20}$$

$$= \frac{500 + 600 + 1400 + 800}{100} = \frac{3300}{100} = 33$$



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After the calculation of mean value we can reject two options that is (B) and (D).

Now the calculation for median, for 100 data (i.e. even)

Median = The average of marks of $50^{th}$, $51^{st}$ observations

$$=\frac{35+35}{2}=35$$

Mode = The marks with highest frequency = 35.

Hence, the correct option is (C).

## 9.21 (D)

Let the probability of rain $P(A)=\frac{7}{10}$

The probability of no rain,

$$P(\bar{A})=1-P(A)=1-\frac{7}{10}=\frac{3}{10}$$

The probability of loss if it rains, $P\left(\frac{B}{A}\right)=\frac{8}{10}$

The probability of loss if no rain, $P\left(\frac{B}{\bar{A}}\right)=\frac{1}{10}$

The probability of no loss on a rainy day,

$$=\frac{7}{10}\times\frac{2}{10}$$

The probability of no loss on a non-rainy day,

$$=\frac{3}{10}\times\frac{9}{10}$$

The probability of no loss on a rainy season,

$$P(B)=\left(\frac{3}{10}\times\frac{9}{10}\right)+\left(\frac{7}{10}\times\frac{2}{10}\right)$$

By Bayes' theorem,

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$$

The required probability is given by,

$$=\frac{\left(\frac{3}{10}\times\frac{9}{10}\right)}{\left(\frac{3}{10}\times\frac{9}{10}\right)+\left(\frac{7}{10}\times\frac{2}{10}\right)}=\frac{27}{41}$$

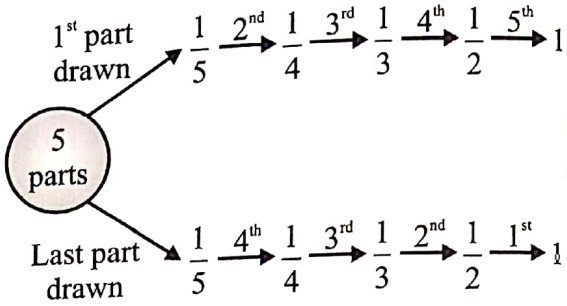
Hence, the correct option is (D).

## 9.22 (A)

There are two ways to correctly assemble the product,

(i) first part to last part

(ii) last part to first part




$P$ (getting correct sequence)

$$=\frac{1}{5}\times\frac{1}{4}\times\frac{1}{3}\times\frac{1}{2}\times1+\frac{1}{5}\times\frac{1}{4}\times\frac{1}{3}\times\frac{1}{2}\times1$$

$$P=\frac{2}{5!}$$

Hence, the correct option is (A).

## 9.23 (C)

**Given :** A fair coin is tossed $N$ times.

$$P(\text{tails})=P(\text{not getting heads})=\left(\frac{1}{2}\right)^N$$

Hence, the correct option is (C).

## 9.24 (B)

**Given :** $f_x(x)=\frac{1}{\sqrt{8\pi}}e^{-\left\{\frac{(x-1)^2}{8}\right\}}$

**Method 1**

$$I=\int_1^\infty f_x(x)\,dx$$

$$I=\frac{1}{2\sqrt{2\pi}}\int_1^\infty e^{-\frac{1}{2}\left(\frac{x-1}{2}\right)^2}dx$$

Let, $\frac{x-1}{2}=t$

$$\frac{dx}{2}=dt$$

$$dx=2dt$$

$$I=\frac{2}{2\sqrt{2\pi}}\int_0^\infty e^{-\left(\frac{1}{2}\right)t^2}dt$$

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Again let, $\frac{1}{2}t^2 = V$

$$t^2 = 2V$$

$$2t\,dt = 2dV$$

$$dt = \frac{dV}{\sqrt{2V}}$$

$$I = \frac{1}{2\sqrt{\pi}} \int_0^\infty e^{-V} V^{-\left(\frac{1}{2}\right)} dV$$

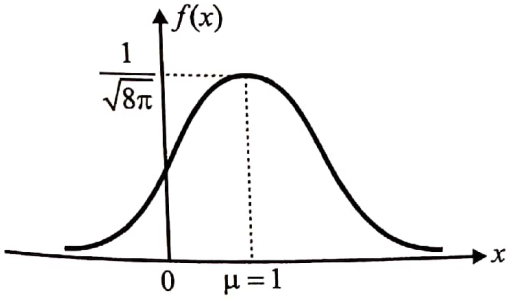
$$\left[\text{Since, } \int_0^\infty e^{-x} x^{n-1} dx = \sqrt{\pi}\right]$$

$$I = \frac{1}{2}$$

Hence, the correct option is (B).

**Method 2**

Given function is a normal distribution function with mean = 1 and variance = 4.




The distribution function is divided into two equal parts which are equiprobable.

Therefore, $\int_1^\infty f_x(x)\,dx = \frac{1}{2}$

Hence, the correct option is (B).

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**9.25 70**

**Given :** Total number of machines = 10

Total number of defective machines = 2

Maximum number of combinations that can be obtained without any defective machine,

$${}^8C_4 \times {}^2C_0 = 70$$

Hence, the maximum number of combinations are **70**.

**9.26 3.5**

**Given :** Data sample in ascending order,

2, 3, 3, 4, 6, 7

Here, the number of samples is even and for even number of samples median is equal to the mean of two middle items.

Therefore, median $= \frac{3+4}{2} = \frac{7}{2} = 3.5$

Hence, the median weight is **3.5 kg**.

**9.27 0.5**

**Given :** $P(F) = 0.3$ and $P(G) = 0.4$

The probability that both event occur simultaneously is

$$P(F \cap G) = 0.2$$

The probability of occurrence of atleast one event is

$$P(F \cup G) = P(F) + P(G) - P(F \cap G)$$

$$P(F \cup G) = 0.3 + 0.4 - 0.2 = 0.5$$

Hence, the probability of occurrence of atleast one event $P(F \cup G)$ is **0.5**.

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